

A Guided Course in Algebraic Topology
from surfaces to transferred structures

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How to use this book

Part I

Spaces, groups, and the first invariants

Chapter 1

Basepoints, conjugacy, and the change-of-basepoint action

Every first course proves that a path from x_0 to x_1 induces an isomorphism $\pi_1(X, x_1) \rightarrow \pi_1(X, x_0)$, and then quietly starts writing $\pi_1(X)$. This chapter is about the word *quietly*. The isomorphism is not canonical: it depends on the path, and two paths differ by conjugation by a loop. That ambiguity is not a blemish to be removed. It is the first appearance of a structure that organises the whole subject — in [Chapter 3](#) it becomes the statement that conjugate subgroups classify isomorphic covering spaces, and in [Chapter 15](#) it returns as the monodromy action attached to a fibration.

Throughout the book we use the standard results of a first course without comment: $\pi_1(S^1) \cong \mathbb{Z}$, the van Kampen theorem, the existence and uniqueness of lifts against a covering space. What such a course tends to leave as folklore — exactly which isomorphisms are canonical, and what happens to a basepoint during a homotopy — is what we set down first, because every later chapter leans on it.

Conventions. A path is a map $\gamma: I \rightarrow X$; we write $\gamma(0)$ for its initial and $\gamma(1)$ for its terminal point. The concatenation $\gamma \cdot \delta$, defined when $\gamma(1) = \delta(0)$, traverses γ first. The reverse path is $\bar{\gamma}(s) = \gamma(1 - s)$, and c_x is the constant path at x . Homotopies of paths are always rel $\{0, 1\}$ unless we say otherwise; homotopies of maps never are.

1.1 The change-of-basepoint isomorphism

Definition 1.1. Let $\gamma: I \rightarrow X$ satisfy $\gamma(0) = x_0$ and $\gamma(1) = x_1$. Define

$$\beta_\gamma: \pi_1(X, x_1) \longrightarrow \pi_1(X, x_0), \quad \beta_\gamma[\alpha] = [\gamma \cdot \alpha \cdot \bar{\gamma}].$$

Note the direction: β_γ runs *backwards* along γ , from the group at the far end to the group at the near end. Getting this consistently right is worth a minute now and saves an hour later.

Proposition 1.2. β_γ is a well-defined group isomorphism, and:

- (a) β_γ depends only on the homotopy class of γ rel $\{0, 1\}$;
- (b) $\beta_{\gamma \cdot \delta} = \beta_\gamma \circ \beta_\delta$ whenever the concatenation is defined;
- (c) $\beta_{c_{x_0}} = \text{id}$, and $\beta_\gamma^{-1} = \beta_{\bar{\gamma}}$.

Proof. Well-definedness and the homomorphism property are the usual computations: a homotopy α_t rel endpoints yields the homotopy $\gamma \cdot \alpha_t \cdot \bar{\gamma}$, and $(\gamma\alpha\bar{\gamma})(\gamma\beta\bar{\gamma}) \simeq \gamma(\alpha\beta)\bar{\gamma}$ because $\bar{\gamma} \cdot \gamma \simeq c_{x_1}$. For (b), $\beta_{\gamma\delta}[\alpha] = [(\gamma\delta)\alpha(\overline{\gamma\delta})] = [\gamma(\delta\alpha\bar{\delta})\bar{\gamma}] = \beta_\gamma\beta_\delta[\alpha]$, using $\overline{\gamma\delta} = \bar{\delta}\bar{\gamma}$. Part (c) is immediate, and with (b) it shows β_γ is invertible. Part (a) is [Exercise 1.3](#). $\square$

Exercise 1.3 (easy). Prove [Proposition 1.2](#)(a): if $\gamma \simeq \gamma'$ rel $\{0, 1\}$ then $\beta_\gamma = \beta_{\gamma'}$.

Solution. Let γ_t be a homotopy rel endpoints from γ to γ' . For a loop α at x_1 , the family $\gamma_t \cdot \alpha \cdot \bar{\gamma}_t$ is a homotopy of loops at x_0 ; it is rel $\{0, 1\}$ because every γ_t starts at x_0 . Hence $[\gamma\alpha\bar{\gamma}] = [\gamma'\alpha\bar{\gamma}']$.

The first thing to extract is that a *loop* changes nothing except by conjugation.

Lemma 1.4. *If γ is a loop at x_0 , then β_γ is conjugation by $[\gamma]$:*

$$\beta_\gamma[\alpha] = [\gamma][\alpha][\gamma]^{-1} \quad \text{for all } [\alpha] \in \pi_1(X, x_0).$$

Consequently $\beta_\gamma = \text{id}$ if and only if $[\gamma]$ lies in the centre of $\pi_1(X, x_0)$.

Proof. The displayed formula is [Definition 1.1](#) read in the group $\pi_1(X, x_0)$, since $[\bar{\gamma}] = [\gamma]^{-1}$. Conjugation by g is the identity map precisely when g commutes with every element. $\square$

Example 1.5. Let $X = S^1 \vee S^1$ with $\pi_1(X, x_0) = F(a, b)$ free on the two circles. Taking $\gamma = a$, a loop at x_0 , [Lemma 1.4](#) gives $\beta_a[b] = [aba^{-1}] \neq [b]$: the group is free, so aba^{-1} and b are distinct words. The change-of-basepoint map is genuinely not the identity even when it is an automorphism of one and the same group.

Corollary 1.6. *Let X be path-connected. The isomorphism $\pi_1(X, x_1) \cong \pi_1(X, x_0)$ is independent of the choice of path for every pair x_0, x_1 if and only if $\pi_1(X, x_0)$ is abelian.*

Proof. Two paths γ, δ from x_0 to x_1 give $\beta_\gamma \circ \beta_\delta^{-1} = \beta_\gamma \circ \beta_{\bar{\delta}} = \beta_{\gamma \cdot \bar{\delta}}$ by [Proposition 1.2](#), and $\gamma \cdot \bar{\delta}$ is a loop at x_0 . By [Lemma 1.4](#) this is conjugation by $[\gamma\bar{\delta}]$. So all choices agree exactly when every loop at x_0 acts trivially by conjugation, i.e. when the group is abelian; and conversely, every class arises as some $[\gamma\bar{\delta}]$ (take δ fixed and $\gamma = \lambda \cdot \delta$ for a loop λ). $\square$

Remark 1.7. This is the precise content of the abuse of notation $\pi_1(X)$: for path-connected X the fundamental group is well defined up to isomorphism, and canonically well defined only up to *inner* automorphism. Any statement one makes about $\pi_1(X)$ without naming a basepoint must therefore be invariant under inner automorphisms — being abelian, being finite, having a given conjugacy class of subgroups. Being *equal* to a particular subgroup is not such a statement, which is exactly why [Chapter 3](#) classifies covering spaces by conjugacy classes of subgroups and not by subgroups.

Exercise 1.8 (medium). Let $f: X \rightarrow Y$ be a map, and let γ join x_0 to x_1 in X . Show that the square

$$\begin{array}{ccc} \pi_1(X, x_1) & \xrightarrow{f_*} & \pi_1(Y, f(x_1)) \\ \beta_\gamma \downarrow & & \downarrow \beta_{f \circ \gamma} \\ \pi_1(X, x_0) & \xrightarrow{f_*} & \pi_1(Y, f(x_0)) \end{array}$$

commutes.

Hint. There is nothing to prove beyond applying f to a concatenation.

Solution. $f_*\beta_\gamma[\alpha] = [f \circ (\gamma\alpha\bar{\gamma})] = [(f\gamma)(f\alpha)(\overline{f\gamma})] = \beta_{f\gamma}f_*[\alpha]$, since f commutes with concatenation and reversal of paths.

1.2 Free versus based homotopy

A loop at x_0 is a map $(S^1, s_0) \rightarrow (X, x_0)$; two such may be compared with or without keeping the basepoint fixed. The two notions differ, and the difference is measured by exactly the maps of the previous section.

Definition 1.9. Loops f, g in X are *freely homotopic* if they are homotopic as maps $S^1 \rightarrow X$, with no condition on basepoints. They are *based homotopic* (at x_0) if $f(s_0) = g(s_0) = x_0$ and the homotopy may be taken to fix s_0 throughout, i.e. if $[f] = [g]$ in $\pi_1(X, x_0)$.

One direction is a construction, and it is worth doing explicitly because the same picture — shrinking a path away — reappears whenever a basepoint has to be moved.

Lemma 1.10. Let h be a path from x_0 to x_1 and let α be a loop at x_1 . Then the loop $h \cdot \alpha \cdot \bar{h}$ at x_0 is freely homotopic to α .

Proof. For $t \in I$ let $h_t(s) = h(t + (1-t)s)$, the tail of h from $h(t)$ to x_1 , and set $\varphi_t = h_t \cdot \alpha \cdot \bar{h}_t$, a loop based at $h(t)$. The map $(s, t) \mapsto \varphi_t(s)$ is continuous, $\varphi_0 = h \cdot \alpha \cdot \bar{h}$, and $\varphi_1 = c_{x_1} \cdot \alpha \cdot c_{x_1}$, which is α up to reparametrisation. This is a homotopy of maps $S^1 \rightarrow X$; its basepoint travels along h , which is precisely what a free homotopy is allowed to do. $\square$

The other direction — what a free homotopy does to a based class — follows from a lemma about homotopies of arbitrary maps, which we will use far more often than the loop case.

Lemma 1.11 (Tracking the basepoint). Let $\varphi_t: X \rightarrow Y$, $t \in I$, be a homotopy, and let $h(t) = \varphi_t(x_0)$ be the path traced by the image of the basepoint. Then

$$\varphi_{0*} = \beta_h \circ \varphi_{1*}: \pi_1(X, x_0) \longrightarrow \pi_1(Y, \varphi_0(x_0)).$$

Proof. Let α be a loop at x_0 and define $H: I \times I \rightarrow Y$ by $H(s, t) = \varphi_t(\alpha(s))$. Reading the four edges of the square: the bottom is $\varphi_0 \circ \alpha$, the top is $\varphi_1 \circ \alpha$, and both vertical edges are h , because $\alpha(0) = \alpha(1) = x_0$.

Now work in the square before mapping it. The bottom edge of $I \times I$ and the concatenation (left edge)·(top edge)·(reversed right edge) are two paths in $I \times I$ with the same endpoints, hence homotopic rel endpoints, $I \times I$ being convex. Composing such a homotopy with H gives a homotopy rel endpoints in Y from $\varphi_0 \circ \alpha$ to $h \cdot (\varphi_1 \circ \alpha) \cdot \bar{h}$. Therefore $\varphi_{0*}[\alpha] = \beta_h \varphi_{1*}[\alpha]$. $\square$

Remark 1.12. Two features of [Lemma 1.11](#) are worth naming, because both get used. First, it is a statement about *any* homotopy, not only one between maps that agree somewhere. Second, the correction term β_h is completely explicit: if one can show the track h is nullhomotopic, or that the target group is abelian, the correction disappears.

Example 1.13. Let $X = Y = S^1$ and let φ_t be rotation by $2\pi t$, so $\varphi_0 = \varphi_1 = \text{id}$. The track h is a loop generating $\pi_1(S^1) \cong \mathbb{Z}$, and [Lemma 1.11](#) asserts $\text{id} = \beta_h$. That is no contradiction: by [Lemma 1.4](#), β_h is conjugation by $[h]$, and $\mathbb{Z}$ is abelian. The lemma has extracted nothing here — as it must not, since the conclusion is about a group in which conjugation is invisible. To see it do work one needs a space with a nonabelian fundamental group, which is the subject of [Chapter 2](#).

The first serious payoff of [Lemma 1.11](#) is that homotopy equivalences behave as one wants them to, even though a homotopy inverse cannot usually be chosen to respect basepoints.

Proposition 1.14. *If $f: X \rightarrow Y$ is a homotopy equivalence, then $f_*: \pi_1(X, x_0) \rightarrow \pi_1(Y, f(x_0))$ is an isomorphism for every x_0 .*

Proof. Let g be a homotopy inverse. Apply [Lemma 1.11](#) to a homotopy from $g \circ f$ to id_X : writing h for the track of x_0 , we get $(gf)_* = \beta_h \circ \text{id}_* = \beta_h$, an isomorphism $\pi_1(X, x_0) \rightarrow \pi_1(X, gf(x_0))$. Since $(gf)_* = g_* \circ f_*$ is injective, f_* is injective at x_0 ; since it is surjective, g_* is surjective at $f(x_0)$.

Both conclusions hold at every basepoint. Apply the same argument to a homotopy from $f \circ g$ to id_Y at the basepoint $f(x_0)$: it shows g_* is injective at $f(x_0)$. Hence $g_*: \pi_1(Y, f(x_0)) \rightarrow \pi_1(X, gf(x_0))$ is an isomorphism, and from $(gf)_* = g_* f_*$ being an isomorphism we conclude that f_* is one too. $\square$

Exercise 1.15 (medium). Show that a loop α at x_0 is freely homotopic to a constant map if and only if $[\alpha] = 1$ in $\pi_1(X, x_0)$.

Hint. One direction is trivial. For the other, a free homotopy from α to a constant map is a homotopy $\varphi_t: S^1 \rightarrow X$; feed it to [Lemma 1.11](#), taking X there to be S^1 .

Solution. If $[\alpha] = 1$ the based nullhomotopy is in particular a free one.

Conversely, let $\varphi_t: S^1 \rightarrow X$ be a homotopy with $\varphi_0 = \alpha$ and φ_1 constant, and let h be the track of s_0 . Let $\iota \in \pi_1(S^1, s_0)$ be the class of the identity loop. By [Lemma 1.11](#),

$$[\alpha] = \varphi_{0*}(\iota) = \beta_h(\varphi_{1*}(\iota)).$$

But φ_1 is constant, so φ_{1*} is the trivial homomorphism and $\varphi_{1*}(\iota) = 1$. As β_h is a homomorphism, $[\alpha] = 1$.

Exercise 1.16 (medium). Suppose $\pi_1(X, x_0)$ is abelian. Show that two loops at x_0 which are freely homotopic are homotopic rel x_0 .

Hint. Run the argument of [Exercise 1.15](#) without assuming the second loop is constant, and then use [Lemma 1.4](#).

Solution. Let $\varphi_t: S^1 \rightarrow X$ be a free homotopy from f to g , both loops at x_0 , and let h be the track of s_0 . Then h is a loop at x_0 , since φ_0 and φ_1 both send s_0 to x_0 . By [Lemma 1.11](#) applied to the generator ι ,

$$[f] = \beta_h[g] = [h][g][h]^{-1},$$

the last equality by [Lemma 1.4](#). If $\pi_1(X, x_0)$ is abelian the right side is $[g]$.

Exercise 1.17 (medium). In $X = S^1 \vee S^1$ exhibit two loops at x_0 that are freely homotopic but not homotopic rel x_0 . Deduce that the hypothesis in [Exercise 1.16](#) cannot be dropped.

Hint. *Lemma 1.10 manufactures free homotopies out of nothing. Apply it with h a loop rather than a path, and use that $F(a, b)$ is free to see the two classes are different.*

Solution. Take $f = b$ and $g = a \cdot b \cdot \bar{a}$, both loops at x_0 . They are freely homotopic by Lemma 1.10 with $h = a$, $\alpha = b$. They are not based homotopic: in the free group $F(a, b)$ the words b and aba^{-1} are distinct, because $F(a, b)$ has trivial centre — or simply because reduced words in a free group are unique. Since $\pi_1(S^1 \vee S^1)$ is nonabelian, this is consistent with Exercise 1.16, and shows the hypothesis there is doing work.

1.3 Homotopies of maps, and the centre

Lemma 1.4 says that β_γ is the identity exactly when $[\gamma]$ is central. So any construction that forces β_γ to be the identity produces central elements. That is a small observation with a long reach: it is the mechanism behind the fact that H-spaces have abelian fundamental group, and — one step further, which we leave for the reader to take when they meet it — behind the constraint on loops of self-maps.

Definition 1.18. An *H-space* is a space X with a basepoint e and a map $\mu: X \times X \rightarrow X$ satisfying $\mu(e, e) = e$, such that $\mu(e, -)$ and $\mu(-, e)$ are homotopic to id_X . It has a *strict unit* if $\mu(e, x) = \mu(x, e) = x$ for all x ; topological groups are the standard example.

Proposition 1.19 (Eckmann–Hilton). *Let X be an H-space with a strict unit e . Then $\pi_1(X, e)$ is abelian.*

Proof. Besides concatenation $*$, the set $\pi_1(X, e)$ carries a second operation: $[f] \odot [g] = [\mu \circ (f, g)]$, the pointwise product, which is well defined because μ takes homotopies to homotopies. The constant loop c_e is a two-sided unit for $\odot$, strictly: $\mu(c_e, g) = g$.

These two operations satisfy the interchange law

$$([f] * [g]) \odot ([p] * [q]) = ([f] \odot [p]) * ([g] \odot [q]),$$

in fact on the nose: both sides are the loop that runs $\mu(f, p)$ on $[0, \frac{1}{2}]$ and $\mu(g, q)$ on $[\frac{1}{2}, 1]$.

Now the Eckmann–Hilton argument. Writing 1 for the class of c_e , which is a unit for both operations,

$$[f] * [g] = ([f] \odot 1) * (1 \odot [g]) = ([f] * 1) \odot (1 * [g]) = [f] \odot [g],$$

so the operations coincide; and

$$[f] * [g] = (1 \odot [f]) * ([g] \odot 1) = (1 * [g]) \odot ([f] * 1) = [g] \odot [f] = [g] * [f].$$

□

Example 1.20. S^1 , S^3 and every topological group are H-spaces with strict units, so all have abelian fundamental group. S^3 is simply connected, so the statement is empty there; the content will come in Chapter 10, where the same multiplication is used on cohomology rather than on π_1 and computes the degree of a power map.

Exercise 1.21 (medium). Prove Proposition 1.19 without assuming the unit is strict, using Lemma 1.11 instead.

Hint. Do not repair the Eckmann–Hilton argument; replace it. The multiplication gives a homomorphism $\mu_*: \pi_1(X \times X, (e, e)) \rightarrow \pi_1(X, e)$, and $\pi_1(X \times X, (e, e)) \cong \pi_1(X, e) \times \pi_1(X, e)$.

Hint. In a product group, $(u, 1)$ and $(1, v)$ commute. Evaluate μ_* on each of them separately — each evaluation is the induced map of a map homotopic to id_X , so [Lemma 1.11](#) applies.

Solution. Since $\mu(e, e) = e$ we have a homomorphism $\mu_*: \pi_1(X \times X, (e, e)) \rightarrow \pi_1(X, e)$, and the projections identify the source with $\pi_1(X, e) \times \pi_1(X, e)$.

Evaluate on the two axes. The class $([f], 1)$ is represented by the loop $s \mapsto (f(s), e)$, so

$$\mu_*([f], 1) = [s \mapsto \mu(f(s), e)] = (\mu(-, e))_*[f].$$

Now $\mu(-, e) \simeq \text{id}_X$, and the track of e under such a homotopy is a loop h' at e , because $\mu(e, e) = e$. By [Lemma 1.11](#), $(\mu(-, e))_* = \beta_{h'}$, an automorphism of $\pi_1(X, e)$. Likewise $\mu_*(1, [g]) = \beta_h[g]$ for a loop h .

In the group $\pi_1(X, e) \times \pi_1(X, e)$ the elements $([f], 1)$ and $(1, [g])$ commute, and μ_* is a homomorphism, so

$$\beta_{h'}[f] \cdot \beta_h[g] = \mu_*([f], [g]) = \beta_h[g] \cdot \beta_{h'}[f].$$

Given arbitrary $u, v \in \pi_1(X, e)$, apply this with $[f] = \beta_{h'}^{-1}(u)$ and $[g] = \beta_h^{-1}(v)$, which is legitimate because $\beta_{h'}$ and β_h are bijections. It gives $uv = vu$.

Remark 1.22. The second proof is shorter, needs no interchange law, and generalises: the only thing it used about μ is that it is a map $X \times X \rightarrow X$ restricting to something invertible on π_1 along each axis. [Proposition 1.19](#) is kept because the interchange argument itself reappears — in [Chapter 10](#), where the same two products act on cohomology instead.

Remark 1.23. One step remains unclimbed, and it is the reader's. Suppose a loop of self-maps of X starts and ends at the identity. [Lemma 1.11](#) computes the induced map on π_1 in terms of the track of the basepoint; [Lemma 1.4](#) converts the answer into a statement about the centre. Putting the two together constrains which loops can arise this way. It is a good test of whether the chapter has been absorbed: the argument is two lines and uses nothing that is not on this page.

1.4 What to carry forward

- β_γ ([Definition 1.1](#)) and its four formal properties ([Proposition 1.2](#)); for a loop it is conjugation ([Lemma 1.4](#)).
- The tracking lemma ([Lemma 1.11](#)): a homotopy that moves the basepoint changes the induced map by β_h , and the correction is explicit. This is used again in [Chapter 11](#) for homotopy extension arguments and in [Chapter 13](#) for maps between towers.
- Homotopy equivalences induce isomorphisms on π_1 ([Proposition 1.14](#)) — the reason we may compute with any convenient model of a space.
- Free and based homotopy of loops differ exactly by conjugation, in the two directions supplied by [Lemma 1.10](#) and [Lemma 1.11](#).

The ambiguity isolated here is about to become an organising principle rather than a nuisance. [Chapter 2](#) builds spaces whose fundamental groups are nonabelian enough for it to be visible, and [Chapter 3](#) shows that conjugacy classes of subgroups — not subgroups — are what covering spaces actually see.

Chapter 2

Presentations from gluing data: van Kampen and CW structures

A space presented by gluing instructions — a polygon with edge identifications, a cube with face identifications, a cylinder with its ends matched by a map — has a fundamental group presented by generators and relations, and the dictionary between the two presentations is mechanical. This chapter sets up that dictionary and then does what a first course usually has no time for: uses it on spaces where the answer is not obvious, and uses it backwards, to prove that certain maps *cannot* exist.

The mechanical part is worth internalising because it is the only computational access to π_1 we have. The subtle part is that the dictionary is not injective: several gluings give the same group, and π_1 will turn out not to determine the space. The chapter ends there, and [Chapters 4](#) and [7](#) supply the finer invariants.

2.1 From cells to generators and relations

We freely use the van Kampen theorem. What we need in practice is its specialisation to CW complexes, and one normalisation.

Proposition 2.1. *Every connected CW complex is homotopy equivalent to a CW complex with a single 0-cell.*

For a graph this is the familiar move of collapsing a spanning tree; in general one collapses a spanning tree of the 1-skeleton. The justification — that collapsing a contractible subcomplex does not change the homotopy type — is a consequence of the homotopy extension property, and we prove it in [Chapter 11](#). We use it now and pay for it later.

Proposition 2.2. *Let X be a connected CW complex with a single 0-cell p , with 1-cells $(a_i)_{i \in I}$ and 2-cells attached along loops w_j , $j \in J$. Then*

$$\pi_1(X, p) \cong \langle a_i \ (i \in I) \mid w_j \ (j \in J) \rangle,$$

where w_j is read as a word in the $a_i^{\pm 1}$ by recording the 1-cells the attaching map traverses.

Proof sketch. The 1-skeleton is a wedge of circles, with free fundamental group on the a_i . Attaching a 2-cell along w kills exactly the normal closure of $[w]$, by van Kampen applied to the union of X^1 (thickened) with the open cell. Cells of dimension ≥ 3 do not change π_1 , again by van Kampen, because the attaching spheres are simply connected. $\square$

Remark 2.3. There is a small subtlety hidden in the phrase “read as a word”. An attaching map $S^1 \rightarrow X^1$ is a *free* loop: it comes with no basepoint, so its class in $\pi_1(X^1, p)$ is only well defined up to conjugation, and up to inversion if one reverses the orientation. This does no harm, because the normal closure of $[w]$ is unchanged by conjugating w . It does mean that a relator is only ever well defined up to conjugacy — which is the reason presentations coming from geometry so often look different from the ones in a group theory book. Compare [Exercise 1.15](#): whether a relator is trivial does not depend on the basepoint either.

Example 2.4 (Surfaces). The closed orientable surface M_g of genus $g \geq 1$ is the quotient of a $4g$ -gon by the edge word $a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}$. All $4g$ vertices are identified to a single point, so this is a CW structure with one 0-cell, $2g$ 1-cells and one 2-cell, and [Proposition 2.2](#) gives

$$\pi_1(M_g) = \left\langle a_1, b_1, \dots, a_g, b_g \mid \prod_{i=1}^g [a_i, b_i] \right\rangle.$$

Abelianising kills the single relator, since it is a product of commutators, so the abelianisation is free abelian of rank $2g$. That one observation — the relator dies on abelianisation — is the whole reason the genus is visible to homology, and it is what makes [Exercise 2.14](#) below possible.

2.2 Mapping tori

Definition 2.5. Let $f: X \rightarrow X$ be a map. The *mapping torus* is

$$T_f = (X \times [0, 1]) / ((x, 0) \sim (f(x), 1)).$$

We fix this convention once and never change it: the *bottom* of the cylinder is glued into the *top* by f . Every sign and every direction below depends on it, and the commonest error in this subject is to prove a correct theorem about the other convention. [Exercise 2.11](#) asks the reader to redo the computation with the roles exchanged, precisely so that the dependence is seen rather than feared.

The seam — the image of $X \times \{0\} \cup X \times \{1\}$ — is a copy of X , namely the image of $X \times \{1\}$, since every bottom point has been identified with a top point. We regard $X \subseteq T_f$ this way, and take the basepoint $p \in X$ with $f(p) = p$.

Proposition 2.6. Let X be a connected CW complex with a single 0-cell p , and let f be cellular with $f(p) = p$. Write $\pi_1(X, p) = \langle a_i \mid r_j \rangle$ as in [Proposition 2.2](#). Then

$$\pi_1(T_f, p) \cong \langle a_i, t \mid r_j, t a_i t^{-1} = f_*(a_i) \rangle,$$

where t is the loop $s \mapsto (p, s)$. If f_* is an automorphism this is the semidirect product $\pi_1(X, p) \rtimes_{f_*} \mathbb{Z}$.

Proof. Give T_f the CW structure whose cells are the cells of the seam X together with one cell $e \times (0, 1)$ for each cell e of X . There is one 0-cell, p ; the 1-cells are the a_i together with $t = p \times (0, 1)$; the 2-cells are those of X , contributing the relators r_j , together with one square $a_i \times (0, 1)$ for each i .

It remains to read the boundary word of that square, and for this it is quicker to use [Lemma 1.11](#) than to stare at the picture. Let α be a loop at p in X and consider the homotopy of loops

$$\varphi_u: S^1 \rightarrow T_f, \quad \varphi_u(s) = (\alpha(s), u), \quad u \in [0, 1].$$

Its track is the path $u \mapsto (p, u)$, which is the loop t . At the top, $u = 1$, the loop φ_1 is α sitting in the seam. At the bottom, $u = 0$, the loop φ_0 is $s \mapsto (\alpha(s), 0) = (f(\alpha(s)), 1)$, which is the loop $f \circ \alpha$ in the seam. So [Lemma 1.11](#) reads

$$f_*[\alpha] = [\varphi_0] = \beta_t[\varphi_1] = [t][\alpha][t]^{-1},$$

using [Lemma 1.4](#) for the last equality. Thus the square $a_i \times (0, 1)$ contributes exactly the relation $t a_i t^{-1} = f_*(a_i)$. Cells of dimension ≥ 3 do not affect π_1 . $\square$

Remark 2.7. The proposition is the algebraic shadow of a geometric fact: T_f fibres over S^1 with fibre X , and t is the monodromy. Conjugation by the monodromy *is* the induced map f_* . In [Chapter 5](#) the same fibration is run through homology instead and produces the Wang sequence, with $1 - f_*$ in place of conjugation by t .

Example 2.8 (The degenerate case). If $f = \text{id}$ then $T_f = X \times S^1$ and the relations say that t is central: $\pi_1 = \pi_1(X) \times \mathbb{Z}$, as it must be. Always check a formula against the case where nothing happens.

Exercise 2.9 (easy). Let $f: S^1 \rightarrow S^1$ have degree d . Show $\pi_1(T_f) \cong \langle a, t \mid tat^{-1} = a^d \rangle$. Identify the space for $d = 1$ and for $d = -1$.

Solution. $\pi_1(S^1) = \langle a \mid \rangle$ and $f_*(a) = a^d$, so [Proposition 2.6](#) gives the presentation at once. For $d = 1$ the group is $\mathbb{Z}^2$ and $T_f = S^1 \times S^1$ by [Example 2.8](#). For $d = -1$ the relation is $tat^{-1} = a^{-1}$ and T_f is the Klein bottle.

Exercise 2.10 (medium). Let $f: \mathbb{RP}^2 \rightarrow \mathbb{RP}^2$ be any map with $f(p) = p$ that induces an isomorphism on π_1 . Show that $\pi_1(T_f) \cong \mathbb{Z}/2 \times \mathbb{Z}$, whatever f is.

Hint. What are the automorphisms of $\mathbb{Z}/2$?

Solution. $\pi_1(\mathbb{RP}^2) = \mathbb{Z}/2 = \langle a \mid a^2 \rangle$, whose only automorphism is the identity, so $f_*(a) = a$. [Proposition 2.6](#) gives $\langle a, t \mid a^2, tat^{-1} = a \rangle$, i.e. $\mathbb{Z}/2 \times \mathbb{Z}$.

Exercise 2.11 (medium). Redo the proof of [Proposition 2.6](#) for the opposite convention $(x, 1) \sim (f(x), 0)$ and show that the relation becomes $t^{-1}a_it = f_*(a_i)$. Conclude that the two conventions give presentations differing by $t \leftrightarrow t^{-1}$, and that the resulting groups are isomorphic.

Solution. With the other convention the seam is the image of $X \times \{0\}$, and in the homotopy $\varphi_u(s) = (\alpha(s), u)$ it is now φ_0 that is α in the seam while φ_1 is $f \circ \alpha$. [Lemma 1.11](#) gives $[\alpha] = \beta_t(f_*[\alpha]) = [t]f_*[\alpha][t]^{-1}$, that is $t^{-1}at = f_*(a)$. Substituting $t \mapsto t^{-1}$ carries one presentation to the other, so the groups are isomorphic; geometrically, the two mapping tori are homeomorphic by reversing the I coordinate.

Exercise 2.12 (hard). Let $G = \langle a, t \mid tat^{-1} = a^2 \rangle$, the group of [Exercise 2.9](#) with $d = 2$, and let $\varphi: G \rightarrow \mathbb{Z}$ be the homomorphism with $\varphi(t) = 1$, $\varphi(a) = 0$.

- (a) Show that $\ker \varphi$ is the increasing union of the cyclic groups generated by $a_n := t^{-n}at^n$, and that $a_{n-1} = a_n^2$.
- (b) Deduce $\ker \varphi \cong \mathbb{Z}[\frac{1}{2}]$, the additive group of dyadic rationals. In particular $\ker \varphi$ is not finitely generated.
- (c) Conclude that $\pi_1(T_f)$ is *not* of the form $\pi_1(S^1) \rtimes \mathbb{Z}$, even though T_f fibres over S^1 with fibre S^1 .

Hint. For (a), conjugate the defining relation by powers of t . For (b), the tower $\mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{2} \dots$ has a familiar colimit.

Hint. For (c), a degree-2 map $S^1 \rightarrow S^1$ is not a homotopy equivalence, so nothing forces the inclusion of the fibre to be injective with normal image; use (b) to see that it is not even surjective onto $\ker \varphi$.

Solution. (a) Set $a_n = t^{-n}at^n$, so $a_0 = a$. Conjugating $tat^{-1} = a^2$ by t^{-n} gives $t^{-(n-1)}at^{n-1} = (t^{-n}at^n)^2$, that is $a_{n-1} = a_n^2$. Each a_n lies in $\ker \varphi$, and $\langle a_{n-1} \rangle \subseteq \langle a_n \rangle$. Every element of $\ker \varphi$ is a word in the $t^{-n}a^{\pm 1}t^n$, hence lies in some $\langle a_n \rangle$; so $\ker \varphi = \bigcup_n \langle a_n \rangle$, and the union is increasing.

(b) The subgroup $\langle a_n \rangle$ is infinite cyclic and contains $\langle a_{n-1} \rangle$ with index 2. So $\ker \varphi$ is the colimit of $\mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{2} \dots$, which is $\mathbb{Z}[\frac{1}{2}]$: send $a_n \mapsto 2^{-n}$. A finitely generated subgroup would lie in some $\langle a_n \rangle$ and be proper.

(c) In a semidirect product $N \rtimes \mathbb{Z}$ with quotient map to $\mathbb{Z}$, the kernel is N . Here the kernel is $\mathbb{Z}[\frac{1}{2}]$, not $\mathbb{Z} = \pi_1(S^1)$. The image of $\pi_1(\text{fibre})$ is the subgroup $\langle a \rangle = \langle a_0 \rangle$, which is proper in the kernel and not normal in G — indeed $t^{-1}at = a_1 \notin \langle a_0 \rangle$.

2.3 Retractions, and how to rule them out

The presentations above are useful in the negative direction too. The basic principle occupies one line.

Lemma 2.13. *If $A \subseteq X$ and $r: X \rightarrow A$ is a retraction, then $i_*: \pi_1(A, a) \rightarrow \pi_1(X, a)$ is injective and split: there is a homomorphism r_* with $r_*i_* = \text{id}$. The same holds after abelianising.*

Proof. Apply π_1 to $r \circ i = \text{id}_A$; functoriality gives $r_*i_* = \text{id}$, and abelianisation is a functor. $\square$

Injectivity alone is a weak conclusion, and the temptation is to look for its failure. Usually it does not fail; what fails is splitting.

Exercise 2.14 (medium). Let M be the Möbius band and ∂M its boundary circle. Show that $\pi_1(\partial M) \rightarrow \pi_1(M)$ is injective, but that M does not retract onto ∂M .

Hint. M deformation retracts onto its core circle c . Under that equivalence, what does the boundary circle become?

Solution. The deformation retraction onto the core gives $\pi_1(M) = \mathbb{Z}\langle c \rangle$, and the boundary circle runs twice around the core, so $i_*: \mathbb{Z} \rightarrow \mathbb{Z}$ is multiplication by 2. That is injective, so no injectivity obstruction exists.

But a retraction would give $r_*: \mathbb{Z} \rightarrow \mathbb{Z}$ with $r_*(2x) = x$ for all x ; taking $x = 1$ forces $2r_*(1) = 1$ in $\mathbb{Z}$, which is impossible. Hence there is no retraction.

The positive direction needs a way to turn a homomorphism into a map. For targets that are aspherical this is possible, and for S^1 it is elementary.

Proposition 2.15. *Let X be a connected CW complex with basepoint p . Every homomorphism $\varphi: \pi_1(X, p) \rightarrow \mathbb{Z}$ is induced by a map $g: X \rightarrow S^1$.*

Proof sketch. Normalise X to have a single 0-cell ([Proposition 2.1](#)) and send it to the basepoint of S^1 . Send the 1-cell a_i to a loop of degree $\varphi(a_i)$. Over a 2-cell with attaching word w_j , the composite $S^1 \rightarrow X^1 \rightarrow S^1$ has degree $\varphi(w_j) = 0$ because φ is a homomorphism killing the relators; a degree-zero self-map of S^1 is nullhomotopic, so the map extends over the cell. For cells of dimension $n \geq 3$ the obstruction to extending lies in $\pi_{n-1}(S^1) = 0$, so the extension continues. $\square$

Remark 2.16. The last two sentences of that proof are obstruction theory in embryo: extend skeleton by skeleton, and at each stage the obstruction is a homotopy class of the attaching map composed with what is already built. [Chapter 11](#) makes the first half precise (the extension lemma) and [Chapter 17](#) makes all of it precise, with the obstructions organised into cohomology classes.

Exercise 2.17 (medium). Let K be the Klein bottle, with $\pi_1(K) = \langle a, b \mid abab^{-1} \rangle$, and let $A \subseteq K$, $B \subseteq K$ be the circles representing a and b .

- (a) Show K does not retract onto A .
- (b) Show that there is a map $K \rightarrow S^1$ restricting to a degree-one map on B .

Hint. Abelianise the presentation. The relator $abab^{-1}$ becomes $2a$.

Solution. Abelianising, $\pi_1(K)^{\text{ab}} = \langle a, b \mid 2a \rangle \cong \mathbb{Z}/2 \oplus \mathbb{Z}$, with a generating the torsion summand.

(a) A retraction onto A would give, after abelianising ([Lemma 2.13](#)), a homomorphism $\mathbb{Z}/2 \oplus \mathbb{Z} \rightarrow \mathbb{Z}$ sending the class of a to a generator of $\mathbb{Z}$. But the class of a has order 2 and $\mathbb{Z}$ is torsion-free, so its image is 0. Contradiction. (Note that a itself has infinite order in $\pi_1(K)$, so once again injectivity is not the obstruction.)

(b) Define $\varphi: \pi_1(K) \rightarrow \mathbb{Z}$ by $\varphi(a) = 0$, $\varphi(b) = 1$. This respects the relator: $\varphi(abab^{-1}) = 0 + 1 + 0 - 1 = 0$. By [Proposition 2.15](#) there is $g: K \rightarrow S^1$ inducing φ , and its restriction to B induces multiplication by 1 on $\pi_1(S^1)$, hence has degree one.

Remark 2.18. Part (b) produces a map that is a retraction *up to homotopy*: $g|_B$ is homotopic to a homeomorphism $B \rightarrow S^1$, not necessarily equal to one. Upgrading a homotopy to an equality is exactly what the homotopy extension property does, and [Chapter 11](#) will let us convert such a g into an honest retraction. Until then we are careful to claim only what has been proved.

2.4 Identification complexes in dimension three

The dictionary of [Proposition 2.2](#) applies verbatim to solids with identified faces, provided one first does the bookkeeping: which vertices become the same point, which edges become the same edge. That step is where the errors happen, and it is worth doing slowly on an example where the answer is known.

Example 2.19 (The three-torus). Let X be the cube I^3 with each face identified with the opposite face by the perpendicular translation.

Vertices. The translations identify all eight vertices to one point: the three translations generate a group acting transitively on the vertices of the cube modulo the identifications, and following any vertex under the three translations reaches every other. So there is one 0-cell.

Edges. The twelve edges fall into three orbits of four, one for each coordinate direction: the four edges parallel to the x -axis are identified by the y - and z -translations. Call the resulting 1-cells a, b, c .

Faces. The six faces are identified in opposite pairs, leaving three 2-cells. The face perpendicular to the z -axis is a square whose boundary reads $aba^{-1}b^{-1}$, and similarly for the others.

So X has one 0-cell, three 1-cells, three 2-cells and one 3-cell, and [Proposition 2.2](#) gives $\pi_1(X) = \langle a, b, c \mid [a, b], [b, c], [a, c] \rangle \cong \mathbb{Z}^3$, as it must, since $X = S^1 \times S^1 \times S^1$.

Remark 2.20. Two features of [Example 2.19](#) are accidents of the translations. First, all vertices became one point; second, each edge orbit had exactly four members. Compose the identifications with a rotation of the face and both counts can change: a rotation carries an edge of one face to a *different* edge of the opposite face, so the edge orbits are permuted in longer cycles and the vertex orbits can split. Any such computation must therefore begin by listing the vertex orbits and then the edge orbits, in that order, before a single relation is written down. [Example 2.21](#) is a first case where the answer is not $\mathbb{Z}^n$.

Example 2.21 (Lens spaces). Fix coprime integers $p > q \geq 1$. Realise S^3 as the unit sphere in $\mathbb{C}^2$ and let $\mathbb{Z}/p$ act by

$$\zeta \cdot (z_1, z_2) = (\zeta z_1, \zeta^q z_2), \quad \zeta = e^{2\pi i/p}.$$

The action is free, so the quotient $L(p, q)$ is a closed 3-manifold with universal cover S^3 , and since S^3 is simply connected, $\pi_1(L(p, q)) \cong \mathbb{Z}/p$.

Equivalently, and this is the picture to keep: $L(p, q)$ is the ball D^3 with its upper hemisphere identified with its lower hemisphere, by the rotation through $2\pi q/p$ about the vertical axis followed by reflection in the equatorial plane. That description gives a CW structure with a single cell in each of the four dimensions, and $L(2, 1) = \mathbb{RP}^3$.

Exercise 2.22 (medium). Verify the cell counts in the ball model of $L(p, q)$: the boundary sphere becomes one 2-cell whose attaching map runs p times around a circle built from one 0-cell and one 1-cell. Deduce the presentation $\pi_1 = \langle a \mid a^p \rangle$ directly, without using the covering space.

Hint. The rotation by $2\pi q/p$ has order p on the equator, so the equator's marked points fall into orbits of size p ; a single 1-cell survives, and the boundary of the 2-cell winds around it once for each of the p sectors.

Solution. Mark p equally spaced points on the equator of D^3 , cutting it into p arcs. The gluing rotates by $2\pi q/p$, which permutes the marked points cyclically and permutes the arcs cyclically; so all p marked points become a single 0-cell and all p arcs a single 1-cell a . The upper hemisphere is glued to the lower, so the two hemispheres become a single 2-cell, and its attaching map runs once along each of the p arcs, i.e. traverses a exactly p times. The interior of the ball is the 3-cell. [Proposition 2.2](#) now gives $\langle a \mid a^p \rangle = \mathbb{Z}/p$, agreeing with [Example 2.21](#).

Remark 2.23 (Where this chapter stops). The lens spaces expose the limit of the method. For fixed p , every $L(p, q)$ has fundamental group $\mathbb{Z}/p$, so π_1 cannot tell them apart. Nor can homology: we will compute in [Chapter 4](#) that $H_*(L(p, q))$ is also independent of q . Yet they are genuinely different spaces — $L(5, 1)$ and $L(5, 2)$ are not even homotopy equivalent. Detecting that needs a product structure rather than a group structure, and we return to it in [Chapter 7](#).

This is the first instance of the pattern that runs through the book: build an invariant, compute with it until it fails, and let the failure name the next invariant.

2.5 What to carry forward

- Cells to presentations ([Proposition 2.2](#)), and the licence to assume a single 0-cell ([Proposition 2.1](#), proved in [Chapter 11](#)).
- Relators are well defined only up to conjugacy ([Remark 2.3](#)) — the chapter-one ambiguity, showing up as a computational convenience.
- The mapping torus presentation ([Proposition 2.6](#)), derived from the tracking lemma, and the warning that it is a semidirect product only when f_* is invertible ([Exercise 2.12](#)).
- Retractions split π_1 ([Lemma 2.13](#)); abelianisation turns that into an arithmetic obstruction, and [Proposition 2.15](#) turns a homomorphism to $\mathbb{Z}$ back into a map.
- Orbit bookkeeping for face identifications: vertices first, then edges, then relations ([Example 2.19](#) and [Remark 2.20](#)).

Chapter 3

Covering spaces as a Galois theory

The classification of covering spaces is the first place where the ambiguity of [Chapter 1](#) becomes an organising principle: covers correspond to subgroups, and isomorphic covers correspond to *conjugate* subgroups, because a cover has no preferred basepoint upstairs. Everything in this chapter is an instance of the same dictionary, and the point of the exercises is to make the dictionary usable in both directions — reading a group off a space, and reading a space off a group.

We take the classification itself as known and concentrate on what a first course usually has no time for: computing with it.

3.1 The dictionary

Theorem 3.1 (Classification of covers; recalled). *Let X be path-connected, locally path-connected and semilocally simply connected, with basepoint x_0 .*

- (a) *Sending $(\tilde{X}, \tilde{x}_0) \mapsto p_*\pi_1(\tilde{X}, \tilde{x}_0)$ is a bijection between isomorphism classes of path-connected pointed covers and subgroups of $\pi_1(X, x_0)$.*
- (b) *Forgetting the basepoint upstairs, isomorphism classes of path-connected covers correspond to conjugacy classes of subgroups.*
- (c) *The cover corresponding to H has deck transformation group $N(H)/H$, where $N(H)$ is the normaliser. It is a normal cover exactly when $H \trianglelefteq \pi_1(X, x_0)$, and then the deck group is $\pi_1(X, x_0)/H$.*

Part (b) is the reason [Corollary 1.6](#) was worth proving. A different choice of basepoint upstairs changes H by conjugation and nothing else; a cover, being a space rather than a pointed space, therefore remembers only the conjugacy class.

Example 3.2 (The model case). $\mathbb{Z}$ acts on $\mathbb{R}$ by translation and $\mathbb{R}/\mathbb{Z} = S^1$. The subgroups of $\pi_1(S^1) = \mathbb{Z}$ are the $n\mathbb{Z}$, and the corresponding covers are $\mathbb{R}/n\mathbb{Z} \rightarrow \mathbb{R}/\mathbb{Z}$, the n -fold covers of the circle, together with $\mathbb{R}$ itself for the trivial subgroup. Every subgroup is normal, every cover is normal, and the deck group of the n -fold cover is $\mathbb{Z}/n\mathbb{Z}$. Nothing in this example is ambiguous, because the group is abelian — which is exactly why it is a poor guide to the general case.

Exercise 3.3 (easy). Classify the connected two-fold covers of $S^1 \vee S^1$ up to isomorphism, and draw them.

Hint. An index-two subgroup of a group G is the kernel of a surjection $G \rightarrow \mathbb{Z}/2$, and index-two subgroups are automatically normal.

Solution. Surjections $F(a, b) \rightarrow \mathbb{Z}/2$ are determined by the images of a and b and must be nonzero, so there are three: $(1, 0)$, $(0, 1)$, $(1, 1)$. Each kernel is normal of index two, and distinct kernels give non-isomorphic covers because the subgroups are distinct and normal, hence in distinct conjugacy classes.

Each cover has two vertices and four edges. For $(1, 0)$: the a -edges join the two vertices, and each vertex carries a b -loop. For $(0, 1)$ the roles are exchanged. For $(1, 1)$: both the a -edges and the b -edges join the two vertices, giving a circle of four edges (alternately a and b).

Exercise 3.4 (medium). Let $H \leq F(a, b)$ be the subgroup corresponding to the three-fold cover with vertices v_1, v_2, v_3 in which a acts by the cycle $(v_1 v_2 v_3)$ and b fixes every vertex. Compute the deck group. Then modify the example so that the deck group is trivial.

Hint. Deck transformations of a connected cover are determined by where they send one point, and they permute the fibre. A deck transformation exists carrying $\tilde{x}$ to $\tilde{x}'$ exactly when the two corresponding subgroups agree.

Solution. The described cover is the one associated with the action of $F(a, b)$ on $\{v_1, v_2, v_3\}$ by $a \mapsto (v_1 v_2 v_3)$, $b \mapsto \text{id}$; it is the cover corresponding to the stabiliser H of v_1 . Since the image of $F(a, b)$ in S_3 is the cyclic group generated by a 3-cycle, which is transitive and abelian, all point stabilisers coincide, so H is normal and the deck group is $\mathbb{Z}/3$.

For a trivial deck group, take instead the action $a \mapsto (v_1 v_2)$, $b \mapsto (v_1 v_3)$. The image is all of S_3 , and the stabiliser of v_1 is a non-normal subgroup of index three; its normaliser is itself, so $N(H)/H$ is trivial and the cover admits no nontrivial deck transformation.

3.2 Covers of graphs

Graphs make the dictionary completely computable, because for them π_1 is free and its rank is an Euler characteristic.

Proposition 3.5. A connected graph Γ is homotopy equivalent to a wedge of circles, and if Γ is finite then $\pi_1(\Gamma)$ is free of rank $1 - \chi(\Gamma) = E - V + 1$. If $\Gamma' \rightarrow \Gamma$ is a k -sheeted cover then $\chi(\Gamma') = k \chi(\Gamma)$.

Proof. Collapsing a spanning tree is a homotopy equivalence (Proposition 2.1) and leaves one vertex and $E - V + 1$ edges, so π_1 is free of that rank by Proposition 2.2. A k -sheeted cover has kV vertices and kE edges. $\square$

Example 3.6 (An index-two subgroup, with a basis). Let $\varphi: F(a, b) \rightarrow \mathbb{Z}/2$ send $a \mapsto 1$, $b \mapsto 0$, and let $H = \ker \varphi$. The corresponding cover is the first graph of Exercise 3.3: vertices v_0, v_1 ; two a -edges, from v_0 to v_1 and from v_1 to v_0 ; a b -loop at each vertex.

Here $V = 2$, $E = 4$, so H is free of rank 3. To name a basis, take the spanning tree consisting of the a -edge from v_0 to v_1 . The three remaining edges give three loops at v_0 , obtained by travelling through the tree, crossing the edge, and returning:

$$b, \quad a b a^{-1}, \quad a^2.$$

So $H = \langle b, aba^{-1}, a^2 \rangle$, free on those three elements. Note that the rank went *up*: a proper subgroup of finite index in F_2 always has rank greater than 2, by [Exercise 3.7](#). Subgroups of free groups are not smaller; they are freer.

Exercise 3.7 (medium). Let $H \leq F_r$ have index $k < \infty$. Show H is free of rank $1 + k(r - 1)$.

Hint. Realise F_r as π_1 of a wedge of r circles and H as the fundamental group of a k -sheeted cover; then use [Proposition 3.5](#).

Solution. The wedge of r circles has $\chi = 1 - r$. The cover corresponding to H is a graph with $\chi = k(1 - r)$, and H is its fundamental group, hence free of rank $1 - k(1 - r) = 1 + k(r - 1)$.

Exercise 3.8 (medium). Deduce that F_2 contains subgroups isomorphic to F_n for every finite n .

Solution. Apply [Exercise 3.7](#) with $r = 2$: a subgroup of index k is free of rank $k + 1$. Index- k subgroups exist for every k — for instance the preimage of the trivial subgroup under $F_2 \rightarrow \mathbb{Z}/k$, $a \mapsto 1, b \mapsto 0$. Taking $k = n - 1$ gives F_n .

The finite-index formula is seductive, and the next exercise is the antidote.

Exercise 3.9 (hard). Let $\varphi: F(a, b) \rightarrow \mathbb{Z}$ send $a \mapsto 1$ and $b \mapsto 0$, and let $K = \ker \varphi$.

- (a) Describe the covering graph of $S^1 \vee S^1$ corresponding to K .
- (b) Show that K is free of infinite rank, with basis $\{a^n b a^{-n} : n \in \mathbb{Z}\}$.
- (c) Conclude that a finitely generated group can have a subgroup that is not finitely generated, and that the formula of [Exercise 3.7](#) has no infinite-index analogue.

Hint. The cover has one vertex for each coset of K , i.e. one for each integer. Where do the a -edges go, and where do the b -edges go?

Hint. Choose the spanning tree to be the whole line of a -edges. What is left?

Solution. (a) The vertices are the cosets Ka^n , indexed by $n \in \mathbb{Z}$. The edge labelled a leaving the vertex n ends at $n + 1$, since $\varphi(a) = 1$; the edge labelled b leaving n returns to n , since $\varphi(b) = 0$. So the cover is a line of a -edges with one b -loop attached at every integer.

(b) The line of a -edges is a spanning tree: it is connected, contains every vertex, and has no circuit. It is contractible, so collapsing it is a homotopy equivalence ([Proposition 2.1](#)), and what remains is a wedge of one circle for each leftover edge — that is, one for each b -loop, indexed by $n \in \mathbb{Z}$. Hence K is free on that set. Travelling out along the tree to the vertex n , round the loop, and back gives the element $a^n b a^{-n}$.

(c) F_2 is generated by two elements, while K has infinite rank, so K is not finitely generated. If a formula of the shape “rank = $1 + k(r - 1)$ ” held for infinite index it would have to give a finite answer for every subgroup of a finitely generated free group, which (b) contradicts.

Remark 3.10. [Exercise 3.9](#) is the easy member of a family. The same method — take the cover determined by a surjection $F_2 \rightarrow A$ onto an abelian group, choose a spanning tree in the resulting graph, and read the leftover edges — computes a free basis for the kernel in every case. The kernel of $F_2 \rightarrow \mathbb{Z}^2$ is the commutator subgroup, and the picture there is the integer grid in the plane. That computation is a good test of the method once this chapter is finished; the only new difficulty is choosing a spanning tree of the grid.

3.3 Group actions and their quotients

A covering space is often presented not by a subgroup but by a group action, and the translation between the two is worth having explicitly.

Definition 3.11. An action of a group G on a space Y is a *covering space action* if every $y \in Y$ has a neighbourhood U such that the translates gU , $g \in G$, are pairwise disjoint.

Proposition 3.12. Let Y be path-connected and locally path-connected, and let G act on it by a covering space action. Then $Y \rightarrow Y/G$ is a normal covering with deck group G , and there is a short exact sequence

$$1 \longrightarrow \pi_1(Y, y_0) \longrightarrow \pi_1(Y/G, \bar{y}_0) \longrightarrow G \longrightarrow 1.$$

In particular, if Y is simply connected then $\pi_1(Y/G) \cong G$.

Proof sketch. Disjointness of the translates makes the quotient map a covering, and G acts by deck transformations transitively on each fibre, so the cover is normal with deck group G by [Theorem 3.1\(c\)](#). The exact sequence is the long exact sequence of the covering: the map to G sends a loop to the deck transformation carrying the start of its lift to the end. $\square$

Example 3.13. The $\mathbb{Z}/p$ action on S^3 of [Example 2.21](#) is free, and a free action of a finite group on a Hausdorff space is a covering space action. Since S^3 is simply connected, $\pi_1(L(p, q)) \cong \mathbb{Z}/p$ — the computation of [Exercise 2.22](#), now obtained without a single cell.

Exercise 3.14 (medium). Classify the connected covering spaces of the torus T^2 .

Hint. $\pi_1(T^2) = \mathbb{Z}^2$ is abelian, so conjugacy classes of subgroups are subgroups, and every cover is normal. Every subgroup of $\mathbb{Z}^2$ is free abelian of rank 0, 1 or 2.

Solution. Write $T^2 = \mathbb{R}^2/\mathbb{Z}^2$. Subgroups $\Lambda \leq \mathbb{Z}^2$ come in three kinds.

- Rank 2: Λ has finite index k , and the cover is the torus $\mathbb{R}^2/\Lambda \rightarrow \mathbb{R}^2/\mathbb{Z}^2$, a k -sheeted cover of T^2 by T^2 .
- Rank 1: $\Lambda = \mathbb{Z}v$ for a primitive vector v times a positive integer; the cover is $\mathbb{R}^2/\Lambda \cong S^1 \times \mathbb{R}$, an infinite cylinder.
- Rank 0: the universal cover $\mathbb{R}^2$.

Since $\mathbb{Z}^2$ is abelian, all of these are normal covers, with deck group $\mathbb{Z}^2/\Lambda$.

Remark 3.15 (Where this chapter stops). Covering space theory is a complete theory of π_1 and nothing else. It cannot distinguish two simply connected spaces, and it says nothing whatever about π_n for $n \geq 2$ — indeed a covering induces isomorphisms on all higher homotopy groups, which is what makes the universal cover such a useful place to work later ([Chapter 13](#)).

The next two chapters therefore change invariant. [Chapter 4](#) builds homology, which sees simply connected spaces; and [Chapter 5](#) produces two spaces with the same homology whose universal covers differ, so that the two invariants of Part I are seen to be genuinely independent.

3.4 What to carry forward

- Covers correspond to subgroups, and without basepoints only to their conjugacy classes ([Theorem 3.1](#)); the deck group is $N(H)/H$.
- For graphs everything is computable: the rank is $1 - \chi$, the Euler characteristic multiplies ([Proposition 3.5](#)), and a basis is read off a spanning tree ([Example 3.6](#)).
- Finite index behaves arithmetically ([Exercise 3.7](#)); infinite index does not ([Exercise 3.9](#)).
- A covering space action gives $1 \rightarrow \pi_1(Y) \rightarrow \pi_1(Y/G) \rightarrow G \rightarrow 1$ ([Proposition 3.12](#)), which is how a group is realised as a fundamental group.

Chapter 4

Homology you can compute: Δ -complexes, cellular chains, degree

Homology is computable in a way that π_1 is not: for a space presented by cells, the chain complex is finite in each degree and the boundary maps are integer matrices. The price is bookkeeping — orientations, orderings, signs — and the purpose of this chapter is to fix conventions and then run them on enough examples that the reader trusts their own arithmetic.

Two threads start here and run through the book. The first is *torsion*: it appears as a cokernel, produced by a gluing that wraps a cell around a cycle more than once, and it is the reason homology is not merely a count of holes. The second is *degree*, which converts a geometric symmetry into an integer, and which will reappear as the cellular boundary, as the Hopf invariant ([Chapter 16](#)) and as the transgression ([Chapter 19](#)).

4.1 Δ -complexes, and the conventions that make them work

Definition 4.1. A Δ -complex structure on X is a collection of maps $\sigma_\alpha: \Delta^n \rightarrow X$, each restricting to a homeomorphism from the interior of Δ^n onto its image, such that every point of X is in exactly one such image, the restriction of each σ_α to a face of Δ^n is one of the maps of the collection, and a set is open iff its preimage under every σ_α is open.

The simplicial chain complex has $\Delta_n(X)$ free abelian on the n -simplices and

$$\partial\sigma = \sum_{i=0}^n (-1)^i \sigma|_{[v_0, \dots, \hat{v}_i, \dots, v_n]}.$$

Everything below depends on the ordering of the vertices of each simplex, so:

Convention. Before computing, write down the vertex ordering of every simplex, and write down which face of which simplex each lower simplex is. The overwhelming majority of errors in this subject are made before any homology is computed, by counting simplices whose identifications have not been recorded.

Example 4.2 (Torus and Klein bottle). Give the square the usual Δ -structure with two triangles U (upper) and L (lower) meeting along a diagonal c , and identify the sides in pairs.

For the torus, all four vertices become one vertex v ; the sides give two edges a, b ; with the diagonal there are three edges and two triangles. With vertices ordered so that $\partial U = b - c + a$ and $\partial L = a - c + b$ we get

$$\Delta_2 = \mathbb{Z}^2 \xrightarrow{\partial_2} \Delta_1 = \mathbb{Z}^3 \xrightarrow{\partial_1=0} \Delta_0 = \mathbb{Z},$$

with $\partial_2(U) = \partial_2(L) = a + b - c$ and $\partial_1 = 0$ because every edge is a loop. Hence $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}^3 / \langle a + b - c \rangle \cong \mathbb{Z}^2$ and $H_2 = \ker \partial_2 \cong \mathbb{Z}$, generated by $U - L$.

For the Klein bottle the same picture with one side reversed gives $\partial_2(U) = a + b - c$ and $\partial_2(L) = a - b + c$, whence ∂_2 has matrix $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ in the basis a, b, c . Now $\ker \partial_2 = 0$, so $H_2 = 0$; and $H_1 = \mathbb{Z}^3 / \langle a + b - c, a - b + c \rangle \cong \mathbb{Z} \oplus \mathbb{Z}/2$, the torsion coming from the sum of the two relators, $2a$.

The last computation is worth pausing over. The two surfaces have identical cell counts and differ only in the sign of one identification, and that sign is the entire difference between $\mathbb{Z}^2$ and $\mathbb{Z} \oplus \mathbb{Z}/2$. Torsion is what a gluing produces when it wraps around with the wrong orientation.

Exercise 4.3 (easy). Compute $H_*(\mathbb{RP}^2; \mathbb{Z})$ from the CW structure that has a single cell in dimensions 0, 1 and 2, with the 2-cell attached along a^2 . Then describe a Δ -complex structure on $\mathbb{RP}^2$ with two triangles, and say — without computing — how many vertex classes it has.

Hint. For the second half: $\mathbb{RP}^2$ is the square with edge word $abab$. Follow the four corners around and see which of them are identified.

Solution. The cellular complex is $\mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z}$, by [Example 4.7](#) with $n = 2$. So $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}/2$ generated by the 1-cell, and $H_2 = \ker(\times 2) = 0$: the surface is closed but carries no fundamental class over $\mathbb{Z}$, which is orientability making its first appearance.

For the Δ -structure, cut the square along a diagonal. Going round the boundary, the corners alternate between two classes, so *two* vertices survive, not one — unlike the torus and the Klein bottle of [Example 4.2](#), where all four corners were identified. That is exactly the kind of fact that must be recorded before any boundary map is written, and missing it is the standard way to get a wrong answer with a plausible shape.

Remark 4.4. The parenthetical in that solution is not an evasion, it is the moral of the section. Δ -complexes are the right setting for *defining* simplicial homology and for proving it agrees with singular homology, and they are a poor setting for computing, because the number of simplices grows quickly and each one carries an ordering. The next section replaces them.

4.2 Cellular chains

Theorem 4.5 (Cellular homology; recalled). *Let X be a CW complex. The groups $C_n^{\text{CW}}(X) = H_n(X^n, X^{n-1})$ are free abelian on the n -cells, and with the boundary maps below their homology is $H_*(X)$.*

Proposition 4.6 (Cellular boundary formula). $d_n(e_\alpha^n) = \sum_\beta d_{\alpha\beta} e_\beta^{n-1}$, where $d_{\alpha\beta}$ is the degree of the composite

$$S^{n-1} \xrightarrow{\varphi_\alpha} X^{n-1} \longrightarrow X^{n-1}/X^{n-2} = \bigvee_\beta S^{n-1} \longrightarrow S_\beta^{n-1},$$

in which φ_α is the attaching map of e_α^n and the last map collapses all wedge summands but the β -th.

So the boundary map is a matrix of degrees, and computing homology of a CW complex is computing degrees of maps between spheres. Two model cases carry most of the weight.

Example 4.7 (Projective spaces). $\mathbb{RP}^n$ has one cell in each dimension $0, \dots, n$, the k -cell attached by the double cover $S^{k-1} \rightarrow \mathbb{RP}^{k-1}$. The composite in [Proposition 4.6](#) is

$$S^{k-1} \longrightarrow \mathbb{RP}^{k-1} \longrightarrow \mathbb{RP}^{k-1}/\mathbb{RP}^{k-2} \cong S^{k-1},$$

which collapses the equator and folds the two hemispheres onto S^{k-1} , the second one composed with the antipodal map. Since the antipodal map of S^{k-1} has degree $(-1)^k$, the degree of the composite is $1 + (-1)^k$. Hence

$$d_k = \begin{cases} 2, & k \text{ even}, \\ 0, & k \text{ odd}, \end{cases}$$

and therefore

$$H_k(\mathbb{RP}^n; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & k = 0, \\ \mathbb{Z}/2, & k \text{ odd}, 0 < k < n, \\ 0, & k \text{ even}, 0 < k < n, \\ \mathbb{Z}, & k = n \text{ odd}, \\ 0, & k = n \text{ even}. \end{cases}$$

The alternation of 0s and 2s is not a curiosity: it is the parity of the antipodal map, and it will return in [Exercise 4.15](#).

Example 4.8 (Lens spaces). Take the CW structure on $L(p, q)$ from [Exercise 2.22](#), with a single cell in each of the dimensions 0 through 3. The 2-cell is attached along a^p , so $d_2 = p$. The 3-cell is attached by a map that carries the upper and lower hemispheres of ∂D^3 each homeomorphically onto the 2-cell, and the identification reverses orientation, so the two contributions have opposite signs and $d_3 = 0$. Finally $d_1 = 0$, there being one 0-cell. The complex

$$\mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{p} \mathbb{Z} \xrightarrow{0} \mathbb{Z}$$

gives

$$H_*(L(p, q); \mathbb{Z}) = (\mathbb{Z}, \mathbb{Z}/p, 0, \mathbb{Z}).$$

Remark 4.9 (The promise of [Remark 2.23](#), kept). The answer does not involve q . So homology joins π_1 in failing to distinguish $L(5, 1)$ from $L(5, 2)$, even though those spaces are not homotopy equivalent. Two invariants have now failed on the same pair of spaces, and both failed for the same reason: they are built from the cells one at a time, and the spaces differ in how the cells are *linked*. [Chapter 7](#) introduces the product that sees linking.

Example 4.10 (Surfaces). The CW structure of [Example 2.4](#) on M_g has one 0-cell, $2g$ 1-cells and one 2-cell attached along $\prod [a_i, b_i]$. Each generator occurs once with each sign in that word, so every degree $d_{\alpha\beta}$ is $1 - 1 = 0$ and $d_2 = 0$. Hence $H_*(M_g) = (\mathbb{Z}, \mathbb{Z}^{2g}, \mathbb{Z})$. Compare [Example 2.4](#): the relator dies on abelianisation, and $d_2 = 0$ is the same statement one dimension up.

Exercise 4.11 (medium). Let $X_n = S^1 \cup_{a^n} e^2$, the complex with one 0-cell, one 1-cell and a 2-cell attached by a degree- n map. Compute $H_*(X_n)$, and compute $\pi_1(X_n)$. Which of the two invariants is stronger here?

Solution. The cellular complex is $\mathbb{Z} \xrightarrow{n} \mathbb{Z} \xrightarrow{0} \mathbb{Z}$, so $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}/n$, $H_2 = \ker(n) = 0$. By [Proposition 2.2](#), $\pi_1(X_n) = \langle a \mid a^n \rangle = \mathbb{Z}/n$. Here the two agree, because the group is abelian; in general H_1 is the abelianisation of π_1 , so π_1 is at least as strong, and [Example 2.4](#) is a case where it is strictly stronger.

Exercise 4.12 (medium). Let X be a finite CW complex with c_n cells of dimension n . Show that $\sum_n (-1)^n c_n = \sum_n (-1)^n \text{rank } H_n(X)$.

Hint. For a chain complex of finitely generated free abelian groups, $\text{rank } C_n = \text{rank } Z_n + \text{rank } B_{n-1}$ where Z and B are cycles and boundaries.

Solution. Write $Z_n = \ker d_n$, $B_n = \text{im } d_{n+1}$. From $0 \rightarrow Z_n \rightarrow C_n \rightarrow B_{n-1} \rightarrow 0$, which splits as B_{n-1} is free, $\text{rank } C_n = \text{rank } Z_n + \text{rank } B_{n-1}$. From $0 \rightarrow B_n \rightarrow Z_n \rightarrow H_n \rightarrow 0$, $\text{rank } Z_n = \text{rank } B_n + \text{rank } H_n$. Hence $\sum (-1)^n \text{rank } C_n = \sum (-1)^n (\text{rank } H_n + \text{rank } B_n + \text{rank } B_{n-1})$, and the B terms cancel in pairs.

4.3 Degree

Definition 4.13. For $f: S^n \rightarrow S^n$ with $n \geq 1$, the *degree* $\deg f$ is the integer with $f_*(z) = (\deg f) z$ for z a generator of $H_n(S^n) \cong \mathbb{Z}$; the definition is independent of the generator.

We use without proof the standard properties: degree is a homotopy invariant and detects homotopy classes ($f \simeq g \iff \deg f = \deg g$); $\deg(fg) = \deg f \cdot \deg g$; the antipodal map of S^n has degree $(-1)^{n+1}$; and a map with a regular value having finitely many preimages has degree the sum of the local degrees.

Exercise 4.14 (easy). Construct, for every $n \geq 1$ and every $k \in \mathbb{Z}$, a map $S^n \rightarrow S^n$ of degree k .

Hint. For $n = 1$ use $z \mapsto z^k$. For larger n , suspend.

Solution. On S^1 , $z \mapsto z^k$ has degree k . Suspension multiplies nothing: the suspension Σf of a degree- k map has degree k , because $H_{n+1}(\Sigma S^n) \cong H_n(S^n)$ naturally. Iterating gives a degree- k map of S^n for every n . Degree 0 is realised by a constant map and degree -1 by a reflection.

The next computation is the one that makes symmetry arguments possible: it says what the double cover does to the top homology class.

Exercise 4.15 (medium). Let $q: S^n \rightarrow \mathbb{RP}^n$ be the quotient map, $n \geq 1$. Show that

$$q_*: H_n(S^n) \longrightarrow H_n(\mathbb{RP}^n)$$

is multiplication by 2 when n is odd, and is zero when n is even.

Hint. Give S^n the CW structure with two cells in each dimension permuted by the antipodal map τ , so that q identifies the two cells of each dimension.

Hint. Write the generator of $H_n(S^n)$ as a cellular cycle $z = e_+ + \varepsilon e_-$ and determine ε using $\tau_* z = (\deg \tau)z$.

Solution. In the antipodal CW structure, $C_n^{\text{CW}}(S^n) = \mathbb{Z}e_+ \oplus \mathbb{Z}e_-$ with τ exchanging e_+ and e_- . A generator of $H_n(S^n) \cong \mathbb{Z}$ is a cycle $z = e_+ + \varepsilon e_-$ for some $\varepsilon = \pm 1$. Applying τ , which has degree $(-1)^{n+1}$,

$$\varepsilon e_+ + e_- = \tau_* z = (-1)^{n+1}(e_+ + \varepsilon e_-),$$

so $\varepsilon = (-1)^{n+1}$.

The map q sends both n -cells of S^n to the single n -cell e of $\mathbb{RP}^n$, so on cellular chains $q_\#(z) = (1 + (-1)^{n+1})e$. For n odd this is $2e$, and $H_n(\mathbb{RP}^n) = \mathbb{Z}\langle e \rangle$ by [Example 4.7](#), so q_* is multiplication by 2. When n is even we have $H_n(\mathbb{RP}^n) = 0$ by [Example 4.7](#), and there is nothing to prove; consistently, $q_\#(z) = 0$.

Remark 4.16. The use of [Exercise 4.15](#) is always the same. Suppose a self-map f of S^n takes the same value at antipodal points. Then f is constant on the fibres of q , so it factors as $f = g \circ q$ for some $g: \mathbb{RP}^n \rightarrow S^n$ — the quotient by a free action being a covering, the induced g is continuous. Applying H_n and [Exercise 4.15](#) turns that factorisation into a divisibility statement about $\deg f$, with the parity of n deciding which statement. Carrying this out, in both parities, and then asking which degrees actually occur, is a good measure of whether this chapter has been absorbed; [Exercise 4.14](#) supplies the construction needed for the second half.

4.4 What to carry forward

- Fix orderings and identifications *before* computing ([Definition 4.1](#) and the convention after it); the standard error is made at that stage.
- Torsion is a cokernel produced by an orientation-reversing gluing ([Example 4.2](#)).
- The cellular boundary is a matrix of degrees ([Proposition 4.6](#)); $\mathbb{RP}^n$ ([Example 4.7](#)) and the lens spaces ([Example 4.8](#)) are the two model computations.
- Homology fails on lens spaces exactly where π_1 failed ([Remark 4.9](#)) — the second invariant of the book to run out, and the reason for [Chapter 7](#).
- Euler characteristic as a free consistency check ([Exercise 4.12](#)).
- The double cover multiplies the top class by 2 in odd dimensions and kills it in even ones ([Exercise 4.15](#)).

Chapter 5

Exact sequences as machines

An exact sequence is not a theorem to be admired but a device to be operated. In each of the three sections below the pattern is the same: a geometric decomposition produces a long exact sequence, and the sequence produces an answer by an entirely mechanical procedure — a cokernel at one end, a kernel at the other, and an extension problem in between. The skill being trained is running the machine confidently, and knowing which of its hypotheses is load-bearing.

The chapter closes by looking at what all this computation cannot see. Homology is about to be the third invariant of Part I to run out of power, and the way it fails — it cannot see the universal cover — points straight at [Chapter 13](#).

5.1 Pairs, quotients, and a hypothesis that matters

The long exact sequence of a pair (X, A) ,

$$\cdots \rightarrow H_n(A) \rightarrow H_n(X) \rightarrow H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \rightarrow \cdots,$$

is useless without a way of identifying $H_n(X, A)$, and the identification one reaches for is $H_n(X, A) \cong \tilde{H}_n(X/A)$. That isomorphism is a theorem with a hypothesis.

Theorem 5.1 (Good pairs; recalled). *If A is a nonempty closed subspace that is a deformation retract of some neighbourhood in X — for instance if (X, A) is a CW pair — then the quotient map induces an isomorphism $H_n(X, A) \cong \tilde{H}_n(X/A)$ for all n .*

Exercise 5.2 (easy). Verify the hypothesis for a CW pair: exhibit a neighbourhood of A in X that deformation retracts to A .

Hint. Work cell by cell: in a cell e^n attached to A , push away from the centre.

Solution. Build N skeleton by skeleton. Having a neighbourhood N^{n-1} of $A \cap X^{n-1}$ in X^{n-1} with a deformation retraction to it, extend over each n -cell: in the characteristic map $D^n \rightarrow X$, take the preimage of N^{n-1} in ∂D^n , thicken it radially inwards to an open collar, and deformation retract that collar radially. The union over all cells is the required neighbourhood, and the deformation retractions assemble because they agree on the boundaries and CW topology is determined cell by cell.

Two facts about the failure of [Theorem 5.1](#) are worth having, because both are counterintuitive and both are used to build counterexamples.

Example 5.3 (Zero-dimensional homology counts components, one at a time). If A is a space with infinitely many path components, then $H_0(A) \cong \bigoplus_{\pi_0(A)} \mathbb{Z}$, a direct *sum*, never a product. The reason is that a singular chain is a *finite* linear combination of simplices. So for the compact space $A = \{0\} \cup \{1/n : n \geq 1\} \subseteq \mathbb{R}$, which has countably many components, $H_0(A)$ is countable — even though A is compact and looks like a limit of finite sets.

Example 5.4 (The shrinking wedge). Let $\mathbb{H} \subseteq \mathbb{R}^2$ be the union of the circles C_n of radius $1/n$ centred at $(1/n, 0)$, all passing through the origin, with the subspace topology. For each n there is a retraction $r_n: \mathbb{H} \rightarrow C_n$ collapsing every other circle to the origin, and it is continuous precisely because the circles shrink.

Together the r_n give a homomorphism

$$\Phi: H_1(\mathbb{H}) \longrightarrow \prod_{n \geq 1} H_1(C_n) = \prod_{n \geq 1} \mathbb{Z},$$

and Φ is *surjective*. To see it, prescribe $(k_n) \in \prod \mathbb{Z}$ and build a single loop which winds k_1 times around C_1 during $[0, \frac{1}{2}]$, then k_2 times around C_2 during $[\frac{1}{2}, \frac{3}{4}]$, and so on. Continuity at the endpoint holds because the circles shrink to the origin, and applying r_{n*} to its class returns k_n .

Exercise 5.5 (hard). Deduce that $\mathbb{H}$ is not homotopy equivalent to a countable wedge of circles, even though it is a countable union of circles glued at a point.

Hint. Compare the cardinality of the two homology groups. What is H_1 of a countable wedge of circles?

Solution. A countable wedge $\bigvee_n S^1$ is a CW complex whose cellular chain complex has $C_1 = \bigoplus_n \mathbb{Z}$ and $C_0 = \mathbb{Z}$ with zero boundary, so $H_1(\bigvee_n S^1) = \bigoplus_n \mathbb{Z}$, which is countable.

By [Example 5.4](#), $H_1(\mathbb{H})$ surjects onto $\prod_n \mathbb{Z}$, which is uncountable. A surjection from a countable group onto an uncountable one is impossible, so $H_1(\mathbb{H})$ is uncountable and the two groups are not isomorphic. Homology being a homotopy invariant, the spaces are not homotopy equivalent.

Remark 5.6. [Examples 5.3](#) and [5.4](#) are the two halves of a machine for breaking [Theorem 5.1](#). Take a pair (X, A) with X contractible and A compact, totally disconnected and infinite. The long exact sequence computes $H_1(X, A) \cong \tilde{H}_0(A)$, a countable direct sum by [Example 5.3](#); while X/A can easily be a space of shrinking-wedge type, with uncountable H_1 by [Example 5.4](#). The hypothesis of [Theorem 5.1](#) is therefore not removable, and the reader now has both computations needed to say so with a specific pair in hand.

5.2 Mayer–Vietoris

Theorem 5.7 (Mayer–Vietoris; recalled). If $X = \mathring{A} \cup \mathring{B}$ then there is a long exact sequence

$$\cdots \rightarrow H_n(A \cap B) \xrightarrow{(i_*, -j_*)} H_n(A) \oplus H_n(B) \xrightarrow{k_* + l_*} H_n(X) \xrightarrow{\partial} H_{n-1}(A \cap B) \rightarrow \cdots$$

Operating it means, in each degree, computing one map of finitely generated abelian groups and taking a cokernel and a kernel. The example to keep in mind is one where the interesting behaviour comes from a gluing of degree two.

Example 5.8 (The Klein bottle from two Möbius bands). The Klein bottle K is the union of two Möbius bands M_1, M_2 glued along their boundary circles; thickening slightly, $A \cap B \simeq S^1$ and $A \simeq B \simeq S^1$, the equivalences being the deformation retractions onto the core circles c_1, c_2 .

The boundary circle of a Möbius band runs twice around the core, so with z a generator of $H_1(A \cap B)$,

$$\varphi: H_1(A \cap B) = \mathbb{Z} \longrightarrow \mathbb{Z}c_1 \oplus \mathbb{Z}c_2, \quad \varphi(z) = (2c_1, -2c_2).$$

This is injective, so $H_2(K) = \ker \varphi = 0$; and

$$H_1(K) = \operatorname{coker} \varphi = \mathbb{Z}^2 / \langle (2, -2) \rangle \cong \mathbb{Z} \oplus \mathbb{Z}/2,$$

since $(2, -2) = 2 \cdot (1, -1)$ with $(1, -1)$ primitive. So $H_*(K) = (\mathbb{Z}, \mathbb{Z} \oplus \mathbb{Z}/2, 0)$, agreeing with the Δ -complex computation of [Example 4.2](#). Two independent routes to the same answer is the cheapest insurance available.

Remark 5.9. The arithmetic in [Example 5.8](#) deserves naming, because it is what decides whether torsion appears. A cokernel $\mathbb{Z}^k / \langle v \rangle$ is free exactly when v is primitive, i.e. when the gcd of its entries is 1; if $v = d \cdot v'$ with v' primitive and $d > 1$, the cokernel has a $\mathbb{Z}/d$ summand. So when gluing along a circle, the question is never “is there a Möbius band?” but “what is the gcd of the entries of the gluing vector?”.

Exercise 5.10 (medium). Let X be the union of two solid tori V_1, V_2 glued along their boundary tori by a homeomorphism carrying the meridian of V_1 to the meridian of V_2 and the longitude to the longitude. Compute $H_*(X)$ and identify X .

Hint. A solid torus deformation retracts onto its core circle. In H_1 of the boundary torus, the meridian bounds a disc in the solid torus and the longitude maps to the core.

Solution. Write $H_1(T^2) = \mathbb{Z}m \oplus \mathbb{Z}l$ and $H_1(V_i) = \mathbb{Z}c_i$. Then $m \mapsto 0$ and $l \mapsto c_i$ in each solid torus, so

$$\varphi(m) = (0, 0), \quad \varphi(l) = (c_1, -c_2).$$

Also $H_2(V_1) \oplus H_2(V_2) = 0$. Mayer-Vietoris gives

$$0 \rightarrow H_3(X) \rightarrow H_2(T^2) = \mathbb{Z} \rightarrow 0, \quad 0 \rightarrow H_2(X) \rightarrow H_1(T^2) \xrightarrow{\varphi} \mathbb{Z}^2 \rightarrow H_1(X) \rightarrow 0.$$

Hence $H_3(X) \cong \mathbb{Z}$; $H_2(X) = \ker \varphi = \mathbb{Z}m \cong \mathbb{Z}$; and $H_1(X) = \operatorname{coker} \varphi = \mathbb{Z}^2 / \langle (1, -1) \rangle \cong \mathbb{Z}$. So $H_*(X) = (\mathbb{Z}, \mathbb{Z}, \mathbb{Z}, \mathbb{Z})$, which is the homology of $S^1 \times S^2$ — and indeed $X \cong S^1 \times S^2$, the two solid tori being the two halves of $S^1 \times S^2$ cut along $S^1 \times S^1$.

Exercise 5.11 (hard). Now glue the two solid tori by a homeomorphism whose effect on H_1 of the boundary carries the meridian m of V_1 to $p l_2 + q m_2$, with $p \geq 1$ and $\gcd(p, q) = 1$. Compute $H_*(X)$.

Hint. Only the images in H_1 matter. Write the gluing on $H_1(T^2)$ as a matrix in $GL_2(\mathbb{Z})$, with $m \mapsto p l_2 + q m_2$ and $l \mapsto r l_2 + s m_2$, where $ps - qr = \pm 1$.

Hint. In V_2 , the meridian m_2 dies and l_2 becomes the core c_2 . So the second component of φ only sees the l_2 -coefficients, namely p and r .

Solution. With the matrix as in the hint, $\varphi(m) = (0, -p c_2)$ and $\varphi(l) = (c_1, -r c_2)$. Then

$$H_1(X) = \operatorname{coker} \varphi = \mathbb{Z}^2 / \langle (0, -p), (1, -r) \rangle \cong \mathbb{Z}/p,$$

because the second generator lets one eliminate the first coordinate, leaving $\mathbb{Z}/\langle p \rangle$. And φ is injective: $a(0, -p) + b(1, -r) = 0$ forces $b = 0$ and then $ap = 0$, so $a = 0$. Hence $H_2(X) = \ker \varphi = 0$, while $H_3(X) \cong H_2(T^2) = \mathbb{Z}$ as before. So

$$H_*(X) = (\mathbb{Z}, \mathbb{Z}/p, 0, \mathbb{Z}),$$

which is the homology of the lens space $L(p, q)$ computed in [Example 4.8](#). Indeed X is $L(p, q)$: the genus-one Heegaard splitting is the standard second description of a lens space.

5.3 The Wang sequence

A mapping torus is glued from one piece along two copies of the fibre, and Mayer–Vietoris turns that into a single long exact sequence in which the monodromy appears as $1 - f_*$.

Proposition 5.12 (Wang sequence). *For the mapping torus T_f of $f: X \rightarrow X$ there is a long exact sequence*

$$\cdots \rightarrow H_n(X) \xrightarrow{1-f_*} H_n(X) \rightarrow H_n(T_f) \rightarrow H_{n-1}(X) \xrightarrow{1-f_*} H_{n-1}(X) \rightarrow \cdots$$

Equivalently, for each n there is a short exact sequence

$$0 \rightarrow \operatorname{coker}(1 - f_* \text{ on } H_n(X)) \rightarrow H_n(T_f) \rightarrow \ker(1 - f_* \text{ on } H_{n-1}(X)) \rightarrow 0, \quad (5.1)$$

which splits whenever the kernel is free abelian.

Proof sketch. Cut T_f along a fibre. Both pieces — a neighbourhood of the seam, and the complement of the seam — deformation retract onto copies of X , and their intersection is two copies of X . The Mayer–Vietoris map $H_n(X) \oplus H_n(X) \rightarrow H_n(X) \oplus H_n(X)$ is $(u, v) \mapsto (u - v, u - f_*v)$ up to sign; changing basis by $(u, v) \mapsto (u - v, v)$ turns it into a map whose cokernel and kernel are those of $1 - f_*$, and the long exact sequence collapses to the one displayed. Exactness of (5.1) is a restatement, and a short exact sequence with free abelian right-hand term splits. $\square$

Exercise 5.13 (medium). Check [Proposition 5.12](#) against $f = \operatorname{id}$, where the answer is known independently.

Solution. For $f = \operatorname{id}$ we have $T_f = X \times S^1$ and $1 - f_* = 0$, so (5.1) reads $0 \rightarrow H_n(X) \rightarrow H_n(X \times S^1) \rightarrow H_{n-1}(X) \rightarrow 0$. That is exactly the Künneth splitting for $X \times S^1$, since $H_*(S^1) = (\mathbb{Z}, \mathbb{Z})$ and there are no Tor terms.

Example 5.14 (Mapping tori of self-maps of a sphere). Let $f: S^k \rightarrow S^k$ have degree d , with $k \geq 2$. Then f_* is the identity on H_0 and multiplication by d on H_k , so $1 - f_*$ is 0 on H_0 and $1 - d$ on H_k . Feeding this into (5.1):

$$H_0(T_f) = \mathbb{Z}, \quad H_1(T_f) = \mathbb{Z}, \quad H_k(T_f) = \mathbb{Z}/(d-1), \quad H_{k+1}(T_f) = \ker(1-d),$$

and all other groups vanish. For $d \neq 1$ the last group is 0; for $d = 1$ it is $\mathbb{Z}$, recovering $H_*(S^k \times S^1)$.

The torsion is the point. The fibre S^k has no torsion, the base S^1 has no torsion, and the total space has $\mathbb{Z}/(d-1)$: it was manufactured by the monodromy, exactly as in [Remark 5.9](#) the torsion was manufactured by the gluing vector.

Exercise 5.15 (medium). Let $X = S^1 \vee S^1$ and let f exchange the two circles. Compute $H_*(T_f)$.

Hint. On $H_1(X) = \mathbb{Z}^2$ the map f_* is the exchange matrix, so $1 - f_*$ has matrix $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$.

Solution. On H_0 , $1 - f_* = 0$; on $H_1 = \mathbb{Z}^2$, $1 - f_*$ has the matrix above, with $\ker \cong \mathbb{Z}$ spanned by $(1, 1)$ and $\text{coker} = \mathbb{Z}^2 / \langle (1, -1) \rangle \cong \mathbb{Z}$.

Degree 0: $H_0(T_f) = \text{coker}(0 \text{ on } H_0) = \mathbb{Z}$. Degree 1: $0 \rightarrow \text{coker}(1 - f_*|_{H_1}) \rightarrow H_1(T_f) \rightarrow \ker(1 - f_*|_{H_0}) \rightarrow 0$ is $0 \rightarrow \mathbb{Z} \rightarrow H_1(T_f) \rightarrow \mathbb{Z} \rightarrow 0$, which splits, so $H_1(T_f) \cong \mathbb{Z}^2$. Degree 2: $0 \rightarrow 0 \rightarrow H_2(T_f) \rightarrow \ker(1 - f_*|_{H_1}) = \mathbb{Z} \rightarrow 0$, so $H_2(T_f) \cong \mathbb{Z}$. All higher groups vanish. Thus $H_*(T_f) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z})$ — the homology of a closed orientable surface of genus one, which is no accident: T_f is a torus.

5.4 What homology cannot see

Every computation above was insensitive to one thing: the way the pieces are attached in a way that is invisible on the level of chains. The sharpest form of this is that homology does not see the universal cover.

Example 5.16 (The universal cover of a wedge). Let $X = S^1 \vee S^2$. Its universal cover $\tilde{X}$ is the real line with a copy of S^2 attached at each integer: the line covers the circle, and each sheet of the covering carries its own copy of the sphere. Hence

$$H_2(\tilde{X}) \cong \bigoplus_{n \in \mathbb{Z}} \mathbb{Z},$$

free abelian of countably infinite rank, and $H_1(\tilde{X}) = 0$.

Note the extra structure: the deck group $\mathbb{Z}$ permutes the spheres, so $H_2(\tilde{X})$ is a free module of rank one over the group ring $\mathbb{Z}[t, t^{-1}] = \mathbb{Z}[\pi_1 X]$. That module structure is invisible downstairs and will be the subject of [Chapter 13](#).

Exercise 5.17 (medium). Using the Hurewicz theorem in the form $\pi_2(Y) \cong H_2(Y)$ for simply connected Y , together with $\pi_n(\tilde{X}) \cong \pi_n(X)$ for $n \geq 2$, show that $\pi_2(S^1 \vee S^2)$ is not finitely generated as an abelian group.

Solution. $\tilde{X}$ is simply connected — it is a universal cover — so Hurewicz gives $\pi_2(\tilde{X}) \cong H_2(\tilde{X}) \cong \bigoplus_{\mathbb{Z}} \mathbb{Z}$ by [Example 5.16](#). A covering induces isomorphisms on π_n for $n \geq 2$, so $\pi_2(X) \cong \bigoplus_{\mathbb{Z}} \mathbb{Z}$, which is not finitely generated.

Remark 5.18 (Where this chapter stops). $H_*(X)$ is a quotient of information about X that has already forgotten the covering. Two spaces can therefore agree in every homology group and disagree about their universal covers, hence about π_2 ; and the two ingredients for producing such a pair are now both available — the universal cover of a wedge ([Example 5.16](#)) and the fact that a torus is aspherical. Constructing the pair, and proving the two spaces are not homotopy equivalent, is the natural exercise to attempt at this point.

The systematic repair is [Chapter 13](#): replace $H_*(X)$ by $H_*(\tilde{X})$ as a module over $\mathbb{Z}[\pi_1 X]$, and the resulting invariant *does* detect homotopy equivalence for CW complexes.

5.5 What to carry forward

- $H_n(X, A) \cong \tilde{H}_n(X/A)$ needs a neighbourhood deformation retract ([Theorem 5.1](#)); [Examples 5.3](#) and [5.4](#) are the standard tools for showing it fails without one.
- Mayer–Vietoris is arithmetic: torsion appears exactly when the gluing vector is imprimitive ([Remark 5.9](#)).
- Heegaard gluings of two solid tori produce $S^1 \times S^2$ and the lens spaces ([Exercises 5.10](#) and [5.11](#)) — the same spaces as [Chapter 2](#), by a different route.
- The Wang sequence ([Proposition 5.12](#)): $\text{coker}(1 - f_*)$ below, $\ker(1 - f_*)$ above, and check the splitting before using it.
- Homology does not see the universal cover ([Remark 5.18](#)); the module structure over $\mathbb{Z}[\pi_1]$ is what is missing.

Chapter 6

Cohomology, coefficients, and the failure of naturality

Cohomology is not merely homology with the arrows reversed. The universal coefficient theorem says how much of H^* is determined by H_* — almost all of it — and the residue, an Ext term sitting one degree up, is exactly what makes cohomology a strictly finer invariant of *maps* than homology.

The chapter is built around one distinction, and it is worth stating before any mathematics: a short exact sequence can be natural, and can split, without the splitting being natural. Readers meet this first as a piece of pedantry and later as the reason a whole subject exists. It recurs in [Chapter 8](#) for the Künneth sequence, and the invariant that non-naturality has been concealing is finally constructed in [Chapter 18](#).

6.1 The universal coefficient theorem

Theorem 6.1 (Universal coefficients; recalled). *For a chain complex of free abelian groups and any abelian group G there is a short exact sequence, natural in the complex,*

$$0 \longrightarrow \text{Ext}(H_{n-1}(X), G) \longrightarrow H^n(X; G) \xrightarrow{h} \text{Hom}(H_n(X), G) \longrightarrow 0,$$

where h evaluates a cocycle on a cycle. The sequence splits, but — see [Section 6.3](#) — not naturally.

The practical content is that H^n is Hom of H_n plus Ext of H_{n-1} : free parts stay where they are, torsion moves up one degree.

Example 6.2 (Projective spaces). Dualise the cellular chain complex of [Example 4.7](#). Since $d_k = 1 + (-1)^k$, the cochain differential d^k is the transpose of d_{k+1} , so $d^k = 2$ for k odd and $d^k = 0$ for k even. Hence

$$H^k(\mathbb{RP}^n; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & k = 0, \\ \mathbb{Z}/2, & k \text{ even}, 0 < k \leq n, \\ 0, & k \text{ odd}, k < n, \\ \mathbb{Z}, & k = n \text{ odd}. \end{cases}$$

Compare with [Example 4.7](#): the $\mathbb{Z}/2$'s have moved up one degree. That is $\text{Ext}(\mathbb{Z}/2, \mathbb{Z}) = \mathbb{Z}/2$ appearing in degree k because H_{k-1} was $\mathbb{Z}/2$.

Exercise 6.3 (easy). Compute $H^*(K; \mathbb{Z})$ for the Klein bottle from [Example 5.8](#), and note where each summand comes from.

Solution. $H_*(K) = (\mathbb{Z}, \mathbb{Z} \oplus \mathbb{Z}/2, 0)$. So

$$H^0 = \text{Hom}(\mathbb{Z}, \mathbb{Z}) = \mathbb{Z}, \quad H^1 = \text{Hom}(\mathbb{Z} \oplus \mathbb{Z}/2, \mathbb{Z}) \oplus \text{Ext}(\mathbb{Z}, \mathbb{Z}) = \mathbb{Z},$$

$$H^2 = \text{Hom}(0, \mathbb{Z}) \oplus \text{Ext}(\mathbb{Z} \oplus \mathbb{Z}/2, \mathbb{Z}) = \mathbb{Z}/2.$$

So $H^*(K; \mathbb{Z}) = (\mathbb{Z}, \mathbb{Z}, \mathbb{Z}/2)$. Note $H^2 \neq 0$ while $H_2 = 0$: the $\mathbb{Z}/2$ is the Ext of the torsion in H_1 , and it lives one degree higher than the torsion that produced it.

6.2 Moore spaces

To probe a theorem about H_* one wants spaces with prescribed homology, and they exist.

Proposition 6.4. *For every abelian group A and every $n \geq 1$ there is a CW complex $M(A, n)$, with cells only in dimensions n and $n + 1$ besides a single 0-cell, such that*

$$\tilde{H}_i(M(A, n)) = \begin{cases} A, & i = n, \\ 0, & i \neq n. \end{cases}$$

Proof. Choose a free resolution $0 \rightarrow F_1 \xrightarrow{\alpha} F_0 \rightarrow A \rightarrow 0$; over $\mathbb{Z}$ every subgroup of a free abelian group is free, so a resolution of length one exists for *every* A , finitely generated or not. Let $X_0 = \bigvee_{F_0} S^n$, a wedge with one sphere for each basis element of F_0 , so that $H_n(X_0) = F_0$. For each basis element of F_1 attach an $(n + 1)$ -cell along a map $S^n \rightarrow X_0$ representing the corresponding element of $\pi_n(X_0) \rightarrow H_n(X_0) = F_0$, namely its image $\alpha(\text{basis element})$; such a map exists because for $n \geq 2$ the Hurewicz map $\pi_n(X_0) \rightarrow H_n(X_0)$ is onto, and for $n = 1$ one uses that H_1 is the abelianisation of a free group.

The cellular chain complex of the result is $F_1 \xrightarrow{\alpha} F_0$ concentrated in degrees $n + 1$ and n , whose homology is A in degree n and $\ker \alpha = 0$ in degree $n + 1$. $\square$

Remark 6.5. Two features will be used repeatedly. First, $M(A, n)$ exists for *arbitrary* A — there is no finite generation hypothesis, because $\mathbb{Z}$ is a principal ideal domain and resolutions have length one. Second, $H_{n+1}(M(A, n)) = 0$ by construction, which is what makes these spaces useful as test objects: they have homology in exactly one degree, so any invariant that is nonzero elsewhere is detecting something other than homology. [Chapter 8](#) uses both facts to prove a theorem of pure algebra by topological means.

Exercise 6.6 (easy). Compute $H^*(M(\mathbb{Z}/m, n); \mathbb{Z})$ for $n \geq 1$, $m \geq 2$.

Hint. The cellular cochain complex is $\mathbb{Z} \xrightarrow{m} \mathbb{Z}$ in degrees n and $n + 1$.

Solution. By [Proposition 6.4](#) the chain complex is $\mathbb{Z} \xrightarrow{m} \mathbb{Z}$ in degrees $n + 1, n$, so dualising gives $\mathbb{Z} \xrightarrow{m} \mathbb{Z}$ in degrees $n, n + 1$. Hence $H^n = \ker(m) = 0$ and $H^{n+1} = \mathbb{Z}/m$, all other reduced groups vanishing. Via [Theorem 6.1](#): H^n has $\text{Hom}(\mathbb{Z}/m, \mathbb{Z}) = 0$ and no Ext contribution, while H^{n+1} is $\text{Ext}(\mathbb{Z}/m, \mathbb{Z}) = \mathbb{Z}/m$.

6.3 A natural sequence with an unnatural splitting

What exactly is being claimed when one says the splitting is not natural? The following turns the claim into something testable.

Proposition 6.7. *Suppose the splitting of Theorem 6.1 could be chosen naturally, i.e. so that for every map $f: X \rightarrow Y$ the induced maps commute with the chosen isomorphisms $H^n(-; \mathbb{Z}) \cong \text{Hom}(H_n, \mathbb{Z}) \oplus \text{Ext}(H_{n-1}, \mathbb{Z})$. Then any map f with $\tilde{H}_*(f) = 0$ would satisfy $\tilde{H}^*(f) = 0$.*

Proof. A natural splitting would identify f^* with $\text{Hom}(f_*, \mathbb{Z}) \oplus \text{Ext}(f_*, \mathbb{Z})$. Each summand is a functor evaluated at f_* ; so if f_* vanishes in all positive degrees, both summands vanish, and therefore so does f^* . $\square$

So it suffices to exhibit one map that kills reduced homology and does not kill reduced cohomology. The smallest such map is already available.

Example 6.8 (A map that homology cannot see). Let $q: \mathbb{RP}^2 \rightarrow \mathbb{RP}^2/\mathbb{RP}^1 \cong S^2$ collapse the 1-skeleton.

On homology q is zero. By Example 4.7, $\tilde{H}_*(\mathbb{RP}^2)$ is $\mathbb{Z}/2$ in degree 1 and 0 elsewhere, while $\tilde{H}_*(S^2)$ is $\mathbb{Z}$ in degree 2 and 0 elsewhere. In degree 1 the target is 0; in degree 2 the source is 0. So $q_* = 0$ in every positive degree, for want of anywhere to go.

On cohomology q is not zero. On cellular chains, q carries the 2-cell of $\mathbb{RP}^2$ homeomorphically to the 2-cell of S^2 , so $q_\#: C_2(\mathbb{RP}^2) \rightarrow C_2(S^2)$ is an isomorphism, and therefore so is $q^\#: C^2(S^2) \rightarrow C^2(\mathbb{RP}^2)$. Passing to cohomology, and using Example 6.2, the map

$$q^*: H^2(S^2; \mathbb{Z}) = \mathbb{Z} \longrightarrow H^2(\mathbb{RP}^2; \mathbb{Z}) = \mathbb{Z}/2$$

is the quotient map $\mathbb{Z} \rightarrow \mathbb{Z}/2$, since a generating cocycle of $C^2(S^2)$ pulls back to a generating cocycle of $C^2(\mathbb{RP}^2)$. In particular $q^* \neq 0$.

Corollary 6.9. *The splitting in Theorem 6.1 cannot be chosen naturally.*

Proof. Example 6.8 contradicts Proposition 6.7. $\square$

Warning 6.10. None of this contradicts Theorem 6.1. What is natural is the *sequence*: for every f , the two squares

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Ext}(H_{n-1}Y, G) & \longrightarrow & H^n(Y; G) & \longrightarrow & \text{Hom}(H_nY, G) \longrightarrow 0 \\ & & \downarrow & & \downarrow f^* & & \downarrow \\ 0 & \longrightarrow & \text{Ext}(H_{n-1}X, G) & \longrightarrow & H^n(X; G) & \longrightarrow & \text{Hom}(H_nX, G) \longrightarrow 0 \end{array}$$

commute, the outer vertical maps being $\text{Ext}(f_*, G)$ and $\text{Hom}(f_*, G)$. And for each fixed X the sequence splits. What fails is the possibility of choosing the splittings compatibly for all X at once. Example 6.8 shows exactly how: there f^* is nonzero although both outer maps are zero, which a splitting compatible with f would forbid.

Exercise 6.11 (medium). Verify directly that the evaluation map $h: H^n(X; G) \rightarrow \text{Hom}(H_n(X), G)$ of Theorem 6.1 is natural, and explain why this does not conflict with Corollary 6.9.

Hint. Write h on cocycles and cycles and chase $f^\#$ and $f_\#$.

Solution. For a cocycle $\varphi \in C^n(Y; G)$ and a cycle $z \in C_n(X)$,

$$h(f^*[\varphi])([z]) = (f^\# \varphi)(z) = \varphi(f_\# z) = h([\varphi])(f_*[z]) = \text{Hom}(f_*, G)(h[\varphi])([z]),$$

so $h \circ f^* = \text{Hom}(f_*, G) \circ h$. This is a statement about the map $H^n \rightarrow \text{Hom}(H_n, G)$, which exists canonically. A splitting is a map $\text{Hom}(H_n, G) \rightarrow H^n$ going the other way; nothing in the computation above produces one, and [Corollary 6.9](#) says no natural choice exists. In [Example 6.8](#) the map h is zero on both spaces — both Hom terms vanish — and all the content of q^* lives in the Ext subgroup, which is precisely where a splitting would have had to place it compatibly.

Remark 6.12 (What to do with the mechanism). [Example 6.8](#) is the smallest instance of a general construction: take a space whose reduced homology is a single torsion group, collapse the part of it carrying the torsion, and compare. [Exercise 6.6](#) has already computed the cohomology of $M(\mathbb{Z}/m, n)$ for every m and n , and [Proposition 6.4](#) supplies the collapse map $M(\mathbb{Z}/m, n) \rightarrow S^{n+1}$; running the argument of [Example 6.8](#) in those dimensions, and then running it backwards along the inclusion of the bottom sphere, is the natural way to consolidate this section.

Remark 6.13 (Forward). The same phenomenon occurs one theorem later. The Künneth sequence for $H_*(X \times Y)$ is natural and splits, and its splitting is not natural either; [Chapter 8](#) treats it, and by then the reader should be able to predict both the statement and the shape of the counterexample before reading them.

The deeper point is that the Ext summand carries information about X that $H_*(X)$ alone does not determine as a functor. Recovering it systematically — finding invariants of a space finer than its cohomology groups — is what cohomology operations do, and [Chapter 18](#) constructs them.

6.4 What to carry forward

- H^n is $\text{Hom}(H_n, G)$ plus $\text{Ext}(H_{n-1}, G)$: torsion moves up one degree ([Theorem 6.1](#) and [Example 6.2](#)).
- Moore spaces exist for every abelian group, with cells in two dimensions only ([Proposition 6.4](#) and [Remark 6.5](#)); they are the standard probes.
- Natural sequence, natural evaluation map, unnatural splitting ([Warning 6.10](#)); the test for non-naturality is [Proposition 6.7](#), and the smallest witness is [Example 6.8](#).
- Cohomology distinguishes maps that homology cannot — the first evidence that the extra structure on H^* is worth having, which [Chapter 7](#) turns into a product.

Chapter 7

Products and duality

Three invariants have now failed on the same pair of spaces. In [Chapter 2](#) the fundamental group did not distinguish $L(5, 1)$ from $L(5, 2)$; in [Chapter 4](#) their homology turned out not to involve q at all. The reason, in both cases, was structural: those invariants are assembled from the cells one at a time, and lens spaces differ in how their cells are *linked*.

This chapter introduces the first invariant with a multiplication. It succeeds where the groups fail — distinguishing spaces with identical cohomology groups — and it also fails, on the same lens spaces, in a way that will take until [Chapter 18](#) to repair. Both halves are the point.

7.1 The cup product

We take the construction as known: $H^*(X; R) = \bigoplus_n H^n(X; R)$ carries an associative product, natural in X , with unit $1 \in H^0$, satisfying

$$\alpha \smile \beta = (-1)^{|\alpha||\beta|} \beta \smile \alpha.$$

Naturality means $f^*(\alpha \smile \beta) = f^*\alpha \smile f^*\beta$: a map of spaces induces a map of rings, and that is where all the power comes from.

Example 7.1 (The torus). $H^*(T^2; \mathbb{Z})$ is the exterior algebra $\Lambda(a, b)$ on two generators of degree one: $a^2 = b^2 = 0$ by graded commutativity, $ab = -ba$, and ab generates $H^2(T^2) \cong \mathbb{Z}$. The last assertion — that the product of the two one-dimensional generators is a generator, not merely some element — is the content, and it is what [Theorem 7.5](#) will explain.

Example 7.2 (Complex projective plane against a wedge). $H^*(\mathbb{CP}^2; \mathbb{Z}) = \mathbb{Z}[x]/(x^3)$ with $|x| = 2$, so the groups are $\mathbb{Z}, 0, \mathbb{Z}, 0, \mathbb{Z}$ in degrees $0, \dots, 4$ and x^2 generates H^4 .

The wedge $S^2 \vee S^4$ has exactly the same cohomology *groups*. But its ring is different: writing y for the generator of $H^2(S^2 \vee S^4)$, we have $y^2 = 0$. Indeed y restricts to a generator on S^2 and to 0 on S^4 , so y^2 restricts to 0 on both — on S^2 because $H^4(S^2) = 0$ — and restriction to the two summands is injective on $\tilde{H}^*(S^2 \vee S^4)$.

So $\mathbb{CP}^2 \not\cong S^2 \vee S^4$: the groups cannot see the difference and the ring can.

The mechanism in that example is worth isolating, because it is the only thing one ever uses about wedges.

Exercise 7.3 (easy). Show that $\tilde{H}^*(X \vee Y; R) \cong \tilde{H}^*(X; R) \oplus \tilde{H}^*(Y; R)$ as graded groups, and that under this isomorphism the product of a class coming from X with a class coming from Y is zero.

Hint. Use the retractions $X \vee Y \rightarrow X$ and $X \vee Y \rightarrow Y$, together with the inclusions, and naturality of the product.

Solution. The inclusions $i: X \rightarrow X \vee Y$, $j: Y \rightarrow X \vee Y$ and the retractions $r: X \vee Y \rightarrow X$, $s: X \vee Y \rightarrow Y$ collapsing the other summand satisfy $ri = \text{id}$, $sj = \text{id}$, $rj = \text{const}$, $si = \text{const}$. Hence (i^*, j^*) is a split surjection with splitting $r^* + s^*$, giving the direct sum decomposition, with the summand “coming from X ” being $\text{im } r^*$.

Now let $\alpha = r^*\alpha'$ and $\beta = s^*\beta'$ with α', β' of positive degree. Then $i^*(\alpha \smile \beta) = i^*\alpha \smile i^*\beta$, and $i^*\beta = i^*s^*\beta' = (si)^*\beta' = 0$ since si is constant and β' has positive degree. So $i^*(\alpha \smile \beta) = 0$, and likewise $j^*(\alpha \smile \beta) = 0$. As (i^*, j^*) is injective on $\tilde{H}^*$, the product vanishes.

Exercise 7.4 (medium). Show that $S^2 \times S^2$ and $S^2 \vee S^2 \vee S^4$ have isomorphic cohomology groups in every degree but non-isomorphic cohomology rings, and deduce that they are not homotopy equivalent.

Hint. By Künneth, $H^*(S^2 \times S^2)$ has generators a, b in degree 2 with $a^2 = b^2 = 0$. What is ab ?

Solution. For the product, Künneth gives $H^*(S^2 \times S^2; \mathbb{Z}) = \mathbb{Z}[a, b]/(a^2, b^2)$ with $|a| = |b| = 2$, so the groups are $\mathbb{Z}, 0, \mathbb{Z}^2, 0, \mathbb{Z}$ and ab generates H^4 . For the wedge, the groups are the same, but by Exercise 7.3 every product of positive-degree classes vanishes: the classes from the two S^2 's multiply to zero, and each squares to zero because $H^4(S^2) = 0$.

A ring isomorphism would have to carry a pair of degree-2 generators to a pair of degree-2 generators and preserve their product, which is a generator of H^4 on one side and 0 on the other. So the rings differ, and since $H^*(-; \mathbb{Z})$ is a homotopy-invariant functor to rings, the spaces are not homotopy equivalent.

7.2 Poincaré duality

Theorem 7.5 (Poincaré duality; recalled). *Let M be a closed connected n -manifold.*

(a) *If M is orientable, capping with the fundamental class gives an isomorphism*

$$H^k(M; \mathbb{Z}) \cong H_{n-k}(M; \mathbb{Z}) \quad \text{for all } k.$$

(b) *With no orientability hypothesis at all, the same holds over $\mathbb{F}_2$:*

$$H^k(M; \mathbb{F}_2) \cong H_{n-k}(M; \mathbb{F}_2).$$

In the orientable case the pairing $H^k(M; \mathbb{Z}) \otimes H^{n-k}(M; \mathbb{Z}) \rightarrow H^n(M; \mathbb{Z}) \cong \mathbb{Z}$ given by the cup product is nondegenerate modulo torsion; with $\mathbb{F}_2$ coefficients it is nondegenerate.

The nondegeneracy statement is what makes the cup product computable on manifolds: it says a nonzero class in H^k must have a partner it pairs nontrivially with. Example 7.1 is the smallest instance.

Exercise 7.6 (easy). Show that $H^1(X; \mathbb{Z})$ is torsion-free for every space X , and free abelian if $H_1(X)$ is finitely generated.

Hint. Apply [Theorem 6.1](#) in degree one.

Solution. By [Theorem 6.1](#), $H^1(X; \mathbb{Z})$ is an extension of $\text{Hom}(H_1 X, \mathbb{Z})$ by $\text{Ext}(H_0 X, \mathbb{Z})$, and $H_0 X$ is free abelian, so $\text{Ext}(H_0 X, \mathbb{Z}) = 0$ and $H^1(X; \mathbb{Z}) \cong \text{Hom}(H_1 X, \mathbb{Z})$. A group of homomorphisms into $\mathbb{Z}$ is torsion-free, since $n\varphi = 0$ with $n \neq 0$ forces $\varphi = 0$; and Hom of a finitely generated group into $\mathbb{Z}$ is finitely generated and torsion-free, hence free.

Corollary 7.7. *For a closed orientable n -manifold M , the group $H_{n-1}(M; \mathbb{Z})$ is free abelian.*

Proof. $H_{n-1}(M; \mathbb{Z}) \cong H^1(M; \mathbb{Z})$ by [Theorem 7.5\(a\)](#), which is free by [Exercise 7.6](#), the homology of a closed manifold being finitely generated. $\square$

Exercise 7.8 (medium). Show that $\chi(M) = 0$ for every closed manifold of odd dimension, orientable or not.

Hint. The Euler characteristic can be computed with coefficients in any field ([Exercise 4.12](#) applies verbatim to the $\mathbb{F}_2$ chain complex). Then use duality in the form that needs no orientation.

Solution. Compute $\chi(M) = \sum_k (-1)^k \dim_{\mathbb{F}_2} H_k(M; \mathbb{F}_2)$, which agrees with the integral Euler characteristic because both equal the alternating sum of the numbers of cells. By [Theorem 7.5\(b\)](#), $\dim_{\mathbb{F}_2} H_k(M; \mathbb{F}_2) = \dim_{\mathbb{F}_2} H^{n-k}(M; \mathbb{F}_2) = \dim_{\mathbb{F}_2} H_{n-k}(M; \mathbb{F}_2)$, the last equality because over a field cohomology is the dual of homology. So the terms k and $n - k$ have equal dimensions, and for n odd they have opposite signs, so they cancel in pairs.

Remark 7.9. Put the last three results together for a closed connected 3-manifold M and almost nothing is left undetermined. Write $H_1(M; \mathbb{Z}) = \mathbb{Z}^r \oplus F$ with F finite. Then $H_0 = \mathbb{Z}$; H_3 is $\mathbb{Z}$ or 0 according to orientability; [Exercise 7.8](#) fixes the rank of H_2 once the others are known; and [Corollary 7.7](#) settles the torsion of H_2 in the orientable case. Only the nonorientable torsion needs a further idea, namely the Bockstein of the coefficient sequence $0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{F}_2 \rightarrow 0$ applied to the $\mathbb{F}_2$ -fundamental class. Assembling this is the natural way to finish the chapter's duality section, and every ingredient is now on the page.

7.3 Where the ring fails

Example 7.10 (Lens spaces, again). By [Example 4.8](#) and [Theorem 6.1](#),

$$H^*(L(p, q); \mathbb{Z}) = (\mathbb{Z}, 0, \mathbb{Z}/p, \mathbb{Z})$$

in degrees 0, 1, 2, 3. The multiplication is forced and carries no information: the only products of positive-degree classes to compute land in $H^4 = 0$ or pair $H^2 \otimes H^2 \rightarrow H^4 = 0$. So the integral cohomology ring of $L(p, q)$, like its homology and its fundamental group, is independent of q .

Yet $L(5, 1)$ and $L(5, 2)$ are not homotopy equivalent. This is a classical theorem: $L(p, q)$ and $L(p, q')$ are homotopy equivalent if and only if $qq' \equiv \pm n^2 \pmod{p}$ for some integer n , and for $p = 5$ the squares mod 5 are $\{0, 1, 4\}$, which is closed under negation, while $qq' = 2$.

Warning 7.11. The book does not prove the classification just quoted, and the reader should not pretend otherwise. What separates these spaces is a refinement of the cup product called the *linking form*, a $\mathbb{Q}/\mathbb{Z}$ -valued pairing on the torsion of H_1 , which is

invisible to the integral ring because that ring is trivial in the relevant degrees. What is worth extracting here is only the shape of the failure: a product can be forced to vanish for degree reasons and therefore carry no information, while a finer operation on the same cohomology does.

Remark 7.12 (The question Part I ends on). [Example 7.2](#) and [Exercise 7.4](#) showed that the cohomology ring is strictly stronger than the cohomology groups. [Example 7.10](#) shows it is not complete. There are even pairs of spaces with *isomorphic cohomology rings*, over every coefficient ring at once, that are not homotopy equivalent; producing one is not hard once one knows where to look, and recognising that the ring has run out is the first step.

What replaces it is an operation that is natural but not induced by any multiplication: a family of maps $H^n(X; \mathbb{F}_2) \rightarrow H^{n+i}(X; \mathbb{F}_2)$ commuting with f^* for every f . These are the Steenrod squares, and [Chapter 18](#) constructs them and uses them to settle a case that the ring provably cannot. Part II builds the machinery they need.

7.4 What to carry forward

- The cup product makes f^* a ring map; that naturality is the whole source of its strength.
- On a wedge, classes from different summands multiply to zero ([Exercise 7.3](#)) — the standard obstruction.
- $\mathbb{CP}^2$ against $S^2 \vee S^4$, and $S^2 \times S^2$ against $S^2 \vee S^2 \vee S^4$: same groups, different rings ([Example 7.2](#) and [Exercise 7.4](#)).
- Duality ([Theorem 7.5](#)): orientable over $\mathbb{Z}$, always over $\mathbb{F}_2$; the cup pairing is nondegenerate, so nonzero classes have partners.
- H^1 is torsion-free ([Exercise 7.6](#)), H_{n-1} of a closed orientable manifold is free ([Corollary 7.7](#)), and $\chi = 0$ in odd dimensions ([Exercise 7.8](#)).
- The ring is not a complete invariant either ([Example 7.10](#) and [Remark 7.12](#)); the repair is [Chapter 18](#).

Part II

Homological algebra and the homotopy toolkit

Chapter 8

Derived functors in practice: Ext and Tor

Part I used Ext and Tor as bookkeeping: a term in the universal coefficient sequence, a term in Künneth. This chapter treats them as what they are — derived functors — and the change of viewpoint pays off twice. It explains which of their properties are theorems and which are accidents of the ring $\mathbb{Z}$; and it makes them computable in situations where no exact sequence is handed to us.

The second half of the chapter runs the traffic in the unusual direction. A space with prescribed homology is an algebraic device: build one, compute its homology in two ways, and out comes a theorem about abelian groups. That manoeuvre is the point of [Section 8.3](#), and it is worth more than either of the particular results proved with it.

8.1 Resolutions, and what is special about $\mathbb{Z}$

Definition 8.1. A *free resolution* of an R -module A is an exact sequence

$$\cdots \rightarrow F_2 \rightarrow F_1 \rightarrow F_0 \rightarrow A \rightarrow 0$$

with each F_i free. Applying $\mathrm{Hom}_R(-, B)$ to the deleted resolution and taking cohomology gives $\mathrm{Ext}_R^n(A, B)$; applying $- \otimes_R B$ and taking homology gives $\mathrm{Tor}_n^R(A, B)$.

Proposition 8.2 (Comparison; recalled). *Any two free resolutions of A are chain homotopy equivalent, by an equivalence lifting the identity of A and unique up to chain homotopy. Consequently $\mathrm{Ext}_R^n(A, B)$ and $\mathrm{Tor}_n^R(A, B)$ do not depend on the resolution chosen.*

Over $\mathbb{Z}$ something special happens, and it is worth naming so that it can later be taken away.

Definition 8.3. A ring is *hereditary* if every submodule of a free module is projective.

Example 8.4 ($\mathbb{Z}$ is hereditary). Every subgroup of a free abelian group is free abelian. Hence for every abelian group A the kernel F_1 of a surjection $F_0 \twoheadrightarrow A$ from a free group is itself free, and

$$0 \longrightarrow F_1 \longrightarrow F_0 \longrightarrow A \longrightarrow 0$$

is a free resolution of length one. Therefore

$$\mathrm{Ext}_{\mathbb{Z}}^n(A, B) = 0 \quad \text{and} \quad \mathrm{Tor}_{\mathbb{Z}}^n(A, B) = 0 \quad \text{for all } n \geq 2,$$

which is why Part I could write Ext and Tor with no index at all.

Example 8.5 (The standard computations). From $0 \rightarrow \mathbb{Z} \xrightarrow{m} \mathbb{Z} \rightarrow \mathbb{Z}/m \rightarrow 0$ one reads off

$$\mathrm{Ext}(\mathbb{Z}/m, \mathbb{Z}) \cong \mathbb{Z}/m, \quad \mathrm{Ext}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/d, \quad \mathrm{Tor}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/d,$$

where $d = \gcd(m, n)$; and both vanish as soon as the first variable is free.

Now remove the hypothesis.

Exercise 8.6 (medium). Let k be a field and $R = k[x]/(x^2)$, the ring of dual numbers, and regard $k = R/(x)$ as an R -module. Show that

$$\mathrm{Ext}_R^n(k, k) \cong k \quad \text{for every } n \geq 0.$$

Hint. Multiplication by x has kernel exactly (x) , which is also its image. What does that say about the sequence $\cdots \rightarrow R \xrightarrow{x} R \xrightarrow{x} R$?

Solution. In R we have $\ker(\cdot x) = (x) = \mathrm{im}(\cdot x)$, so

$$\cdots \longrightarrow R \xrightarrow{\cdot x} R \xrightarrow{\cdot x} R \longrightarrow k \longrightarrow 0$$

is exact: the augmentation $R \rightarrow k = R/(x)$ has kernel (x) , which is the image of the first $\cdot x$, and exactness at every later spot is the displayed equality. So k admits a free resolution which repeats itself with period one.

Apply $\mathrm{Hom}_R(-, k)$. Each $\mathrm{Hom}_R(R, k) = k$, and the induced map is multiplication by x on k , which is zero because x acts as 0 on $R/(x)$. So the complex computing Ext is $k \xrightarrow{0} k \xrightarrow{0} k \rightarrow \cdots$, and $\mathrm{Ext}_R^n(k, k) = k$ for every $n \geq 0$.

Remark 8.7. A periodic resolution produces periodic Ext, and periodicity is common: it happens for $k[x]/(x^2)$ above, for cyclic groups over $\mathbb{Z}$ in group cohomology (Chapter 13 uses $H^*(\mathbb{Z}/p)$ being nonzero in every degree), and for any ring where some element has annihilator equal to its own ideal. The general lesson is that “how long is a resolution” is a question about the ring, and Example 8.4 is the accident that lets one forget it over $\mathbb{Z}$.

8.2 Tor, flatness, and balance

Tor may be computed by resolving either variable, and the resolution need not be free.

Theorem 8.8 (Balance and flat resolutions; recalled). $\mathrm{Tor}_n^R(A, B)$ may be computed from a projective resolution of A or of B , with canonically isomorphic results. More generally it may be computed from a resolution of either variable by flat modules.

Example 8.9. Compute $\mathrm{Tor}(\mathbb{Z}/m, \mathbb{Z}/n)$ twice. Resolving the first variable by $0 \rightarrow \mathbb{Z} \xrightarrow{m} \mathbb{Z} \rightarrow \mathbb{Z}/m \rightarrow 0$ and tensoring with $\mathbb{Z}/n$ gives $\mathbb{Z}/n \xrightarrow{m} \mathbb{Z}/n$, whose kernel is $\mathbb{Z}/\gcd(m, n)$. Resolving the second variable gives $\mathbb{Z}/m \xrightarrow{n} \mathbb{Z}/m$, whose kernel is again $\mathbb{Z}/\gcd(m, n)$. The two computations look different and agree, which is what Theorem 8.8 promises.

The flat-resolution clause is not a technicality; it is what makes the non-finitely-generated cases computable at all, because the useful resolutions there are rarely free.

Exercise 8.10 (medium). Show that a torsion-free abelian group A is flat, and deduce $\operatorname{Tor}(A, B) = 0$ for all B . In particular $\mathbb{Q}$ is flat, so

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

is a flat resolution of $\mathbb{Q}/\mathbb{Z}$ of length one.

Hint. A finitely generated torsion-free abelian group is free. And A is the filtered union of its finitely generated subgroups.

Solution. Write $A = \varinjlim A_i$ over its finitely generated subgroups, a filtered colimit. Each A_i is finitely generated and torsion-free, hence free, hence flat. Tensor products commute with filtered colimits and filtered colimits are exact, so $A \otimes -$ is exact: A is flat. Equivalently $\operatorname{Tor}_1(A, B) = \varinjlim \operatorname{Tor}_1(A_i, B) = 0$.

$\mathbb{Q}$ is torsion-free, so it is flat, and the displayed sequence is then a resolution of $\mathbb{Q}/\mathbb{Z}$ by flat modules, of length one. By [Theorem 8.8](#) it computes $\operatorname{Tor}(-, \mathbb{Q}/\mathbb{Z})$.

Exercise 8.11 (easy). Show that the kernel of $A \rightarrow A \otimes \mathbb{Q}$, $a \mapsto a \otimes 1$, is exactly the torsion subgroup of A .

Solution. If $na = 0$ with $n \neq 0$ then $a \otimes 1 = a \otimes \frac{n}{n} = na \otimes \frac{1}{n} = 0$, so torsion elements die. Conversely $A \otimes \mathbb{Q}$ is the localisation of A at $\mathbb{Z} \setminus \{0\}$, in which $a \otimes 1 = 0$ precisely when $na = 0$ for some $n \neq 0$; so the kernel consists of torsion elements only.

Remark 8.12. [Exercises 8.10](#) and [8.11](#) fit together. Applying $A \otimes -$ to the flat resolution $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$ makes $\operatorname{Tor}(A, \mathbb{Q}/\mathbb{Z})$ the kernel of $A \otimes \mathbb{Z} \rightarrow A \otimes \mathbb{Q}$, and [Exercise 8.11](#) identifies that kernel. Carrying this out, and then deducing a characterisation of torsion-freeness in terms of the vanishing of Tor , is the natural exercise to set oneself here; both ingredients are on this page.

8.3 Spaces as algebraic devices

Moore spaces ([Proposition 6.4](#)) have homology concentrated in one degree, for an arbitrary prescribed group. That makes them instruments: compute an invariant of a Moore space in two different ways, and the comparison is a theorem of algebra.

Example 8.13 (Tor realised as a homology group). Let $n, m \geq 1$ and put $X = M(A, n)$, $Y = M(B, m)$. Künneth gives

$$H_k(X \times Y) \cong \bigoplus_{i+j=k} H_i(X) \otimes H_j(Y) \oplus \bigoplus_{i+j=k-1} \operatorname{Tor}(H_i(X), H_j(Y)).$$

The only nonzero reduced homology of X is A in degree n , and of Y is B in degree m . In degree $k = n + m + 1$ the tensor summands all vanish — the available degrees are 0 , n , m and $n + m$, and $n + m + 1$ is none of them — while the Tor summands require $i + j = n + m$, leaving exactly $\operatorname{Tor}(A, B)$. Hence

$$H_{n+m+1}(M(A, n) \times M(B, m)) \cong \operatorname{Tor}(A, B).$$

Exercise 8.14 (medium). Deduce that $\operatorname{Tor}(A, B) \cong \operatorname{Tor}(B, A)$ for all abelian groups A and B , without using a resolution.

Hint. $X \times Y$ and $Y \times X$ are homeomorphic.

Solution. Take $n = m$ in [Example 8.13](#), so that $\text{Tor}(A, B) \cong H_{2n+1}(M(A, n) \times M(B, n))$ and $\text{Tor}(B, A) \cong H_{2n+1}(M(B, n) \times M(A, n))$. The swap map $(u, v) \mapsto (v, u)$ is a homeomorphism between the two spaces, so it induces an isomorphism on homology in every degree, and in particular in degree $2n + 1$.

Remark 8.15 (The method, stated plainly). The manoeuvre has three steps: realise an algebraic functor as a homology group of a space built to order; find a second computation of that homology group; and read the comparison as an algebraic identity. [Example 8.13](#) realises Tor ; the second computation in [Exercise 8.14](#) was a homeomorphism. Other second computations are available — coefficients in a field, where the universal coefficient theorem loses its Ext term, is the most useful one, and it converts a vanishing statement about the cohomology of a Moore space directly into the vanishing of an Ext group. [Proposition 6.4](#) builds the space for any group at all, so nothing here requires finite generation.

8.4 Künneth, and non-naturality again

Theorem 8.16 (Künneth; recalled). *For CW complexes X, Y there is a short exact sequence, natural in both variables,*

$$0 \rightarrow \bigoplus_{i+j=k} H_i(X) \otimes H_j(Y) \rightarrow H_k(X \times Y) \rightarrow \bigoplus_{i+j=k-1} \text{Tor}(H_i(X), H_j(Y)) \rightarrow 0,$$

and it splits — but not naturally.

Example 8.17 (A Künneth drill). $H_*(\mathbb{RP}^2) = (\mathbb{Z}, \mathbb{Z}/2, 0)$, so for $X = Y = \mathbb{RP}^2$:

$$H_*(\mathbb{RP}^2 \times \mathbb{RP}^2) = (\mathbb{Z}, (\mathbb{Z}/2)^2, \mathbb{Z}/2, \mathbb{Z}/2, 0).$$

Degrees 1 and 2 come from tensor summands, and the $\mathbb{Z}/2$ in degree 3 comes from $\text{Tor}(\mathbb{Z}/2, \mathbb{Z}/2)$ alone: there is no tensor contribution in that degree at all. A class living in a degree where the tensor part vanishes is exactly what a natural splitting would have to keep track of.

Proposition 8.18. *Suppose the splitting of [Theorem 8.16](#) could be chosen naturally in both variables. Then for maps $f: X \rightarrow X'$ and $g: Y \rightarrow Y'$ the induced map $(f \times g)_*$ would decompose as $(\bigoplus f_* \otimes g_*) \oplus (\bigoplus \text{Tor}(f_*, g_*))$. In particular, were f_* to vanish on $\tilde{H}_*$, every summand built from positive-degree homology of X would be killed by $(f \times g)_*$.*

Proof. A natural splitting identifies the middle term with the sum of the two outer ones, compatibly with induced maps. On each outer term the induced map is the functor $\otimes$ or Tor evaluated at the pair (f_*, g_*) , and both vanish as soon as f_* does. $\square$

Remark 8.19 (What to do with it). [Proposition 8.18](#) is the exact analogue of [Proposition 6.7](#), and it is refuted in exactly the same way: exhibit one map that is zero on reduced homology and one product in which it is not zero. [Chapter 6](#) supplied both the collapse map $M(\mathbb{Z}/m, n) \rightarrow S^{n+1}$ and the computation showing what it does to cohomology; [Example 8.17](#) shows where in a product a Tor class can sit with no tensor class beside it. Crossing such a collapse with the identity and asking what happens to

the Tor summand is the natural experiment, and the reader who ran [Remark 6.12](#) has already done its harder half.

The pattern is now established twice, and it is worth stating as a slogan: an Ext or Tor summand records information about a space that its homology groups do not determine *as a functor*. Everything in [Chapter 18](#) is an attempt to name that information.

8.5 What to carry forward

- Resolutions compute derived functors and the answer is resolution-independent ([Proposition 8.2](#)); over $\mathbb{Z}$ they have length one, and that is a property of $\mathbb{Z}$ ([Example 8.4](#) and [Exercise 8.6](#)).
- Tor is balanced and may be computed by *flat* resolutions ([Theorem 8.8](#)); torsion-free abelian groups are flat ([Exercise 8.10](#)).
- A Moore space realises Tor as a homology group ([Example 8.13](#)); comparing two computations of that group proves algebra ([Remark 8.15](#)).
- Künneth is natural and splits, unnaturally ([Theorem 8.16](#) and [Proposition 8.18](#)) — the second instance of the pattern opened in [Chapter 6](#).

Chapter 9

Towers, inverse limits, and Bocksteins

Two constructions in this chapter look unrelated and are not. Both concern a sequence of approximations to an object, and both measure the failure of an obvious guess.

The inverse limit of a tower of groups is not right exact, and its first derived functor $\varprojlim^1$ is the error term. It appears whenever the cohomology of an infinite complex is assembled from its skeleta, and it is usually zero — but only for a reason, and the reason is worth knowing.

The Bockstein is the error term of a different reduction: $H^*(X; \mathbb{F}_p)$ does not determine $H^*(X; \mathbb{Z})$, and β is the first approximation to the difference. It is a differential on mod- p cohomology whose homology detects which torsion has order exactly p . [Chapter 18](#) will identify β with the first Steenrod square, and [Chapter 26](#) will use that identification to write an integral characteristic class as a Bockstein.

9.1 The derived functor of the inverse limit, as a cokernel

Let (G_i, α_i) be a tower of abelian groups,

$$\cdots \longrightarrow G_2 \xrightarrow{\alpha_2} G_1 \xrightarrow{\alpha_1} G_0.$$

Definition 9.1. Let $\Delta: \prod_i G_i \rightarrow \prod_i G_i$ be given by $\Delta(x)_i = x_i - \alpha_{i+1}(x_{i+1})$. Then $\varprojlim G_i = \ker \Delta$ and $\varprojlim^1 G_i = \operatorname{coker} \Delta$.

So $\varprojlim^1$ measures the failure of the system of equations $x_i - \alpha_{i+1}(x_{i+1}) = y_i$ to be solvable. That is the only description we need: every computation below either solves those equations or shows they cannot be solved.

Exercise 9.2 (easy). Show that $\varprojlim^1 G_i = 0$ if every α_i is surjective.

Hint. Solve the equations recursively, choosing a preimage at each stage.

Solution. Given $y \in \prod G_i$, set $x_0 = 0$ and choose inductively x_{i+1} with $\alpha_{i+1}(x_{i+1}) = x_i - y_i$, which is possible because α_{i+1} is onto. Then $\Delta(x) = y$, so Δ is surjective and its cokernel vanishes.

Exercise 9.3 (medium). Call a tower *pro-zero* if for every i some composite $G_{i+n} \rightarrow G_i$ is zero. Show that a pro-zero tower has $\varprojlim G_i = 0$ and $\varprojlim^1 G_i = 0$.

Hint. Write $\alpha^{(n)}$ for the n -fold composite $G_{i+n} \rightarrow G_i$ and try $x_i = \sum_{n \geq 0} \alpha^{(n)}(y_{i+n})$.

Solution. The sum is finite, since $\alpha^{(n)}$ into G_i vanishes for large n . With that definition,

$$\alpha_{i+1}(x_{i+1}) = \sum_{n \geq 0} \alpha^{(n+1)}(y_{i+1+n}) = \sum_{m \geq 1} \alpha^{(m)}(y_{i+m}),$$

so $x_i - \alpha_{i+1}(x_{i+1}) = \alpha^{(0)}(y_i) = y_i$, and Δ is onto. For the limit, a compatible family satisfies $x_i = \alpha^{(n)}(x_{i+n}) = 0$ once n is large.

Theorem 9.4 (Six-term sequence; recalled). *A short exact sequence of towers $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ induces an exact sequence*

$$0 \rightarrow \varprojlim A \rightarrow \varprojlim B \rightarrow \varprojlim C \rightarrow \varprojlim^1 A \rightarrow \varprojlim^1 B \rightarrow \varprojlim^1 C \rightarrow 0.$$

Definition 9.5. Write $G_i^{(n)} = \text{im}(G_{i+n} \rightarrow G_i)$, a descending chain of subgroups of G_i . The tower satisfies the *Mittag-Leffler condition* if for each i that chain is eventually constant, with stable value G_i^∞ .

Warning 9.6. The condition does not say that the maps are eventually surjective. It says that inside each *fixed* G_i the images of the groups far above stop shrinking. A tower whose maps are eventually zero satisfies it, with $G_i^\infty = 0$, and no map in it need be surjective.

Remark 9.7 (The step that remains). The subgroups G_i^∞ form a subtower whose transition maps are surjective, and under Mittag-Leffler the quotient tower G_i/G_i^∞ is pro-zero: for large n the image of G_{i+n} in G_i is already G_i^∞ , so the composite into G_i/G_i^∞ vanishes. Feeding that short exact sequence of towers into [Theorem 9.4](#), and applying [Exercises 9.2](#) and [9.3](#) to the two outer terms, yields the standard vanishing criterion in a line. Assembling it is the natural exercise to set oneself here; the only thing to watch is [Warning 9.6](#).

[Exercise 9.2](#) might suggest that $\varprojlim^1$ is a formality.

Example 9.8 (A $\varprojlim^1$ that is uncountable). Fix a prime p , let $G_i = \mathbb{Z}$ for all i , and let every transition map be multiplication by p . Then $\varprojlim G_i = 0$, since a compatible family has x_0 divisible by p^n for every n .

For the derived functor, consider $\Sigma: \prod_i \mathbb{Z} \rightarrow \mathbb{Z}_p$ into the p -adic integers, $\Sigma(y) = \sum_i p^i y_i$. The series converges p -adically, and Σ is surjective because every p -adic integer has such an expansion. Composing with Δ telescopes:

$$\Sigma(\Delta(x)) = \sum_i p^i (x_i - px_{i+1}) = \lim_{N \rightarrow \infty} (x_0 - p^N x_N) = x_0,$$

because $p^N x_N \rightarrow 0$ in $\mathbb{Z}_p$. So Σ maps $\text{im } \Delta$ onto $\mathbb{Z} \subseteq \mathbb{Z}_p$, and it descends to an isomorphism

$$\varprojlim^1 \mathbb{Z} \cong \mathbb{Z}_p / \mathbb{Z},$$

an uncountable group. A tower of copies of $\mathbb{Z}$ with the simplest possible maps already has an enormous $\varprojlim^1$.

Theorem 9.9 (Milnor sequence; recalled). *Let the CW complex X be the union of an increasing sequence of subcomplexes $X_0 \subseteq X_1 \subseteq \dots$. For every n there is a short exact sequence*

$$0 \rightarrow \varprojlim^1 H^{n-1}(X_i; G) \rightarrow H^n(X; G) \rightarrow \varprojlim H^n(X_i; G) \rightarrow 0.$$

Exercise 9.10 (easy). Use [Theorem 9.9](#) to compute $H^*(\mathbb{CP}^\infty; \mathbb{Z})$ from the groups $H^*(\mathbb{CP}^n; \mathbb{Z})$, and say which hypothesis makes the $\varprojlim^1$ term disappear.

Solution. Each restriction $H^k(\mathbb{CP}^{n+1}; \mathbb{Z}) \rightarrow H^k(\mathbb{CP}^n; \mathbb{Z})$ is surjective: for k even and $k \leq 2n$ both groups are $\mathbb{Z}$ and the generator restricts to the generator, and otherwise the target is 0. So the tower of cohomology groups has surjective transition maps and $\varprojlim^1 = 0$ by [Exercise 9.2](#). Hence $H^k(\mathbb{CP}^\infty; \mathbb{Z}) \cong \varprojlim_n H^k(\mathbb{CP}^n; \mathbb{Z})$, which is $\mathbb{Z}$ for k even and 0 for k odd.

9.2 The Bockstein as a differential

Definition 9.11. The coefficient sequence $0 \rightarrow \mathbb{Z}/p \rightarrow \mathbb{Z}/p^2 \rightarrow \mathbb{Z}/p \rightarrow 0$ induces a natural connecting homomorphism

$$\beta: H^n(X; \mathbb{F}_p) \longrightarrow H^{n+1}(X; \mathbb{F}_p),$$

the *Bockstein*. The sequence $0 \rightarrow \mathbb{Z} \xrightarrow{p} \mathbb{Z} \rightarrow \mathbb{F}_p \rightarrow 0$ induces the *integral Bockstein* $\tilde{\beta}: H^n(X; \mathbb{F}_p) \rightarrow H^{n+1}(X; \mathbb{Z})$, of which β is the mod- p reduction.

Exercise 9.12 (medium). Show that $\beta^2 = 0$ and that β is a derivation: $\beta(xy) = \beta(x)y + (-1)^{|x|}x\beta(y)$.

Hint. For the first, the composite $\mathbb{Z}/p \hookrightarrow \mathbb{Z}/p^2 \twoheadrightarrow \mathbb{Z}/p$ occurring in two consecutive connecting maps is zero. For the second, lift cocycles to $\mathbb{Z}/p^2$ -cochains and differentiate the product of the lifts.

Solution. Applying the connecting map twice factors through $\mathbb{Z}/p \hookrightarrow \mathbb{Z}/p^2 \twoheadrightarrow \mathbb{Z}/p$, which is zero; hence $\beta^2 = 0$.

For the derivation rule, represent x, y by $\mathbb{F}_p$ -cocycles a, b and choose $\mathbb{Z}/p^2$ -cochain lifts $\tilde{a}, \tilde{b}$, so that $\delta\tilde{a} = pa'$ and $\delta\tilde{b} = pb'$ with a', b' representing $\beta x, \beta y$. The product $\tilde{a}\tilde{b}$ lifts ab , and

$$\delta(\tilde{a}\tilde{b}) = (\delta\tilde{a})\tilde{b} + (-1)^{|\tilde{a}|}\tilde{a}(\delta\tilde{b}) = p(a'\tilde{b} + (-1)^{|\tilde{a}|}\tilde{a}b').$$

Reducing mod p gives the stated formula.

Since $\beta^2 = 0$, one may take homology with respect to it.

Definition 9.13. The *Bockstein homology* of X is $H(H^*(X; \mathbb{F}_p), \beta)$.

Proposition 9.14 (What β detects; recalled). *If $H^*(X; \mathbb{Z})$ is finitely generated in each degree, the Bockstein homology of X is isomorphic to $(H^*(X; \mathbb{Z})/\text{torsion}) \otimes \mathbb{F}_p$. It therefore vanishes in positive degrees exactly when the free part of $H^*(X; \mathbb{Z})$ sits in degree 0 and every p -torsion class has order exactly p .*

Example 9.15 (β on a Moore space). For $X = M(\mathbb{Z}/p, n)$ the reduced mod- p cohomology is $\mathbb{F}_p$ in degrees n and $n+1$, and $\beta: H^n \rightarrow H^{n+1}$ is an isomorphism: by [Exercise 6.6](#) the integral cohomology is $\mathbb{Z}/p$ in degree $n+1$, on which multiplication by p is zero, so the connecting map of $0 \rightarrow \mathbb{Z} \xrightarrow{p} \mathbb{Z} \rightarrow \mathbb{F}_p \rightarrow 0$ is nonzero.

For $X = M(\mathbb{Z}/p^2, n)$ the mod- p cohomology consists of exactly the same two groups, but now $\beta = 0$: the integral torsion has order p^2 , and multiplication by p on $\mathbb{Z}/p^2$ is not injective, which kills the connecting map.

Two spaces with identical mod- p cohomology groups are thus separated by β . It sees the *order* of the torsion, not merely its presence.

Example 9.16 (Infinite projective space). $H^*(\mathbb{RP}^\infty; \mathbb{F}_2) = \mathbb{F}_2[x]$ with $|x| = 1$, and $\beta x = x^2$ — [Example 9.15](#) in degree one, $\mathbb{RP}^2 = M(\mathbb{Z}/2, 1)$ being the 2-skeleton. By [Exercise 9.12](#),

$$\beta(x^k) = k x^{k+1},$$

which is x^{k+1} for k odd and 0 for k even. So $\ker \beta$ has basis $1, x^2, x^4, \dots$ and $\operatorname{im} \beta$ has basis $x^2, x^4, \dots$, and the Bockstein homology is $\mathbb{F}_2$ in degree 0 and zero in every positive degree.

Read through [Proposition 9.14](#), that says the free part of $H^*(\mathbb{RP}^\infty; \mathbb{Z})$ lives in degree 0 and every torsion class has order exactly 2 — both true by [Example 6.2](#).

Exercise 9.17 (medium). For p odd, $H^*(L(p, q); \mathbb{F}_p)$ has basis $1, x, y, xy$ in degrees $0, 1, 2, 3$, with $\beta x = y$. Compute the Bockstein homology, and check it against [Proposition 9.14](#) and the integral computation of [Example 4.8](#).

Hint. Get βy from $\beta^2 = 0$, and $\beta(xy)$ from the derivation rule. Note which degree y^2 would live in.

Solution. From $\beta x = y$ and $\beta^2 = 0$ we get $\beta y = 0$. By [Exercise 9.12](#), $\beta(xy) = \beta(x)y - x\beta(y) = y^2$, which vanishes because it lies in $H^4(L(p, q); \mathbb{F}_p) = 0$.

So $\ker \beta$ has basis $1, y, xy$ while $\operatorname{im} \beta$ has basis y . The Bockstein homology is $\mathbb{F}_p$ in degree 0, zero in degrees 1 and 2, and $\mathbb{F}_p$ in degree 3.

This agrees with [Proposition 9.14](#): by [Example 4.8](#) the integral cohomology is $\mathbb{Z}, 0, \mathbb{Z}/p, \mathbb{Z}$, whose free part sits in degrees 0 and 3, and whose only torsion is one copy of $\mathbb{Z}/p$ — of order exactly p , which is why degree 2 contributes nothing.

Remark 9.18 (Three continuations). First, β is only the first approximation. Torsion of order p^r is detected by an r -th Bockstein defined on the previous Bockstein homology, and the family assembles into the *Bockstein spectral sequence*, whose E_∞ -page is the free part mod p and whose r -th differential counts torsion of order p^r . [Example 9.15](#) is the case $r = 1$ against $r = 2$.

Second, because β is a derivation squaring to zero, the mod- p cohomology of a product is a tensor product of complexes and its Bockstein homology is the tensor product of the Bockstein homologies. Applied to a smash power of a Moore space, and read back through [Proposition 9.14](#), that determines the integral cohomology of the smash power completely. It is the natural consolidating computation for this chapter, and everything it needs is now available.

Third, β is not merely a connecting homomorphism: it is a stable cohomology operation, natural in X and commuting with suspension. [Chapter 18](#) identifies it with Sq^1 , and that identification is what lets [Chapter 26](#) express an integral Euler class as a Bockstein.

9.3 What to carry forward

- $\varprojlim^1$ is the cokernel of $1 - \text{shift}$ ([Definition 9.1](#)). It vanishes for surjective towers ([Exercise 9.2](#)) and for pro-zero ones ([Exercise 9.3](#)); those two facts and [Theorem 9.4](#) give the general criterion ([Remark 9.7](#)).
- Mittag-Leffler is about images stabilising, not about surjectivity ([Warning 9.6](#)).
- $\varprojlim^1$ can be uncountable ([Example 9.8](#)), and it is the error term in the Milnor sequence ([Theorem 9.9](#)).

- β is a derivation with $\beta^2 = 0$ (Exercise 9.12), so mod- p cohomology is a complex, and its homology detects the free part together with the order of the torsion (Proposition 9.14, Examples 9.15 and 9.16, and Exercise 9.17).

Chapter 10

H-spaces and Hopf algebras

A multiplication on a space induces a comultiplication on its cohomology, and the resulting structure — a Hopf algebra — is rigid enough to answer questions that look purely geometric. This chapter is the first place in the book where an algebraic structure *forbids* something: Hopf’s theorem will show that most spheres carry no multiplication at all.

The other half of the chapter concerns maps rather than spaces. The multiplication on Y makes $[X, Y]$ into an algebraic object, and the k -th power map is then k times the identity in a precise sense. Where exactly that “ k times” survives into cohomology is the delicate point, and [Example 10.14](#) shows that the naive guess is false.

10.1 The cohomology of an H-space

Recall [Definition 1.18](#): an H-space is a space X with basepoint e and a map $\mu: X \times X \rightarrow X$, with $\mu(e, e) = e$, for which $\mu(e, -)$ and $\mu(-, e)$ are homotopic to the identity.

Let k be a field and suppose $H^*(X; k)$ is finite-dimensional in each degree, so that the Künneth theorem gives $H^*(X \times X; k) \cong H^*(X; k) \otimes H^*(X; k)$. Then μ induces

$$\mu^*: H^*(X; k) \longrightarrow H^*(X; k) \otimes H^*(X; k),$$

a *coproduct*. The unit conditions say that μ^* is counital, and $H^*(X; k)$ becomes a graded Hopf algebra: a graded-commutative algebra with a coassociative coproduct that is a map of algebras.

Example 10.1 (Odd spheres). $H^*(S^n; k) = \Lambda(x)$ with $|x| = n$. There is nothing in degrees strictly between 0 and n , so the only possible terms in μ^*x are $x \otimes 1$ and $1 \otimes x$, and the counit conditions force both coefficients to be 1:

$$\mu^*x = x \otimes 1 + 1 \otimes x.$$

For n odd this is consistent: $x^2 = 0$ automatically by graded commutativity. For n even it is not, as [Exercise 10.4](#) shows.

Example 10.2 (The torus and infinite projective space). T^2 is a topological group, and $H^*(T^2; k) = \Lambda(a, b)$ with $|a| = |b| = 1$ and $\mu^*a = a \otimes 1 + 1 \otimes a$, similarly for b . Since μ^* is a ring map,

$$\mu^*(ab) = (a \otimes 1 + 1 \otimes a)(b \otimes 1 + 1 \otimes b) = ab \otimes 1 + a \otimes b - b \otimes a + 1 \otimes ab,$$

the sign coming from moving b of degree one past a of degree one. So ab has extra terms: not every class behaves like a and b do.

$\mathbb{CP}^\infty$ is also an H-space — it is $K(\mathbb{Z}, 2)$, and the multiplication classifies the addition of degree-two cohomology classes — with $H^*(\mathbb{CP}^\infty; k) = k[c]$, $|c| = 2$, and $\mu^*c = c \otimes 1 + 1 \otimes c$. Again $\mu^*(c^2) = c^2 \otimes 1 + 2c \otimes c + 1 \otimes c^2$ has an extra term.

Theorem 10.3 (Hopf; recalled). *Let A be a connected graded Hopf algebra over a field of characteristic zero which is finite-dimensional as a vector space. Then A is an exterior algebra $\Lambda(x_1, \dots, x_r)$ on generators of odd degree.*

Exercise 10.4 (medium). Deduce that S^n is not an H-space for any even $n \geq 2$.

Hint. What would $H^*(S^n; \mathbb{Q})$ have to be, as an algebra, if [Theorem 10.3](#) applied?

Solution. Suppose S^n were an H-space with n even and $n \geq 2$. Then $A = H^*(S^n; \mathbb{Q}) = \mathbb{Q}\{1, x\}$ with $|x| = n$ even and $x^2 = 0$ would be a connected finite-dimensional graded Hopf algebra over $\mathbb{Q}$, so by [Theorem 10.3](#) it would be an exterior algebra on odd generators. But A has a nonzero element in the even degree n and none in odd degrees, so its generators would have to include one of even degree n , which [Theorem 10.3](#) forbids.

Equivalently and more concretely: for a free graded-commutative algebra a generator of even degree generates a polynomial algebra, which is infinite-dimensional, contradicting $x^2 = 0$.

Remark 10.5. S^1 , S^3 and S^7 are H-spaces, being the unit spheres of the complex numbers, the quaternions and the octonions. [Exercise 10.4](#) excludes the even spheres, and a much harder theorem — proved with the machinery of [Chapter 16](#) and completed by Adams — excludes every odd sphere except those three.

10.2 Primitives

Definition 10.6. An element $\alpha \in H^*(X; k)$ of positive degree is *primitive* if

$$\mu^*\alpha = \alpha \otimes 1 + 1 \otimes \alpha.$$

The primitives form a graded subspace $PH^*(X; k)$.

Lemma 10.7 (Primitivity for degree reasons). *If $H^i(X; k) = 0$ for all $0 < i < n$, then every element of $H^n(X; k)$ is primitive.*

Proof. Write $\mu^*\alpha = \sum \alpha' \otimes \alpha''$ with the terms graded by $(|\alpha'|, |\alpha''|)$ summing to n . The counit conditions supply the terms $\alpha \otimes 1$ and $1 \otimes \alpha$; every other term has both factors of positive degree less than n , hence lies in a group that vanishes. $\square$

Example 10.8. By [Lemma 10.7](#), the generator x of $H^n(S^n; k)$ is primitive. By the computations of [Example 10.2](#), the primitives of $\Lambda(a, b) = H^*(T^2; k)$ are exactly the span of a and b : the class ab is not primitive, since $\mu^*(ab)$ contains $a \otimes b$. Likewise the primitives of $k[c] = H^*(\mathbb{CP}^\infty; k)$ in the case $\text{char } k = 0$ are exactly the span of c : for $r \geq 2$,

$$\mu^*(c^r) = \sum_{i=0}^r \binom{r}{i} c^i \otimes c^{r-i},$$

which has nonzero middle terms.

Exercise 10.9 (easy). Let $f: X \rightarrow Y$ be a map of H-spaces which commutes with the multiplications up to homotopy, i.e. $f \circ \mu_X \simeq \mu_Y \circ (f \times f)$. Show that f^* carries primitives to primitives.

Solution. Let $\alpha \in PH^*(Y; k)$. Then

$$\mu_X^*(f^*\alpha) = (f \circ \mu_X)^*\alpha = (\mu_Y \circ (f \times f))^*\alpha = (f^* \otimes f^*)\mu_Y^*\alpha,$$

using homotopy invariance for the middle equality and the Künneth identification for the last. Since $\mu_Y^*\alpha = \alpha \otimes 1 + 1 \otimes \alpha$, the right-hand side is $f^*\alpha \otimes 1 + 1 \otimes f^*\alpha$.

10.3 Maps into an H-space, and power maps

Definition 10.10. For maps $f, g: X \rightarrow Y$ into an H-space Y , their *pointwise product* is

$$fg = \mu \circ (f, g): X \rightarrow Y, \quad (fg)(x) = \mu(f(x), g(x)).$$

The k -th *power map* of Y is $p_k: Y \rightarrow Y$, the k -fold pointwise product of the identity with itself.

The pointwise product descends to homotopy classes, so $[X, Y]$ carries a binary operation with a two-sided unit. If the multiplication is homotopy-associative and admits a homotopy inverse — as it does for a topological group — then $[X, Y]$ is a group, naturally in X .

Remark 10.11. The case to keep in mind is $Y = K(G, n)$. There $[X, K(G, n)] \cong H^n(X; G)$, a fact proved in [Chapter 18](#), and under that identification the pointwise product is the addition of cohomology classes. The group structure on cohomology and the H-space structure on an Eilenberg–MacLane space are the same thing seen from two sides.

Example 10.12 (The baby case). For $Y = S^1$ with its group structure, p_k is $z \mapsto z^k$. On $\pi_1(S^1) = \mathbb{Z}$ it is multiplication by k : the loop $t \mapsto e^{2\pi it}$ is sent to $t \mapsto e^{2\pi ikt}$, which winds k times. Hence $\deg p_k = k$.

That computation generalises, and the general statement is about homotopy groups.

Exercise 10.13 (medium). Let G be a topological group (or any homotopy-associative H-space with a homotopy inverse). Show that for every $n \geq 1$ the k -th power map induces multiplication by k on $\pi_n(G)$.

Hint. $\pi_n(G)$ carries two products, concatenation and the pointwise one, and by the Eckmann–Hilton argument of [Chapter 1](#) they coincide. Which one is $p_k \circ f$ obviously built from?

Solution. Let $\alpha \in \pi_n(G)$ be represented by $f: S^n \rightarrow G$. By definition of the power map, $p_k \circ f$ is the k -fold pointwise product $f \cdot f \cdots f$, so $[p_k \circ f]$ is the k -fold product of $[f]$ under the pointwise operation on $\pi_n(G)$. By [Proposition 1.19](#) — whose proof applies verbatim to π_n — the pointwise operation agrees with concatenation, which is the group law of $\pi_n(G)$. Hence $p_{k*}(\alpha) = k\alpha$.

Now the point of the chapter. The corresponding statement in cohomology is *false* in general, and it fails in the smallest example.

Example 10.14 (The power map on a torus). Take $G = T^2$ and $H^*(T^2; \mathbb{Z}) = \Lambda(a, b)$ as in [Example 10.2](#). The power map p_k is $(z, w) \mapsto (z^k, w^k)$, so by [Example 10.12](#) applied in each factor,

$$p_k^*a = ka, \quad p_k^*b = kb, \quad p_k^*(ab) = (ka)(kb) = k^2 ab.$$

So p_k acts by k in degree one and by k^2 in degree two; its degree as a self-map of the surface T^2 is k^2 , not k .

Compare [Example 10.8](#): the classes on which p_k^* is multiplication by k are exactly a and b , the primitives, and ab — the one class that failed to be primitive — is exactly the one where the factor is k^2 instead.

Remark 10.15 (The pattern, and what to do with it). [Example 10.14](#) is not an accident of the torus. The pointwise product of two maps is $\mu \circ (f, g)$, so computing $(fg)^*$ means evaluating μ^* and then $f^* \otimes g^*$; and $\mu^*\alpha$ has exactly two terms precisely when α is primitive. On a primitive class the two contributions add; on a non-primitive class the cross terms survive and produce something else — k^2 in the example above.

Turning that observation into a proposition, and then reading off what the k -th power map does to the primitives of a specific H-space, is the natural continuation of this chapter. [Lemma 10.7](#) identifies the primitives of a sphere in one line, and [Example 10.12](#) shows what the answer must specialise to. Note that [Exercise 10.13](#) gives no information here beyond the top degree, since the higher homotopy of an H-space and its cohomology are quite different objects; the cohomological statement has to be proved cohomologically.

Remark 10.16 (Forward). The Hopf algebra structure of this chapter is used three more times. In [Chapter 26](#) the H-space structure of $\mathbb{CP}^\infty = K(\mathbb{Z}, 2)$ is what makes the Euler class of a tensor product of line bundles additive. In [Chapter 16](#) the failure of most spheres to be H-spaces is refined into the Hopf invariant one problem. And in [Chapter 31](#) the same structure reappears rationally, where Hopf's theorem becomes the statement that the rational homotopy of an H-space is its space of primitives.

10.4 What to carry forward

- A multiplication on X makes $H^*(X; k)$ a Hopf algebra ([Examples 10.1](#) and [10.2](#)); Hopf's theorem then constrains which spaces admit one ([Theorem 10.3](#) and [Exercise 10.4](#)).
- Primitives ([Definition 10.6](#)); low degrees are primitive for free ([Lemma 10.7](#)); H-maps preserve them ([Exercise 10.9](#)).
- The k -th power map is multiplication by k on homotopy groups ([Exercise 10.13](#)) but not on cohomology ([Example 10.14](#)); primitivity is exactly the condition that repairs the discrepancy ([Remark 10.15](#)).

Chapter 11

Cofibrations, approximation, and connectivity

This chapter pays a debt. Twice in Part I we collapsed a contractible subcomplex and asserted that nothing was lost — once to normalise a CW complex to a single 0-cell (Proposition 2.1), and once to read a free basis off a spanning tree (Example 3.6). The justification is the homotopy extension property, and it is proved in Section 11.1.

The rest of the chapter is about replacing a space by a better model of itself. Two theorems do all the work — cellular approximation and the extension lemma — and between them they turn statements about maps into statements about cell dimensions. That translation is the technique of Chapter 17, met here in its simplest form.

11.1 The homotopy extension property

Definition 11.1. A pair (X, A) has the *homotopy extension property* (HEP) if every map $X \times \{0\} \cup A \times I \rightarrow Y$ extends to $X \times I \rightarrow Y$. An inclusion $A \hookrightarrow X$ with the HEP is a *cofibration*.

Concretely: a homotopy defined on A and a map defined on X that agree at time zero can be combined into a homotopy on all of X .

Exercise 11.2 (easy). Show that every CW pair has the homotopy extension property.

Hint. It suffices to construct a retraction $D^n \times I \rightarrow D^n \times \{0\} \cup \partial D^n \times I$, and then to induct over cells.

Solution. For a single cell, project radially from the point $(0, 2) \in D^n \times \mathbb{R}$: the ray from that point through any $(x, t) \in D^n \times I$ meets $D^n \times \{0\} \cup \partial D^n \times I$ in exactly one point, and this defines a retraction r .

Now let (X, A) be a CW pair and suppose given F on $X \times \{0\} \cup A \times I$. Extend over the cells of $X \setminus A$ by induction on dimension: for an n -cell with characteristic map $\Phi: D^n \rightarrow X$, the map F is already defined on $D^n \times \{0\} \cup \partial D^n \times I$ after composing with $\Phi \times \text{id}$, and precomposing with the retraction r extends it over $D^n \times I$. The extensions agree on overlaps because they are defined cell by cell, and the CW topology makes the resulting function continuous.

Proposition 11.3. Let (X, A) be a CW pair with A contractible. Then the quotient map $q: X \rightarrow X/A$ is a homotopy equivalence.

Proof. Let $h_t: A \rightarrow A$ be a contraction, $h_0 = \text{id}_A$ and h_1 constant at a_0 . Regard it as a homotopy $A \times I \rightarrow X$ and combine it with id_X at time zero; by [Exercise 11.2](#) there is a homotopy $H_t: X \rightarrow X$ with $H_0 = \text{id}_X$ and $H_t|_A = h_t$.

At time one, H_1 maps A to the single point a_0 , so it factors through the quotient: $H_1 = g \circ q$ for a map $g: X/A \rightarrow X$. We claim g is a homotopy inverse for q .

In one direction, $g \circ q = H_1 \simeq H_0 = \text{id}_X$.

In the other, consider $q \circ g: X/A \rightarrow X/A$. For each t the map H_t sends A into A , so $q \circ H_t$ factors through q , giving a homotopy $\bar{H}_t: X/A \rightarrow X/A$ with $q \circ H_t = \bar{H}_t \circ q$. Then $\bar{H}_0 = \text{id}$ and $\bar{H}_1 \circ q = q \circ H_1 = q \circ g \circ q$; since q is surjective this gives $\bar{H}_1 = q \circ g$, so $q \circ g \simeq \text{id}_{X/A}$. $\square$

Example 11.4 (The debt, paid). A spanning tree T of a connected graph Γ is a contractible subcomplex, so $\Gamma \rightarrow \Gamma/T$ is a homotopy equivalence and Γ/T is a wedge of circles — one for each edge outside T . That is exactly the move made in [Example 3.6](#) and [Exercise 3.9](#), and in the proof of [Proposition 3.5](#).

More generally, a spanning tree of the 1-skeleton of a connected CW complex is a contractible subcomplex, and collapsing it produces a homotopy equivalent complex with a single 0-cell. This is [Proposition 2.1](#).

Warning 11.5. The proof used the HEP exactly once, to promote the contraction of A to a homotopy of X . Nothing else in the argument would survive its absence: without some hypothesis relating A to X , the contraction cannot be made to move the rest of the space, and H_1 never gets defined. The natural generalisation — replacing “CW pair” by “cofibration” throughout — costs nothing, since [Exercise 11.2](#) was the only place the CW structure was used.

Exercise 11.6 (medium). Let (X, A) be a CW pair. Show that the mapping cone $X \cup_A CA$ is homotopy equivalent to X/A .

Hint. CA is contractible. Which pair is [Proposition 11.3](#) being applied to?

Solution. Inside $X \cup_A CA$ the cone CA is a contractible subcomplex, so by [Proposition 11.3](#) the quotient map $X \cup_A CA \rightarrow (X \cup_A CA)/CA$ is a homotopy equivalence. Collapsing the cone identifies all of A to a point, so the target is X/A .

11.2 Better models

Theorem 11.7 (Cellular approximation; recalled). *Every map between CW complexes is homotopic to a cellular one, and a homotopy between cellular maps may be taken cellular.*

Exercise 11.8 (easy). Deduce that the inclusion $X^n \hookrightarrow X$ of the n -skeleton induces an isomorphism on π_k for $k < n$ and a surjection for $k = n$.

Solution. Let $k < n$. Any map $S^k \rightarrow X$ is homotopic to a cellular one by [Theorem 11.7](#), hence lands in $X^k \subseteq X^n$: this gives surjectivity. If two maps $S^k \rightarrow X^n$ become homotopic in X , the homotopy $S^k \times I \rightarrow X$ may be taken cellular, so it lands in $X^{k+1} \subseteq X^n$ when $k+1 \leq n$: this gives injectivity for $k < n$. For $k = n$ only the surjectivity argument applies.

Theorem 11.9 (Connectivity-adapted models; recalled). *Every m -connected CW complex is homotopy equivalent to one whose cells occupy only dimension 0 — a single cell there — and dimensions strictly greater than m .*

Proof sketch. Build a CW approximation from the bottom. Start with a point mapping to the basepoint; having built a complex Z_k with a map to X inducing isomorphisms on π_i for $i < k$ and a surjection on π_k , attach cells of dimension $k + 1$ to kill the kernel on π_k and cells of dimension $k + 1$ to hit new generators of π_{k+1} . When X is m -connected the first m stages require no cells at all, because there is nothing to hit and nothing to kill; the construction therefore produces cells only in dimension 0 and in dimensions $> m$. $\square$

Example 11.10. S^n is $(n - 1)$ -connected, and the standard CW structure — one 0-cell and one n -cell — is already of the shape promised by [Theorem 11.9](#). A simply connected complex may always be modelled with no 1-cells, which is why arguments about simply connected spaces so often begin “we may assume there are no cells in dimension one”.

11.3 The extension lemma

Everything above is preparation for a single statement, which converts a question about extending maps into a question about the dimensions of cells.

Theorem 11.11 (Extension lemma). *Let (X, A) be a CW pair and Y a space with $\pi_{k-1}(Y) = 0$ for every k such that $X \setminus A$ contains a cell of dimension k . Then every map $A \rightarrow Y$ extends to X .*

Proof sketch. Extend over cells by induction on dimension. To extend over a k -cell with attaching map $\varphi: S^{k-1} \rightarrow X^{k-1}$, one must show the composite $S^{k-1} \rightarrow X^{k-1} \rightarrow Y$ is nullhomotopic; its class lies in $\pi_{k-1}(Y)$, which vanishes by hypothesis, and a nullhomotopy is precisely an extension over the cell. $\square$

Remark 11.12. That proof is obstruction theory in embryo, and it is worth naming the shape now: an extension is built skeleton by skeleton; at each stage the obstruction to continuing is the homotopy class of a composite; and if the relevant homotopy group vanishes, the obstruction vanishes with it. [Chapter 17](#) keeps track of the obstructions when they do *not* vanish, and organises them into cohomology classes with coefficients in $\pi_{k-1}(Y)$. The proof of [Proposition 2.15](#) in [Chapter 2](#) was the same argument, run against $Y = S^1$ with $\pi_k(S^1) = 0$ for $k \geq 2$.

Exercise 11.13 (medium). Let Y be n -connected and let X be a CW complex of dimension at most n . Show that every map $X \rightarrow Y$ is nullhomotopic.

Hint. A nullhomotopy of $f: X \rightarrow Y$ is an extension over $X \times I$ of the map defined on $X \times \{0, 1\}$ by f and by a constant. What are the cells of $X \times I$ relative to $X \times \{0, 1\}$, and what dimensions do they have?

Solution. Put $W = X \times I$ and $A = X \times \{0, 1\}$, a CW pair. Define g on A by f on $X \times \{0\}$ and by the constant map on $X \times \{1\}$. A cell of $W \setminus A$ is $e \times (0, 1)$ for a cell e of X , so its dimension is $\dim e + 1 \leq n + 1$. The extension lemma therefore requires $\pi_{k-1}(Y) = 0$ for $k \leq n + 1$, i.e. $\pi_j(Y) = 0$ for $j \leq n$, which is exactly n -connectedness. So g extends over $X \times I$, and the extension is a homotopy from f to a constant map.

Remark 11.14 (Where connectivity arguments go next). Two directions open here, and both are used repeatedly later.

The first is to feed a space into [Exercise 11.13](#) as both source and target, or as source and target after replacing one of them by a connectivity-adapted model from

Theorem 11.9. Such arguments turn a numerical coincidence between connectivity and dimension into a statement about the space itself, and they need nothing beyond this page.

The second is to estimate the connectivity of a pair by counting the dimensions of its relative cells. **Theorem 11.9** is what makes this work: after replacing X and Y by good models, the relative cells of a product are products of cells, and their dimensions add. That is the standard route to connectivity estimates for products and smash products, and the elementary shadow of the Blakers–Massey theorem.

11.4 What to carry forward

- CW pairs have the HEP ([Exercise 11.2](#)); collapsing a contractible subcomplex is a homotopy equivalence ([Proposition 11.3](#)), which is the promise made in [Chapters 2](#) and [3](#).
- The HEP is used exactly once in that proof, to move a homotopy of A out into X ([Warning 11.5](#)).
- Mapping cone $\simeq$ quotient for a CW pair ([Exercise 11.6](#)).
- Cellular approximation makes homotopy groups skeletal ([Exercise 11.8](#)); connectivity can be built into the model ([Theorem 11.9](#)).
- The extension lemma ([Theorem 11.11](#)) trades homotopy groups against cell dimensions; it is obstruction theory with all obstructions assumed to vanish ([Remark 11.12](#)).

Chapter 12

Vector bundles and Stiefel–Whitney classes

Characteristic classes attach cohomology to a bundle, turning geometric questions — is this bundle trivial? does this manifold admit q independent vector fields? — into computations in a ring one already knows. This chapter sets up the mod-2 theory, which is the easiest and the sharpest of its kind, and shows it deciding a question about projective spaces.

The pattern to watch is that a single structural identity about the tangent bundle, fed through a multiplicative axiom, converts a topological question into an arithmetic one about binomial coefficients. That conversion is the whole technique, and [Chapter 26](#) runs it again over $\mathbb{Z}$ where the arithmetic is harder and the conclusions are stronger.

12.1 Bundles

Definition 12.1. A *real vector bundle* of rank n over X is a map $p: E \rightarrow X$ with a real vector space structure on each fibre $p^{-1}(x)$, locally trivial: every point of X has a neighbourhood U with a homeomorphism $p^{-1}(U) \cong U \times \mathbb{R}^n$ that is linear on fibres. We write ε^n for the trivial bundle $X \times \mathbb{R}^n$.

Bundles pull back along maps, add under Whitney sum $E \oplus F$, and are classified: over a paracompact base, isomorphism classes of rank- n real bundles correspond to homotopy classes of maps $X \rightarrow BO(n)$, with the universal bundle pulling back to the given one.

Example 12.2 (The tautological line bundle). Over $\mathbb{R}P^n$ let γ be the bundle whose fibre over a line $\ell \subseteq \mathbb{R}^{n+1}$ is ℓ itself. It is the basic example, and it is not trivial: a nowhere-zero section would be a continuous choice of nonzero vector in each line, i.e. a map $S^n \rightarrow \mathbb{R}^{n+1} \setminus \{0\}$ that is equivariant for the antipodal action up to sign in a way no map can be.

Proposition 12.3. $T\mathbb{R}P^q \oplus \varepsilon^1 \cong (q+1)\gamma$, the Whitney sum of $q+1$ copies of the tautological line bundle.

Proof. Write $\gamma^\perp$ for the orthogonal complement bundle, so $\gamma \oplus \gamma^\perp = \varepsilon^{q+1}$. At a line ℓ , a tangent vector to $\mathbb{R}P^q$ is a linear map $\ell \rightarrow \ell^\perp$: moving the line means tilting it, and

the tilt is recorded by such a map. Hence $T\mathbb{RP}^q \cong \text{Hom}(\gamma, \gamma^\perp)$. Adding $\text{Hom}(\gamma, \gamma) \cong \varepsilon^1$ gives

$$T\mathbb{RP}^q \oplus \varepsilon^1 \cong \text{Hom}(\gamma, \gamma^\perp \oplus \gamma) = \text{Hom}(\gamma, \varepsilon^{q+1}) \cong (q+1) \text{Hom}(\gamma, \varepsilon^1),$$

and $\text{Hom}(\gamma, \varepsilon^1) \cong \gamma$ because a real line bundle is isomorphic to its dual, via any fibrewise inner product. $\square$

12.2 Stiefel–Whitney classes

Theorem 12.4 (Existence and axioms; recalled). *There are classes $w_i(E) \in H^i(X; \mathbb{F}_2)$, for a real bundle E over X , with $w_0(E) = 1$ and $w_i(E) = 0$ for $i > \text{rank } E$, such that:*

- (a) Naturality: $w_i(f^*E) = f^*w_i(E)$.
- (b) Whitney formula: $w(E \oplus F) = w(E)w(F)$, where $w = 1 + w_1 + w_2 + \cdots$ is the total class.
- (c) Normalisation: $w(\gamma) = 1 + a$ over $\mathbb{RP}^\infty$, where a generates $H^1(\mathbb{RP}^\infty; \mathbb{F}_2)$.

These properties determine the classes, and $H^*(BO(n); \mathbb{F}_2) = \mathbb{F}_2[w_1, \dots, w_n]$.

The last statement is what makes the axioms legitimate rather than optimistic: the classes exist because the cohomology of the classifying space is a polynomial ring on them, and every characteristic class is a polynomial in them.

Example 12.5 (Trivial bundles, and orientability). A trivial bundle is pulled back from a point, so by naturality $w(\varepsilon^n) = 1$; and by the Whitney formula, adding a trivial summand does not change the total class. Consequently w is an invariant of the *stable* isomorphism class of a bundle, which is exactly the right level of generality for [Proposition 12.3](#) to be useful.

In rank one, w_1 detects orientability: $w_1(E) = 0$ if and only if E is orientable. Over $\mathbb{RP}^1 = S^1$ the bundle γ is the Möbius line bundle and $w(\gamma) = 1 + a$ with $a \neq 0$, so it is non-orientable — the Möbius band having one side is a statement about w_1 .

Example 12.6 (Projective spaces). $H^*(\mathbb{RP}^q; \mathbb{F}_2) = \mathbb{F}_2[a]/(a^{q+1})$. Restricting the normalisation of [Theorem 12.4](#) along $\mathbb{RP}^q \subseteq \mathbb{RP}^\infty$ gives $w(\gamma) = 1 + a$ there, and [Proposition 12.3](#) with the Whitney formula gives the total class of the tangent bundle:

$$w(\mathbb{RP}^q) := w(T\mathbb{RP}^q) = w(T\mathbb{RP}^q \oplus \varepsilon^1) = (1 + a)^{q+1} \in \mathbb{F}_2[a]/(a^{q+1}).$$

Everything about the tangent bundle that w can see is now a question about expanding a power of $1 + a$ in a truncated polynomial ring.

Warning 12.7. The truncation is not decoration. In $\mathbb{F}_2[a]$ the element $(1 + a)^{q+1}$ is never 1 for $q \geq 1$; it is only after killing a^{q+1} that the question becomes interesting. Every computation below must be carried out in $\mathbb{F}_2[a]/(a^{q+1})$, and the commonest error is to expand in the polynomial ring and forget to truncate.

Example 12.8 (The first three cases). In $\mathbb{F}_2[a]/(a^{q+1})$:

$$q = 1 : (1 + a)^2 = 1 + a^2 = 1; \quad q = 2 : (1 + a)^3 = 1 + a + a^2; \quad q = 3 : (1 + a)^4 = 1.$$

For $q = 1$ we used $a^2 = 0$ and $2a = 0$; for $q = 2$, that $3 \equiv 1 \pmod{2}$ and $a^3 = 0$; for $q = 3$, that all of $\binom{4}{1}, \binom{4}{2}, \binom{4}{3}$ are even.

So $w(\mathbb{RP}^2) = 1 + a + a^2 \neq 1$, while $w(\mathbb{RP}^1) = w(\mathbb{RP}^3) = 1$.

Exercise 12.9 (easy). Deduce from [Example 12.8](#) that $\mathbb{RP}^2$ is not parallelisable, and indeed that $T\mathbb{RP}^2$ does not become trivial after adding any trivial bundle.

Solution. If $T\mathbb{RP}^2 \oplus \varepsilon^k$ were trivial then its total class would be 1 by [Example 12.5](#); but adding a trivial summand does not change w , so $w(\mathbb{RP}^2)$ would be 1, contradicting $w(\mathbb{RP}^2) = 1 + a + a^2$. In particular $T\mathbb{RP}^2$ itself is not trivial, so $\mathbb{RP}^2$ is not parallelisable.

Note that $\mathbb{RP}^1 = S^1$ and $\mathbb{RP}^3$ pass this test, which does not prove they are parallelisable — only that w raises no objection. Both in fact are, being Lie groups.

Exercise 12.10 (easy). Show that $(1+x)^{2^k} = 1 + x^{2^k}$ in $\mathbb{F}_2[x]$, and deduce that $\binom{2^k}{i}$ is even for $0 < i < 2^k$.

Hint. Squaring is a ring homomorphism in characteristic 2.

Solution. In characteristic 2 the Frobenius $y \mapsto y^2$ is a ring homomorphism, so $(1+x)^2 = 1+x^2$, and iterating k times gives $(1+x)^{2^k} = 1+x^{2^k}$. Comparing coefficients with the binomial expansion, $\binom{2^k}{i} \equiv 0 \pmod{2}$ for $0 < i < 2^k$.

Remark 12.11 (The question this sets up). [Example 12.6](#) says that $w(\mathbb{RP}^q) = 1$ exactly when $(1+a)^{q+1} = 1$ in $\mathbb{F}_2[a]/(a^{q+1})$, i.e. exactly when $\binom{q+1}{i}$ is even for all $0 < i \leq q$. [Exercise 12.10](#) settles one direction of the resulting arithmetic; the converse needs the general statement about binomial coefficients modulo 2 — Lucas’ theorem, which reads off the parity of $\binom{m}{i}$ from a digitwise comparison of the binary expansions of i and m , the coefficient being odd exactly when no digit of i exceeds the corresponding digit of m . Assembling the two halves determines exactly which projective spaces pass the test of [Exercise 12.9](#), and [Example 12.8](#) already shows the pattern in the first three cases.

Two cautions for whoever carries that out. The conclusion concerns *stable* triviality, since w cannot distinguish $T\mathbb{RP}^q$ from $T\mathbb{RP}^q \oplus \varepsilon$; promoting it to parallelisability is a genuinely harder problem, solved by Adams. And the computation must respect [Warning 12.7](#).

12.3 What the mod-2 theory cannot do

Remark 12.12. Stiefel–Whitney classes live in $\mathbb{F}_2$ -cohomology, so they are blind to everything of odd order and to all sign information. Three examples of what they miss, each repaired later:

- An oriented bundle has an *Euler class* $e(E) \in H^n(X; \mathbb{Z})$ whose mod-2 reduction is $w_n(E)$ but which carries strictly more information; [Chapter 26](#) computes with it.
- Complex bundles have Chern classes in integral cohomology, and the relation between c_i and w_{2i} is again reduction mod 2.
- The relation between the two theories is itself a cohomology operation: $\text{Sq}^1 w_i$ is expressible in terms of the w_j by the Wu formula, and Sq^1 is a stable operation of degree one, so it converts mod-2 data into information about integral torsion — compare [Chapter 9](#). That triangle — Stiefel–Whitney, Bockstein, Euler — is what [Chapter 26](#) closes.

12.4 What to carry forward

- $T\mathbb{RP}^q \oplus \varepsilon \cong (q+1)\gamma$ ([Proposition 12.3](#)) — one structural identity, from which everything about $w(\mathbb{RP}^q)$ follows.
- The axioms ([Theorem 12.4](#)), and the fact that w is a stable invariant ([Example 12.5](#)).
- Compute in the truncated ring ([Warning 12.7](#)); the arithmetic is Frobenius and Lucas ([Exercise 12.10](#)).
- w_1 detects orientability; the mod-2 theory is blind to signs and to odd torsion ([Remark 12.12](#)), which is why [Chapter 26](#) exists.

Chapter 13

Whitehead theorems, Postnikov towers, and what the invariants miss

Chapter 5 ended by observing that homology cannot see the universal cover. This chapter says what does. Whitehead's theorem is the statement that homotopy groups, *together with a map realising them*, form a complete invariant of CW homotopy type; the rest of the chapter is about how much of that completeness survives when one of those two ingredients is weakened.

Every result here has a hypothesis that looks removable and is not. The reader should finish the chapter able to name, for each theorem, the space or map that goes wrong when the hypothesis is dropped.

13.1 Whitehead's theorem, and the map it needs

Theorem 13.1 (Whitehead; recalled). *A map $f: X \rightarrow Y$ between connected CW complexes that induces isomorphisms $\pi_n(X, x_0) \rightarrow \pi_n(Y, f(x_0))$ for all $n \geq 1$ is a homotopy equivalence.*

Warning 13.2 (The map is not optional). The theorem is about a map. Two spaces can have isomorphic homotopy groups in every degree and fail to be homotopy equivalent, and the book has already met such a pair.

Take the lens spaces $L(5, 1)$ and $L(5, 2)$ of [Example 2.21](#). Both have fundamental group $\mathbb{Z}/5$ and universal cover S^3 , so for $n \geq 2$

$$\pi_n(L(5, q)) \cong \pi_n(S^3)$$

by the covering-space isomorphism, independently of q ; and π_1 agrees too. So all their homotopy groups are isomorphic. Their homology agrees ([Example 4.8](#)) and so does their cohomology ring ([Example 7.10](#)). They are nevertheless not homotopy equivalent.

What is missing is not an invariant but a map: no map between them induces those isomorphisms.

Exercise 13.3 (easy). Let X be a connected CW complex with $\pi_n(X) = 0$ for all $n \geq 1$. Show X is contractible.

Solution. Apply [Theorem 13.1](#) to the inclusion of a point $\{x_0\} \hookrightarrow X$, which induces isomorphisms $0 \rightarrow \pi_n(X)$ in every degree by hypothesis. Both spaces are CW complexes, so the inclusion is a homotopy equivalence.

13.2 Homology, and the universal cover

Homotopy groups are hard to compute and homology is easy, so one wants a homological criterion. There is one, with a hypothesis.

Theorem 13.4 (Homological Whitehead; recalled). *Let $f: X \rightarrow Y$ be a map of simply connected CW complexes inducing an isomorphism on $H_n(-; \mathbb{Z})$ for every n . Then f is a homotopy equivalence.*

Proof sketch. Replace f by an inclusion using the mapping cylinder. The homology of the pair vanishes, and the pair is simply connected, so the relative Hurewicz theorem gives $\pi_n(Y, X) = 0$ for all n ; the long exact sequence of the pair then makes f_* an isomorphism on homotopy groups, and [Theorem 13.1](#) applies. $\square$

Warning 13.5 (Simple connectivity is essential). Let P be the Poincaré homology sphere: a closed orientable 3-manifold with $\pi_1(P)$ the binary icosahedral group of order 120, and with $H_*(P; \mathbb{Z}) \cong H_*(S^3; \mathbb{Z})$. Collapsing the complement of an open ball gives a degree-one map

$$c: P \longrightarrow P/(P \setminus B^3) \cong S^3,$$

which is an isomorphism on H_3 because it has degree one, and trivially so in the other degrees, both spaces having the homology of a sphere. So c_* is an isomorphism on all integral homology.

But c is not a homotopy equivalence: $\pi_1(P)$ has order 120 and $\pi_1(S^3)$ is trivial. The single hypothesis that fails is simple connectivity, and it fails on the source.

The repair promised in [Remark 5.18](#) is to work upstairs.

Proposition 13.6. *Let $f: X \rightarrow Y$ be a map of connected CW complexes inducing an isomorphism on π_1 , and let $\tilde{f}: \tilde{X} \rightarrow \tilde{Y}$ be a lift to the universal covers. If $\tilde{f}_*$ is an isomorphism on homotopy groups, then f is a homotopy equivalence.*

Proof. A covering induces isomorphisms on π_n for $n \geq 2$, and the square formed by f , $\tilde{f}$ and the two covering projections commutes. So f_* is an isomorphism on π_n for $n \geq 2$, and by hypothesis also on π_1 . Apply [Theorem 13.1](#). $\square$

Exercise 13.7 (easy). Prove the fact used twice above: a covering $p: \tilde{X} \rightarrow X$ induces isomorphisms $\pi_n(\tilde{X}) \rightarrow \pi_n(X)$ for all $n \geq 2$.

Hint. Lift maps $S^n \rightarrow X$ and homotopies $S^n \times I \rightarrow X$, using that S^n is simply connected for $n \geq 2$.

Solution. For surjectivity, a map $S^n \rightarrow X$ with $n \geq 2$ lifts to $\tilde{X}$ because S^n is simply connected, so the lifting criterion applies. For injectivity, if two lifts become homotopic downstairs, the homotopy $S^n \times I \rightarrow X$ lifts (again by simple connectivity of $S^n \times I$) to a homotopy upstairs starting at the given lift, and uniqueness of lifts identifies its other end with the second lift.

Remark 13.8 (The invariant that works). [Proposition 13.6](#) is stated in terms of homotopy groups upstairs, but the useful version replaces them by homology. The homology of $\tilde{X}$ is not merely a graded abelian group: the deck group $\pi_1(X)$ acts on it, making it a module over the group ring $\mathbb{Z}[\pi_1 X]$ — this is what [Example 5.16](#) exhibited for $S^1 \vee S^2$, where $H_2(\tilde{X})$ was free of rank one over $\mathbb{Z}[t, t^{-1}]$. That module is homology with local coefficients, $H_*(X; \mathbb{Z}[\pi_1 X])$.

Combining [Proposition 13.6](#) with [Theorem 13.4](#) applied upstairs — the universal cover being simply connected — turns a homological hypothesis on $\tilde{X}$ into a homotopy equivalence downstairs. Assembling that statement, and checking exactly which hypothesis on π_1 it needs, is the natural exercise at this point, and it is the invariant [Chapter 5](#) promised.

13.3 Postnikov towers

Whitehead's theorem says the homotopy groups determine the homotopy type once a map is given. Postnikov towers organise the homotopy groups so that the *glue* between them becomes visible as well.

Theorem 13.9 (Postnikov towers; recalled). *For a connected CW complex X there are spaces X^n and maps forming a commutative tower under X ,*

$$X \longrightarrow \cdots \longrightarrow X^n \longrightarrow X^{n-1} \longrightarrow \cdots \longrightarrow X^1,$$

such that $X \rightarrow X^n$ induces isomorphisms on π_i for $i \leq n$, and $\pi_i(X^n) = 0$ for $i > n$. Each map $X^n \rightarrow X^{n-1}$ is a fibration with fibre $K(\pi_n X, n)$, and it is pulled back from the path fibration over $K(\pi_n X, n+1)$ along a map

$$k^n: X^{n-1} \longrightarrow K(\pi_n X, n+1),$$

the k -invariant, a class in $H^{n+1}(X^{n-1}; \pi_n X)$. The stages and the tower are unique up to homotopy over X .

Dually, the n -connected cover $X\langle n \rangle \rightarrow X$ kills the homotopy below degree $n+1$ and leaves the rest alone.

Exercise 13.10 (easy). Show that the n -connected cover $X\langle n \rangle \rightarrow X$ induces isomorphisms on π_i for $i > n$ and satisfies $\pi_i(X\langle n \rangle) = 0$ for $i \leq n$. Then identify $X\langle 2 \rangle$ when X is simply connected.

Solution. The defining property of the connected cover is exactly the pair of statements about π_i ; they are what the construction (attach cells of dimension $\geq n+2$ to kill nothing, or take the fibre of $X \rightarrow X^n$) achieves. Taking the fibre of $X \rightarrow X^n$ realises $X\langle n \rangle$, and the long exact sequence of that fibration gives both claims.

For X simply connected and $n = 2$: $X^2 = K(\pi_2 X, 2)$, so the fibre sequence $X\langle 2 \rangle \rightarrow X \rightarrow K(\pi_2 X, 2)$ exhibits $X\langle 2 \rangle$ as the 2-connected cover, with π_i equal to $\pi_i(X)$ for $i \geq 3$ and zero below.

Example 13.11 (A two-stage tower). Let X be simply connected with $\pi_2 X = \pi_3 X = \mathbb{Z}$ and all other homotopy groups zero. Then $X^2 = K(\mathbb{Z}, 2) = \mathbb{CP}^\infty$, and $X = X^3$ sits in a fibration

$$K(\mathbb{Z}, 3) \longrightarrow X \longrightarrow \mathbb{CP}^\infty$$

classified by a single k -invariant

$$k \in H^4(\mathbb{CP}^\infty; \mathbb{Z}) = \mathbb{Z} \cdot \iota^2,$$

where $\iota \in H^2(\mathbb{CP}^\infty; \mathbb{Z})$ is the fundamental class and ι^2 its square. So such spaces are classified by an integer. The value $k = 0$ gives the product $\mathbb{CP}^\infty \times K(\mathbb{Z}, 3)$; the value $k = \iota^2$ gives the 3-type of S^2 , and the appearance of a *cup square* as a k -invariant is the homotopy-theoretic shadow of the Hopf map.

Exercise 13.12 (medium). Show that a two-stage Postnikov tower with vanishing k -invariant is a product of Eilenberg–MacLane spaces.

Hint. The stage is the pullback of the path fibration along k . What is the pullback of a fibration along a constant map?

Solution. By [Theorem 13.9](#) the stage $X^n \rightarrow X^{n-1}$ is the pullback of the path fibration $PK(\pi, n+1) \rightarrow K(\pi, n+1)$ along k^n . If k^n is nullhomotopic the pullback is, up to homotopy, the pullback along a constant map, which is the product of the base with the fibre. With $X^{n-1} = K(\pi_{n-1}, n-1)$ that gives $X^n \simeq K(\pi_{n-1}, n-1) \times K(\pi_n, n)$.

Remark 13.13 (Manipulating the two truncations). Postnikov stages cut the homotopy groups off above a degree; connected covers cut them off below. Both are determined by universal properties — a map from X inducing isomorphisms in a range, respectively a map to X doing so — and both are unique up to homotopy under or over X by [Theorem 13.9](#). Two such constructions applied in the two possible orders therefore have the same homotopy groups, namely those of X in a window and zero outside it; and by uniqueness each is characterised by a universal property that the other also satisfies.

Writing that argument out is the natural exercise here, and it is a good test of whether [Warning 13.2](#) has been absorbed: isomorphic homotopy groups are not enough, so the argument must produce a comparison *map* and then invoke [Theorem 13.1](#). The uniqueness clause of [Theorem 13.9](#) is what supplies it.

13.4 What the invariants miss

Remark 13.14 (Spaces). [Warning 13.2](#) is worth restating as a summary of the book so far. For $L(5, 1)$ and $L(5, 2)$:

- the fundamental groups agree ([Example 2.21](#));
- the homology groups agree ([Example 4.8](#));
- the cohomology rings agree ([Example 7.10](#));
- all higher homotopy groups agree ([Warning 13.2](#));

and the spaces are still different. Every invariant built so far is a functor applied to one space at a time; what distinguishes these two is how their cells are linked, which is visible only to an operation that mixes degrees. [Chapter 18](#) constructs the first such operation.

Remark 13.15 (Maps). There is a second failure, one degree more subtle, and it concerns maps rather than spaces. A map $f: X \rightarrow Y$ induces f_* on homotopy groups and on homology. Both can vanish identically with f still essential.

The place to look for such an f is a two-stage Postnikov tower. If $\dim X \leq n$ then $[X, Y] \rightarrow [X, Y^n]$ is a bijection — by [Exercise 11.8](#) and the extension lemma, nothing above stage n can be detected by a complex of dimension n — so classifying maps out of X reduces to classifying maps into a tower with two stages. Such a map is a cohomology class on X together with a lift, the obstruction to lifting is the k -invariant evaluated on that class, and the ambiguity in the lift is a quotient of a cohomology group of X . When the k -invariant is a cup square, as in [Example 13.11](#), that obstruction is a quadratic condition, and quadratic conditions have solutions that linear invariants cannot see.

Carrying this out on a specific low-dimensional X is the natural place to test the chapter. [Chapter 17](#) develops the machinery properly; what should be clear already is why π_* and H_* cannot be expected to suffice.

13.5 What to carry forward

- Whitehead's theorem needs a map ([Theorem 13.1](#)); [Warning 13.2](#) is the standing counterexample to forgetting it.
- The homological form needs simple connectivity ([Theorem 13.4](#)), and the Poincaré homology sphere shows why ([Warning 13.5](#)).
- Work on the universal cover ([Proposition 13.6](#)); the right invariant is $H_*(\tilde{X})$ as a $\mathbb{Z}[\pi_1 X]$ -module ([Remark 13.8](#)) — the repair promised in [Chapter 5](#).
- Postnikov towers and k -invariants ([Theorem 13.9](#)); the k -invariant measures non-splitting ([Exercise 13.12](#)), and a cup square as k -invariant is the Hopf map in disguise ([Example 13.11](#)).
- Neither π_* nor H_* classifies spaces ([Remark 13.14](#)) or maps ([Remark 13.15](#)).

Chapter 14

Fibrations and the first homotopy groups of spheres

A fibration converts a space into two simpler ones at the cost of a long exact sequence, and that trade is the only systematic access to higher homotopy groups we have. This chapter sets up the sequence, isolates the mechanism by which it splits, and runs it on the two bundles where the answer is a theorem rather than a tautology.

Everything here is preparation for [Chapter 15](#), which replaces the long exact sequence by a spectral sequence and thereby handles fibrations whose fibre is not a sphere.

14.1 The long exact sequence

Definition 14.1. A map $p: E \rightarrow B$ is a *fibration* if it has the homotopy lifting property: every homotopy $X \times I \rightarrow B$ whose restriction to $X \times \{0\}$ lifts to E lifts entirely.

Fibre bundles over paracompact bases are fibrations, and that is how almost every example arises.

Theorem 14.2 (Relative homotopy; recalled). *Let $p: E \rightarrow B$ be a fibration with B path-connected, $b_0 \in B$, $F = p^{-1}(b_0)$ and $x_0 \in F$. Then*

$$p_*: \pi_n(E, F, x_0) \longrightarrow \pi_n(B, b_0)$$

is an isomorphism for all $n \geq 1$.

Corollary 14.3 (The long exact sequence).

$$\cdots \rightarrow \pi_n(F) \xrightarrow{i_*} \pi_n(E) \xrightarrow{p_*} \pi_n(B) \xrightarrow{\partial} \pi_{n-1}(F) \rightarrow \cdots \rightarrow \pi_0(E).$$

Proof. Substitute [Theorem 14.2](#) into the long exact sequence of the pair (E, F) . □

Exercise 14.4 (easy). Let $F \rightarrow E \rightarrow B$ be a fibration with E contractible. Show $\pi_n(B) \cong \pi_{n-1}(F)$ for all $n \geq 1$.

Solution. In [Corollary 14.3](#) the terms $\pi_n(E)$ and $\pi_{n-1}(E)$ vanish, so exactness of $\pi_n(E) \rightarrow \pi_n(B) \xrightarrow{\partial} \pi_{n-1}(F) \rightarrow \pi_{n-1}(E)$ forces ∂ to be injective and surjective.

Example 14.5 (Path-loop). For any based space B the path space PB of paths starting at b_0 is contractible, and evaluation at the endpoint is a fibration $PB \rightarrow B$ with fibre ΩB . By [Exercise 14.4](#),

$$\pi_n(B) \cong \pi_{n-1}(\Omega B) \quad (n \geq 1),$$

which is the statement that looping shifts homotopy groups down by one.

Exercise 14.6 (medium). The unit sphere $S^\infty \subseteq \mathbb{C}^\infty$ is contractible, and the circle acts freely on it with quotient $\mathbb{CP}^\infty$. Deduce that $\mathbb{CP}^\infty$ is a $K(\mathbb{Z}, 2)$.

Hint. A free action of a compact Lie group gives a fibre bundle, hence a fibration, with fibre the group.

Solution. The action gives a fibre bundle $S^1 \rightarrow S^\infty \rightarrow \mathbb{CP}^\infty$, hence a fibration. Since S^∞ is contractible, [Exercise 14.4](#) gives

$$\pi_n(\mathbb{CP}^\infty) \cong \pi_{n-1}(S^1).$$

Now $\pi_0(S^1) = 0$, $\pi_1(S^1) = \mathbb{Z}$ and $\pi_k(S^1) = 0$ for $k \geq 2$ — the last because the universal cover $\mathbb{R}$ is contractible and covers induce isomorphisms on π_k for $k \geq 2$ ([Exercise 13.7](#)). So $\pi_2(\mathbb{CP}^\infty) = \mathbb{Z}$ and all other homotopy groups vanish: $\mathbb{CP}^\infty$ is a $K(\mathbb{Z}, 2)$.

14.2 When the sequence splits

A long exact sequence is most useful when it breaks into short pieces, and there are two standard hypotheses that make it do so. One is easy and is below; the other is harder and is the point of [Remark 14.9](#).

Exercise 14.7 (medium). Suppose the fibration $F \rightarrow E \xrightarrow{p} B$ admits a section, i.e. a map $s: B \rightarrow E$ with $p \circ s = \text{id}_B$. Show that for $n \geq 2$

$$\pi_n(E) \cong \pi_n(F) \oplus \pi_n(B).$$

Hint. $p_* s_* = \text{id}$ already gives a splitting of one map in [Corollary 14.3](#). What does that force ∂ to be?

Solution. From $p \circ s = \text{id}$ we get $p_* s_* = \text{id}$ on $\pi_n(B)$, so p_* is surjective with a chosen splitting s_* . Exactness at $\pi_n(B)$ then makes $\partial = 0$, so the sequence breaks into

$$0 \rightarrow \pi_n(F) \xrightarrow{i_*} \pi_n(E) \xrightarrow{p_*} \pi_n(B) \rightarrow 0,$$

exact because ∂ vanishes on both sides. For $n \geq 2$ these are abelian groups and the sequence splits via s_* , giving $\pi_n(E) \cong \pi_n(F) \oplus \pi_n(B)$.

Warning 14.8 (Surjective does not mean split). A short exact sequence of abelian groups splits when the right-hand term is free, and otherwise only when a splitting is produced. In [Exercise 14.7](#) the splitting came from the section; with no section in sight, exhibiting one is work, and asserting it because a map happens to be surjective is an error.

Remark 14.9 (The other splitting mechanism). The dual hypothesis to a section is that the fibre inclusion is nullhomotopic. It also collapses [Corollary 14.3](#) into short exact sequences — $i_* = 0$ makes p_* injective and ∂ surjective — but now no splitting is handed over, and by [Warning 14.8](#) one has to be built.

The construction is worth naming, because it is used again in [Chapter 17](#). A nullhomotopy of a map $g: A \rightarrow E$ is exactly a map $CA \rightarrow E$ out of the cone extending g . So if $\alpha: S^{n-1} \rightarrow F$ is given and the inclusion $F \hookrightarrow E$ is nullhomotopic, composing the nullhomotopy with α produces a map of pairs

$$(D^n, S^{n-1}) \longrightarrow (E, F),$$

that is, a class in $\pi_n(E, F)$ — which [Theorem 14.2](#) identifies with $\pi_n(B)$. Checking that this is a homomorphism, and that it is a section of ∂ , is the natural exercise here, and it turns a structural hypothesis about the fibre into a direct sum decomposition of the homotopy groups of the base.

14.3 First computations

Example 14.10 (The Hopf bundle). The Hopf map $p: S^3 \rightarrow S^2$, $(z_1, z_2) \mapsto [z_1 : z_2]$, is a fibre bundle with fibre S^1 . Since $\pi_n(S^1) = 0$ for $n \geq 2$, the long exact sequence

$$\pi_n(S^1) \rightarrow \pi_n(S^3) \rightarrow \pi_n(S^2) \rightarrow \pi_{n-1}(S^1)$$

has vanishing outer terms for $n \geq 3$, giving

$$\pi_n(S^3) \cong \pi_n(S^2) \quad (n \geq 3).$$

In particular $\pi_3(S^2) \cong \pi_3(S^3) = \mathbb{Z}$, generated by the Hopf map itself.

That is already a surprise worth registering: S^2 has nontrivial homotopy in a degree above its dimension, and it has infinitely many distinct homotopy classes of maps from S^3 , while $H_3(S^2) = 0$ sees none of them.

Exercise 14.11 (easy). Show that $\pi_n(T^k) = 0$ for all $n \geq 2$, and more generally that a space with contractible universal cover has vanishing higher homotopy groups.

Solution. The universal cover of T^k is $\mathbb{R}^k$, which is contractible. By [Exercise 13.7](#), $\pi_n(T^k) \cong \pi_n(\mathbb{R}^k) = 0$ for $n \geq 2$. The same argument applies whenever the universal cover is contractible; such a space is a $K(\pi_1, 1)$.

Theorem 14.12 (Freudenthal; recalled). *The suspension map $\pi_n(S^k) \rightarrow \pi_{n+1}(S^{k+1})$ is an isomorphism for $n < 2k - 1$ and a surjection for $n = 2k - 1$.*

Example 14.13 (The stable range). Freudenthal makes $\pi_{n+k}(S^k)$ independent of k once $k > n + 1$; the common value is the n -th *stable stem* π_n^s . The first few are

$$\pi_0^s = \mathbb{Z}, \quad \pi_1^s = \mathbb{Z}/2, \quad \pi_2^s = \mathbb{Z}/2, \quad \pi_3^s = \mathbb{Z}/24.$$

Outside the stable range the groups are wilder and depend on k ; the computation of [Example 14.10](#) is unstable, $\pi_3(S^2)$ living well below the range where Freudenthal applies.

Remark 14.14 (What the long exact sequence cannot do). [Example 14.10](#) worked because the fibre was S^1 , whose homotopy groups vanish above degree one, so two of every three terms in the sequence were zero. That is a special circumstance. For a fibration with fibre S^3 , say, the outer terms do not vanish and the sequence determines its middle terms only up to extension — and in general not even that, since the maps are unknown.

There are two ways forward, and the book takes both. [Chapter 17](#) keeps track of the ambiguity by organising it into cohomology classes. [Chapter 15](#) replaces the long exact sequence by a spectral sequence, which computes the homology of the total space from that of the base and the fibre whatever they are, at the price of differentials that must themselves be determined.

Before leaving the long exact sequence it is worth noticing what it *can* do when a structural hypothesis is available: [Exercise 14.7](#) and [Remark 14.9](#) each convert a single geometric statement into a direct sum decomposition, and applied to the classical Hopf bundles $S^3 \rightarrow S^7 \rightarrow S^4$ and $S^7 \rightarrow S^{15} \rightarrow S^8$ they produce infinite cyclic summands in homotopy groups of spheres that no elementary argument reaches.

14.4 What to carry forward

- $\pi_n(E, F) \cong \pi_n(B)$ for a fibration ([Theorem 14.2](#)) — the source of the long exact sequence and of every splitting argument.
- Contractible total space turns the connecting map into an isomorphism ([Exercise 14.4](#)); this computes Ω and identifies $\mathbb{CP}^\infty = K(\mathbb{Z}, 2)$ ([Example 14.5](#) and [Exercise 14.6](#)).
- A section splits the sequence ([Exercise 14.7](#)); a nullhomotopic fibre inclusion does too, but the splitting must be constructed from a cone ([Warning 14.8](#) and [Remark 14.9](#)).
- $\pi_3(S^2) = \mathbb{Z}$ ([Example 14.10](#)); homotopy above the dimension is the normal state of affairs, and homology sees none of it.
- Freudenthal and the stable range ([Theorem 14.12](#) and [Example 14.13](#)).

Part III

Spectral sequences, operations, and the rational beginning

Chapter 15

The Serre spectral sequence: setting it up and running it

Chapter 14 ended by admitting that the long exact sequence of a fibration computes almost nothing once the fibre is complicated. The replacement computes the homology of the total space from that of the base and the fibre, for any fibre at all — at the cost that the answer arrives in instalments, with differentials that have to be determined by other means.

Learning to use it is learning three habits. Draw the E^2 -page and see which entries are forced to vanish. Decide what the answer *must* be, and read the differentials backwards from that. And keep the extension problem in view, since the spectral sequence computes an associated graded and not a group.

15.1 The machine

Theorem 15.1 (Serre; recalled). *Let $F \rightarrow E \xrightarrow{p} B$ be a fibration with B path-connected. There is a first-quadrant spectral sequence with*

$$E_{p,q}^2 = H_p(B; H_q(F)) \implies H_{p+q}(E),$$

with differentials $d^r: E_{p,q}^r \rightarrow E_{p-r, q+r-1}^r$, the coefficients being local if $\pi_1(B)$ acts non-trivially on $H_(F)$. Convergence means: there is a filtration of $H_n(E)$ whose associated graded is $\bigoplus_{p+q=n} E_{p,q}^\infty$.*

There is a cohomological version,

$$E_2^{p,q} = H^p(B; H^q(F)) \implies H^{p+q}(E),$$

with differentials $d_r: E_r^{p,q} \rightarrow E_r^{p+r, q-r+1}$.

Example 15.2 (The degenerate case). For a trivial bundle $E = B \times F$ with B simply connected, the spectral sequence collapses at E^2 : it must, since the answer is already known by Künneth to be $\bigoplus H_p(B) \otimes H_q(F)$ plus Tor terms, and that is the size of the E^2 -page. Collapse is the normal state of affairs only for products; expecting it in general is the standard error, and Example 15.5 is the antidote.

The first real computation uses the shape of the page rather than any knowledge of the differentials: when only two columns are nonzero, everything is forced.

Example 15.3 (The homology of a loop space). Apply [Theorem 15.1](#) to the path-loop fibration $\Omega S^n \rightarrow PS^n \rightarrow S^n$ of [Example 14.5](#), with $n \geq 2$. The base is simply connected, so the coefficients are untwisted, and $H_p(S^n)$ vanishes except for $p = 0, n$. So the E^2 -page has only two nonzero columns, and the only possible differential is

$$d^n: E_{n,q}^n = A_q \longrightarrow E_{0,q+n-1}^n = A_{q+n-1}, \quad A_q := H_q(\Omega S^n).$$

The total space is contractible, so E^∞ must vanish outside total degree 0, where it is $\mathbb{Z}$. In total degree $k > 0$ the only possible entries are $E_{0,k}$ and $E_{n,k-n}$; the first must be hit and the second must die, so every d^n is an isomorphism. Starting from $A_0 = \mathbb{Z}$ and iterating,

$$H_q(\Omega S^n) = \begin{cases} \mathbb{Z}, & q \equiv 0 \pmod{n-1}, \\ 0, & \text{otherwise.} \end{cases}$$

No differential was computed; the answer was forced by what had to vanish.

Warning 15.4. That is the characteristic move of the subject, and it is worth stating explicitly because it inverts the usual logic. One does not compute differentials and deduce the answer. One knows something about the answer — often only that certain groups vanish — and deduces the differentials. A reader waiting for a formula for d^r will wait indefinitely.

Example 15.5 (A spectral sequence that does not collapse). Take the Hopf fibration $S^1 \rightarrow S^3 \rightarrow S^2$. The E^2 -page is concentrated in columns $p = 0, 2$ and rows $q = 0, 1$, with $\mathbb{Z}$ in each of the four spots $(0, 0), (2, 0), (0, 1), (2, 1)$.

The total space is S^3 , with homology $\mathbb{Z}, 0, 0, \mathbb{Z}$. In total degree 1 the sole entry is $E_{0,1} = \mathbb{Z}$, and it has to disappear. Just one differential is capable of removing it,

$$d^2: E_{2,0}^2 \longrightarrow E_{0,1}^2,$$

which must therefore be surjective. Total degree 2 contains only $E_{2,0}$, which must also die, so d^2 must be injective. Hence d^2 is an isomorphism $\mathbb{Z} \rightarrow \mathbb{Z}$, and what survives is $E_{0,0}^\infty = \mathbb{Z}$ and $E_{2,1}^\infty = \mathbb{Z}$, giving $H_*(S^3) = (\mathbb{Z}, 0, 0, \mathbb{Z})$ as required.

Had one assumed collapse, the answer would have been $H_*(S^2 \times S^1)$, which is wrong. The Hopf fibration is not a product, and the spectral sequence says so by having a nonzero differential.

15.2 Local coefficients

When $\pi_1(B)$ acts nontrivially on $H_*(F)$ the E^2 -page is homology with local coefficients, and ignoring this gives wrong answers. The smallest example is one the book has already computed twice by other means.

Example 15.6 (The Klein bottle, a third time). The Klein bottle fibres over S^1 with fibre S^1 ; it is the mapping torus of $z \mapsto \bar{z}$, so the monodromy acts on $H_1(F) = \mathbb{Z}$ by -1 . Write $\mathbb{Z}^-$ for $\mathbb{Z}$ with that action of $\pi_1(S^1) = \mathbb{Z}$.

The E^2 -page has two columns. Row $q = 0$ is untwisted: $H_0(S^1; \mathbb{Z}) = \mathbb{Z}$ and $H_1(S^1; \mathbb{Z}) = \mathbb{Z}$. Row $q = 1$ is twisted:

$$H_0(S^1; \mathbb{Z}^-) = \mathbb{Z}/(x + x) = \mathbb{Z}/2, \quad H_1(S^1; \mathbb{Z}^-) = \ker(1 - (-1)) = \ker(\times 2) = 0.$$

No differential is possible, the base having no homology above degree one, so $E^2 = E^\infty$ and

$$H_0(K) = \mathbb{Z}, \quad H_1(K) = \mathbb{Z} \oplus \mathbb{Z}/2, \quad H_2(K) = 0,$$

agreeing with the Δ -complex computation of [Example 4.2](#) and the Mayer–Vietoris computation of [Example 5.8](#). Note that the $\mathbb{Z}/2$ arose as a group of *coinvariants* — the twisting is what produced the torsion, and an untwisted E^2 -page would have given $H_1 = \mathbb{Z}^2$, the answer for the torus.

Exercise 15.7 (medium). Let $T_f \rightarrow S^1$ be the mapping torus fibration of [Definition 2.5](#). Show that the action of $\pi_1(S^1)$ on $H_*(F)$ is generated by f_* .

Hint. Transport the fibre once around the base circle. In the mapping torus, going once around t is exactly applying f .

Solution. The action of a loop γ in the base on $H_*(F)$ is induced by the homotopy equivalence of F obtained by lifting γ to a homotopy of the fibre. For T_f with the convention of [Definition 2.5](#), the loop t lifts to the homotopy $\varphi_u(x) = (x, u)$ carrying the fibre at the seam to itself, and at $u = 1$ the identification $(x, 0) \sim (f(x), 1)$ makes that self-map f . Hence t acts by f_* , and $\pi_1(S^1) = \langle t \rangle$ acts through the powers of f_* .

Remark 15.8 (When the action is trivial). Three common situations give untwisted coefficients: B simply connected; F having homology on which every automorphism in the image of the monodromy acts trivially; and, for an aspherical fibration coming from a group extension, the case where the monodromy — which is conjugation — is inner-trivial. That last case is worth working out for oneself, since the monodromy of a fibration $K(A, 1) \rightarrow K(B, 1) \rightarrow K(C, 1)$ can be identified with the conjugation action of B on A by exactly the transport argument of [Exercise 15.7](#), and the condition on the extension that makes it trivial is then a purely group-theoretic one.

15.3 Forcing differentials, and the extension problem

When the groups are finite, counting orders replaces knowing maps.

Exercise 15.9 (easy). Suppose all groups $E_{p,q}^\infty$ with $p+q = n$ are finite. Show $|H_n(E)| = \prod_{p+q=n} |E_{p,q}^\infty|$.

Solution. Convergence gives a finite filtration $0 = F_{-1} \subseteq F_0 \subseteq \cdots \subseteq F_n = H_n(E)$ with $F_p/F_{p-1} \cong E_{p,n-p}^\infty$. For finite groups the order is multiplicative in short exact sequences, so $|H_n(E)| = \prod_p |F_p/F_{p-1}|$.

Warning 15.10 (E^∞ is not the answer). The spectral sequence computes the associated graded of a filtration, not the group. Two groups with the same associated graded need not agree: filter $\mathbb{Z}/4$ by $\{0\} \subseteq \mathbb{Z}/2 \subseteq \mathbb{Z}/4$ and the associated graded is $\mathbb{Z}/2 \oplus \mathbb{Z}/2$. So an E^∞ -page consisting of two copies of $\mathbb{Z}/2$ in one total degree is consistent with $H_n(E)$ being either $\mathbb{Z}/2 \oplus \mathbb{Z}/2$ or $\mathbb{Z}/4$, and the spectral sequence alone cannot decide.

The ambiguity is real and it is often where the interesting content sits: a non-split extension in E^∞ is the spectral sequence’s way of recording that the total space is glued together in a way the pieces do not see. When the answer is known independently — as for a fibration of Eilenberg–MacLane spaces coming from a group extension — the comparison runs the other way, and the extension problem is solved by the group theory rather than by the topology.

Exercise 15.11 (medium). Let $F \rightarrow E \rightarrow S^1$ be a fibration over the circle. Show that the Serre spectral sequence collapses at E^2 and reduces to short exact sequences

$$0 \rightarrow H_0(S^1; H_n(F)) \rightarrow H_n(E) \rightarrow H_1(S^1; H_{n-1}(F)) \rightarrow 0,$$

and identify these with the Wang sequence of [Proposition 5.12](#).

Hint. How many nonzero columns does the E^2 -page have? And for a local system M over S^1 with monodromy φ , what are $H_0(S^1; M)$ and $H_1(S^1; M)$?

Solution. $H_p(S^1; -)$ vanishes for $p \geq 2$, so only the columns $p = 0, 1$ survive. Every differential d^r with $r \geq 2$ has source or target in a vanishing column, so $E^2 = E^\infty$, and the two-step filtration of $H_n(E)$ gives the displayed short exact sequence.

For a local system M over S^1 with monodromy φ , homology is computed by the complex $M \xrightarrow{1-\varphi} M$, so $H_0 = \text{coker}(1 - \varphi)$ and $H_1 = \ker(1 - \varphi)$. By [Exercise 15.7](#) the monodromy of a mapping torus is f_* , so the sequence reads

$$0 \rightarrow \text{coker}(1 - f_* \text{ on } H_n(F)) \rightarrow H_n(E) \rightarrow \ker(1 - f_* \text{ on } H_{n-1}(F)) \rightarrow 0,$$

which is (5.1).

Remark 15.12 (How a computation of this kind is organised). For a fibration whose total space has known homology — typically because base, fibre and total space are all classifying spaces of groups in a short exact sequence — the procedure is fixed:

- (i) decide whether the coefficients are twisted ([Remark 15.8](#));
- (ii) write out the E^2 -page from the homology of the base and fibre;
- (iii) compare $\prod |E_{p,q}^2|$ in each total degree with the known $|H_n(E)|$ using [Exercise 15.9](#);
- (iv) where the orders exceed the target, differentials must act, and their ranks are determined by how much has to be killed;
- (v) where the orders match, no differential acts — but the extension problem remains, and [Warning 15.10](#) says it must be settled separately.

Step (v) is the one readers skip. A total degree in which the orders already agree is precisely a total degree in which the spectral sequence has told you nothing about how the group is assembled.

15.4 Serre classes

Often one wants a conclusion only up to some class of groups that is being ignored — finite groups, torsion groups, finitely generated groups. The axioms that make such ignoring legitimate are these.

Definition 15.13. Call a class $\mathcal{C}$ of abelian groups a *Serre class* if it contains the zero group and is closed under subgroups, quotients and extensions — all three at once, which is to say that in any exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ the middle group belongs to $\mathcal{C}$ exactly when the two outer ones do.

Exercise 15.14 (easy). Show that the finitely generated abelian groups form a Serre class, and that the free abelian groups do not.

Solution. Let $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ be exact. If B is finitely generated then so is C , being a quotient, and so is A , being a subgroup of a finitely generated abelian group. Conversely if A and C are finitely generated, lift generators of C and adjoin generators of A to generate B . So the class is a Serre class.

Free abelian groups are not: $0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{Z}/2 \rightarrow 0$ has both outer terms free on the left and middle but $\mathbb{Z}/2$ is not free, so the “only if” direction fails.

Remark 15.15 (What they are for). The mod- $\mathcal{C}$ Hurewicz theorem says that if X is simply connected and $\pi_i(X) \in \mathcal{C}$ for $i < n$, then $H_i(X) \in \mathcal{C}$ for $i < n$ and the Hurewicz map $\pi_n(X) \rightarrow H_n(X)$ is an isomorphism modulo $\mathcal{C}$. With $\mathcal{C}$ the finitely generated groups this yields, via the Serre spectral sequence applied to Postnikov towers, that the homotopy groups of a simply connected finite complex are all finitely generated — a theorem with no elementary proof, and one that [Chapter 14](#) could not have reached.

Verifying the axioms for a proposed class is routine and worth doing once for each class one intends to use; the only trap is the two-sidedness noted in [Exercise 15.14](#).

15.5 What to carry forward

- $E_{p,q}^2 = H_p(B; H_q F) \Rightarrow H_{p+q}(E)$ ([Theorem 15.1](#)); two nonzero columns force everything ([Example 15.3](#)).
- Read differentials backwards from what must vanish ([Warning 15.4](#)); do not expect collapse ([Example 15.5](#)).
- Twisted coefficients are not optional ([Example 15.6](#)); the monodromy of a mapping torus is f_* ([Exercise 15.7](#)).
- E^∞ gives the associated graded, not the group ([Warning 15.10](#)); order counting ([Exercise 15.9](#)) is the standard way to force differentials.
- The Wang sequence is the case of a circle base ([Exercise 15.11](#)).
- Serre classes let whole families of groups be ignored ([Definition 15.13](#) and [Remark 15.15](#)).

Chapter 16

Multiplicative structure: Leray–Hirsch, Gysin, Thom

The cohomology Serre spectral sequence carries a multiplication, and that changes what it can do. Degree bookkeeping now constrains not only which groups appear but which products are possible, and a single differential propagates through a whole ring by the Leibniz rule. Most of the classical computations in this subject are one differential plus multiplicativity.

The chapter then extracts three named tools from the machine — Leray–Hirsch, Gysin, Thom — and ends with the invariant that measures how far a map $S^{2n-1} \rightarrow S^n$ is from being inessential.

16.1 Products

Theorem 16.1 (Multiplicativity; recalled). *For a fibration $F \rightarrow E \rightarrow B$ with B path-connected and untwisted coefficients, the cohomology Serre spectral sequence is multiplicative: each E_r is a bigraded ring, each differential is a derivation,*

$$d_r(xy) = d_r(x)y + (-1)^{|x|}x d_r(y),$$

the product on E_{r+1} is induced from that on E_r , and the product on $E_2 = H^(B; H^*(F))$ is the cup product on the base tensored with that on the fibre. The filtration of $H^*(E)$ is multiplicative, and E_∞ is its associated graded ring.*

The last sentence contains the standing caveat: E_∞ gives the associated graded *ring*, so products landing in higher filtration are invisible.

Example 16.2 (The cohomology of $K(\mathbb{Z}, 2)$). Run the machine backwards: base unknown, fibre and total space known. Take the path–loop fibration

$$K(\mathbb{Z}, 1) \longrightarrow PK(\mathbb{Z}, 2) \longrightarrow K(\mathbb{Z}, 2),$$

whose fibre is S^1 and whose total space is contractible. Write $B = K(\mathbb{Z}, 2)$, which is simply connected, so coefficients are untwisted. The E_2 -page has two rows, $q = 0$ and $q = 1$, with $E_2^{p,q} = H^p(B) \otimes H^q(S^1)$.

Everything except $E_\infty^{0,0} = \mathbb{Z}$ must die. The only differential available is $d_2: E_2^{p,1} \rightarrow E_2^{p+2,0}$. Let x generate $H^1(S^1) = \mathbb{Z}$, sitting in $E_2^{0,1}$, and put

$$c = d_2(x) \in H^2(B).$$

This d_2 must be injective on $E_2^{0,1}$ and onto $E_2^{2,0}$, so c generates $H^2(B) \cong \mathbb{Z}$.

Now multiplicativity does the rest. Elements of the top row are $x \cdot b$ with $b \in H^p(B)$, and since $d_2b = 0$,

$$d_2(xb) = d_2(x)b = cb.$$

So $d_2: E_2^{p,1} \rightarrow E_2^{p+2,0}$ is multiplication by c , and it must be an isomorphism for every $p \geq 0$ — injective so the top row dies, surjective so the bottom row dies above degree 0. Hence multiplication by c is an isomorphism $H^p(B) \rightarrow H^{p+2}(B)$ for all $p \geq 0$, and starting from $H^0(B) = \mathbb{Z}$, $H^1(B) = 0$ we get

$$H^*(K(\mathbb{Z}, 2); \mathbb{Z}) = \mathbb{Z}[c], \quad |c| = 2.$$

Combined with [Exercise 14.6](#), this is the computation of $H^*(\mathbb{CP}^\infty)$ — now derived rather than quoted.

Exercise 16.3 (medium). Compute $H^*(\Omega S^{2n+1}; \mathbb{Q})$ as a ring, and say what changes over $\mathbb{Z}$.

Hint. Use the path-loop fibration over S^{2n+1} . Only one differential can act; call the transgressing class x and apply the Leibniz rule to x^k .

Solution. The fibration $\Omega S^{2n+1} \rightarrow P \rightarrow S^{2n+1}$ has contractible total space and simply connected base. The base has cohomology only in degrees 0 and $2n+1$, so only d_{2n+1} can be nonzero. By [Example 15.3](#) the fibre has homology $\mathbb{Q}$ in degrees $2nk$; let x generate $H^{2n}(\Omega S^{2n+1}; \mathbb{Q})$, and let ι generate $H^{2n+1}(S^{2n+1}; \mathbb{Q})$, with $d_{2n+1}(x) = \iota$.

By the Leibniz rule, $d_{2n+1}(x^k) = kx^{k-1}\iota$. Over $\mathbb{Q}$ this is an isomorphism from the class x^k in column 0 onto $x^{k-1}\iota$ in column $2n+1$ for every $k \geq 1$, so everything above total degree 0 cancels, as it must. Hence $x^k \neq 0$ for all k , and

$$H^*(\Omega S^{2n+1}; \mathbb{Q}) = \mathbb{Q}[x], \quad |x| = 2n.$$

Over $\mathbb{Z}$ the factor k is an obstruction: the differential is multiplication by k , so cancellation forces x^k to be $k!$ times a generator. The integral cohomology is the divided power algebra on x , additively $\mathbb{Z}$ in each degree $2nk$ but with $x^k = k! \gamma_k$.

16.2 Leray–Hirsch

Theorem 16.4 (Leray–Hirsch; recalled). *Let $F \rightarrow E \xrightarrow{p} B$ be a fibration with B path-connected, and suppose each $H^k(F; R)$ is a finitely generated free R -module. If there are classes $c_1, \dots, c_m \in H^*(E; R)$ whose restrictions i^*c_j form a basis of $H^*(F; R)$, then $H^*(E; R)$ is a free module over $H^*(B; R)$ with basis $c_1, \dots, c_m$; equivalently*

$$H^*(F; R) \otimes_R H^*(B; R) \longrightarrow H^*(E; R), \quad i^*c_j \otimes b \longmapsto c_j \smile p^*b,$$

is an isomorphism of $H^(B; R)$ -modules.*

Warning 16.5. This is an isomorphism of *modules*, not of rings. It says nothing about the products $c_i c_j$, which have to be computed separately and which are exactly where the interesting information sits. A bundle whose Leray–Hirsch decomposition looks like a product can have a ring structure that is not the tensor product, and comparing two bundles with the same module structure and different rings is a standard way to produce spaces with identical cohomology groups and different homotopy type.

Exercise 16.6 (easy). Show that [Theorem 16.4](#) does not apply to the Hopf fibration $S^1 \rightarrow S^3 \rightarrow S^2$, and reconcile this with [Example 15.5](#).

Solution. One would need some $c \in H^1(S^3)$ whose restriction generates $H^1(S^1) = \mathbb{Z}$. Since $H^1(S^3) = 0$, there is no candidate at all. Consistently, [Example 15.5](#) found $d^2 \neq 0$: in the language of the spectral sequence, the generator of $E_2^{0,1}$ is not a permanent cycle, and Leray–Hirsch is exactly the hypothesis that makes the whole fibre row consist of permanent cycles.

Example 16.7 (The projective bundle formula). Let $E \rightarrow B$ be a complex vector bundle of rank n and let $\mathbb{P}(E) \rightarrow B$ be its projectivisation, with fibre $\mathbb{CP}^{n-1}$. The tautological line bundle on $\mathbb{P}(E)$ has first Chern class $x \in H^2(\mathbb{P}(E); \mathbb{Z})$ restricting on each fibre to the generator of $H^2(\mathbb{CP}^{n-1})$. Then $1, x, \dots, x^{n-1}$ restrict to a basis of $H^*(\mathbb{CP}^{n-1}; \mathbb{Z})$, so by [Theorem 16.4](#)

$$H^*(\mathbb{P}(E); \mathbb{Z}) \text{ is free over } H^*(B; \mathbb{Z}) \text{ on } 1, x, \dots, x^{n-1}.$$

The ring structure is then determined by the single relation expressing x^n in terms of the lower powers, and its coefficients are by definition the Chern classes of E . [Chapter 26](#) takes that as the definition.

Exercise 16.8 (easy). Check [Example 16.7](#) in the case $B = \text{pt}$.

Solution. With B a point, $E = \mathbb{C}^n$ and $\mathbb{P}(E) = \mathbb{CP}^{n-1}$, and $H^*(B) = \mathbb{Z}$ concentrated in degree 0. The theorem says $H^*(\mathbb{CP}^{n-1})$ is free on $1, x, \dots, x^{n-1}$ over $\mathbb{Z}$, which together with $x^n = 0$ — there being nothing in $H^{2n}(\mathbb{CP}^{n-1})$ — gives $\mathbb{Z}[x]/(x^n)$.

16.3 Gysin and Thom

A sphere bundle has a fibre with cohomology in only two degrees, so its spectral sequence has two rows and collapses into a long exact sequence.

Theorem 16.9 (Gysin sequence). *Let $E \rightarrow B$ be an oriented real vector bundle of rank k over a path-connected base, with sphere bundle $S(E) \xrightarrow{p} B$ and Euler class $e \in H^k(B; \mathbb{Z})$. There is a long exact sequence*

$$\dots \rightarrow H^{n-k}(B) \xrightarrow{\smile e} H^n(B) \xrightarrow{p^*} H^n(S(E)) \rightarrow H^{n-k+1}(B) \xrightarrow{\smile e} H^{n+1}(B) \rightarrow \dots$$

Exercise 16.10 (medium). Derive [Theorem 16.9](#) from the Serre spectral sequence of the sphere bundle, and identify the Euler class in spectral-sequence terms.

Hint. The fibre is S^{k-1} , so only the rows $q = 0$ and $q = k - 1$ are nonzero, and only d_k can act. Write out the two-row page and see what the exactness of a two-step filtration says.

Solution. With fibre S^{k-1} the E_2 -page has $E_2^{p,0} = H^p(B)$ and $E_2^{p,k-1} = H^p(B)$, all other rows zero. The only possible differential is

$$d_k: E_k^{p,k-1} \rightarrow E_k^{p+k,0},$$

and by multiplicativity it is determined by its value on the fibre generator $u \in E_2^{0,k-1}$: writing $e := d_k(u) \in H^k(B)$, the Leibniz rule gives $d_k(ub) = eb$, so d_k is cup product with e .

Then $E_\infty^{p,0} = \text{coker}(\smile e)$ and $E_\infty^{p,k-1} = \ker(\smile e)$, and the two-step filtration of $H^n(S(E))$ gives short exact sequences

$$0 \rightarrow \text{coker} \left(H^{n-k}(B) \xrightarrow{\smile e} H^n(B) \right) \rightarrow H^n(S(E)) \rightarrow \ker \left(H^{n-k+1}(B) \xrightarrow{\smile e} H^{n+1}(B) \right) \rightarrow 0,$$

which splice into the displayed long exact sequence. The Euler class is the transgression of the fibre's fundamental class.

Example 16.11 (Projective space by Gysin). Apply [Theorem 16.9](#) to $S^1 \rightarrow S^{2n+1} \rightarrow \mathbb{CP}^n$, the sphere bundle of the tautological complex line bundle, so $k = 2$ and $e = c \in H^2(\mathbb{CP}^n)$. Since $H^m(S^{2n+1}) = 0$ for $0 < m < 2n + 1$, exactness forces

$$\smile c: H^{m-2}(\mathbb{CP}^n) \longrightarrow H^m(\mathbb{CP}^n)$$

to be an isomorphism for $0 < m \leq 2n$. Starting from $H^0 = \mathbb{Z}$ gives $H^*(\mathbb{CP}^n; \mathbb{Z}) = \mathbb{Z}[c]/(c^{n+1})$, and the truncation appears exactly where the sphere's top class interrupts the sequence.

Theorem 16.12 (Thom isomorphism; recalled). *For an oriented rank- k vector bundle $E \rightarrow B$ there is a class $u \in H^k(\text{Th}(E))$, the Thom class, such that*

$$H^n(B) \xrightarrow{\cong} \tilde{H}^{n+k}(\text{Th}(E)), \quad b \mapsto p^*b \smile u,$$

is an isomorphism for all n . The Euler class is the restriction of u to the zero section.

16.4 The Hopf invariant

Definition 16.13. Let $f: S^{2n-1} \rightarrow S^n$ with $n \geq 2$, and let $C_f = S^n \cup_f e^{2n}$ be its mapping cone. Then $H^*(C_f; \mathbb{Z})$ is $\mathbb{Z}$ in degrees $0, n, 2n$; choose generators $\alpha \in H^n$ and $\beta \in H^{2n}$. The *Hopf invariant* $H(f)$ is the integer with

$$\alpha \smile \alpha = H(f) \beta.$$

Example 16.14 (The Hopf map). For the Hopf map $p: S^3 \rightarrow S^2$ the mapping cone is $\mathbb{CP}^2$, since attaching a 4-cell to $S^2 = \mathbb{CP}^1$ along p is precisely how $\mathbb{CP}^2$ is built. By [Example 16.11](#), $H^*(\mathbb{CP}^2) = \mathbb{Z}[c]/(c^3)$ and c^2 generates H^4 , so $H(p) = \pm 1$.

Exercise 16.15 (medium). Show that $H(\Sigma g) = 0$ for every $g: S^{2n-2} \rightarrow S^{n-1}$; that is, the Hopf invariant vanishes on suspensions.

Hint. What is the mapping cone of a suspension, and what do cup products of positive-degree classes do on a suspension?

Solution. The mapping cone of Σg is ΣC_g . On a suspension every product of positive-degree reduced classes vanishes: the reduced diagonal $\Sigma Z \rightarrow \Sigma Z \wedge \Sigma Z$ is nullhomotopic, so the cup product of two reduced classes is zero. Hence $\alpha^2 = 0$ in $H^*(\Sigma C_g)$ and $H(\Sigma g) = 0$.

Exercise 16.16 (medium). Show that $H(f) = 0$ whenever n is odd.

Solution. Graded commutativity gives $\alpha \smile \alpha = (-1)^{n \cdot n} \alpha \smile \alpha$, which for n odd reads $\alpha^2 = -\alpha^2$, so $2\alpha^2 = 0$. But α^2 lies in $H^{2n}(C_f) \cong \mathbb{Z}$, which is torsion-free, so $\alpha^2 = 0$ and $H(f) = 0$.

Remark 16.17 (What the three tools do together). The tools of this chapter interlock in a way worth naming, because the combination is the standard route to the hardest classical results.

Suppose one is given a fibre bundle whose fibre, total space and base are all spheres. [Theorem 16.9](#) constrains the dimensions immediately: the sequence relates the cohomology of the base to that of the total space through cup product with the Euler class, and a sphere has cohomology in only two degrees, so most of the maps are forced to be zero or isomorphisms and the possible dimensions are pinned down by arithmetic alone.

Then [Theorem 16.12](#) converts the geometry into the Hopf invariant: the mapping cone of the bundle projection is the Thom space of the associated vector bundle, so the Thom class and the Euler class identify the cup square of [Definition 16.13](#) with a characteristic class of the bundle. That identification turns “which sphere bundles over spheres exist” into “which Hopf invariants occur”.

At which point the answer is not elementary: Adams proved that $H(f) = \pm 1$ is possible only for $n = 1, 2, 4, 8$. The book stops there and says so. What it does supply is everything up to that boundary — and [Example 16.14](#) shows the boundary is attained.

16.5 What to carry forward

- The spectral sequence is a ring and its differentials are derivations ([Theorem 16.1](#)); one differential plus Leibniz computes a whole ring ([Example 16.2](#) and [Exercise 16.3](#)).
- Leray–Hirsch ([Theorem 16.4](#)) gives a module isomorphism and never a ring isomorphism ([Warning 16.5](#)); what it assumes is that the fibre row consists of permanent cycles ([Exercise 16.6](#)).
- The projective bundle formula ([Example 16.7](#)) is where Chern classes come from.
- Gysin is the two-row spectral sequence, with the Euler class as the transgression ([Exercise 16.10](#)); Thom converts bundles into cones ([Theorem 16.12](#)).
- The Hopf invariant ([Definition 16.13](#)) is unstable ([Exercise 16.15](#)) and vanishes in odd degrees ([Exercise 16.16](#)); it equals ± 1 for the Hopf map ([Example 16.14](#)).

Chapter 17

Obstruction theory and Postnikov towers, operationally

[Theorem 11.11](#) extended a map over a CW pair by assuming that every obstruction vanished for want of anywhere to live. This chapter keeps the obstructions instead of assuming them away, and organises them into cohomology classes. The result is a machine that answers questions of the form “does this map extend?”, “does this bundle have a section?”, “does this map lift?” with a computable class rather than a hope.

Two habits carry over from [Chapter 15](#). The obstruction is defined cell by cell but only its *cohomology class* is meaningful, and the passage from cocycle to class is exactly the freedom to modify what has already been built. And the whole theory is stated for a target whose fundamental group acts trivially on its higher homotopy; without that, everything below needs local coefficients.

17.1 The obstruction cocycle

Let (W, A) be a CW pair and Y a simple space — one where $\pi_1(Y)$ acts trivially on every $\pi_n(Y)$, so that $\pi_n(Y)$ needs no basepoint.

Construction 17.1. Suppose f is defined on $A \cup W^n$. For each $(n+1)$ -cell e of W with attaching map $\varphi_e: S^n \rightarrow W^n$, the composite $f \circ \varphi_e$ is a map $S^n \rightarrow Y$; let

$$c(f)(e) = [f \circ \varphi_e] \in \pi_n(Y).$$

This defines a cellular cochain $c(f) \in C^{n+1}(W, A; \pi_n Y)$, the *obstruction cochain*.

Theorem 17.2 (Obstruction theory; recalled). *With notation as above:*

- (a) $c(f)$ is a cocycle, and it vanishes identically if and only if f extends over $A \cup W^{n+1}$;
- (b) its class $[c(f)] \in H^{n+1}(W, A; \pi_n Y)$ vanishes if and only if the restriction of f to $A \cup W^{n-1}$ extends over $A \cup W^{n+1}$ — that is, if and only if f can be altered on the n -cells so as to extend.

Warning 17.3 (Cochain versus class). The two statements differ, and the difference is the whole point. The cochain $c(f)$ answers “does *this* f extend?”; its class answers “does *some* map agreeing with f away from the n -cells extend?”. Only the second is a cohomological invariant, and only the second is stable under the modifications one always wants to make. A reader who computes $c(f) \neq 0$ and concludes that no extension exists has proved nothing.

Example 17.4 (One cell). If $W = A \cup e^{n+1}$ is a single cell attached along φ , then $H^{n+1}(W, A; \pi_n Y) \cong \pi_n(Y)$ and the obstruction is literally $[f \circ \varphi]$. So f extends over the cell precisely when the composite $S^n \rightarrow A \rightarrow Y$ is nullhomotopic, and a nullhomotopy is the extension. This is the entire content of [Theorem 17.2](#) in the smallest case, and it was the inductive step in the proof of [Theorem 11.11](#).

Exercise 17.5 (easy). Deduce [Theorem 11.11](#) from [Theorem 17.2](#).

Solution. Suppose the vanishing hypothesis of [Theorem 11.11](#) holds, so that $\pi_{k-1}(Y)$ is trivial in every dimension k occupied by a relative cell. Then the coefficient group $\pi_n(Y)$ appearing in [Theorem 17.2](#) is zero in each relevant degree, hence $H^{n+1}(W, A; \pi_n Y) = 0$ and every obstruction class vanishes. Extending skeleton by skeleton, the map extends over all of W .

Exercise 17.6 (medium). Let $Y = K(\pi, n)$ and let (W, A) be a CW pair. Show that a map $A \rightarrow K(\pi, n)$ extends over W if and only if its class in $H^n(A; \pi)$ lies in the image of the restriction $H^n(W; \pi) \rightarrow H^n(A; \pi)$.

Hint. Use the identification $[X, K(\pi, n)] \cong H^n(X; \pi)$ and the naturality of that bijection.

Solution. Under the natural bijection $[X, K(\pi, n)] \cong H^n(X; \pi)$, a map $A \rightarrow K(\pi, n)$ corresponds to a class $\alpha \in H^n(A; \pi)$, and extending it over W corresponds to finding $\tilde{\alpha} \in H^n(W; \pi)$ restricting to α , since the bijection is natural in the space. Hence extendability is exactly membership in the image of the restriction map.

Consistently with [Theorem 17.2](#): $\pi_k(K(\pi, n))$ vanishes except for $k = n$, so there is only one possibly nonzero obstruction, and it is the primary one.

17.2 Lifting

The same argument runs against a fibration instead of a space.

Theorem 17.7 (Obstructions to a lift; recalled). *Let $F \rightarrow X \xrightarrow{p} B$ be a fibration with F simple, let (W, A) be a CW pair, let $g: W \rightarrow B$, and suppose a lift of $g|_A$ to X is given. The obstructions to extending that lift over W lie in the groups*

$$H^{n+1}(W, A; \pi_n F), \quad n \geq 1,$$

one for each skeleton, each defined once the previous ones have been arranged to vanish.

Example 17.8 (Sections in low dimensions). Take $A = \emptyset$ and $g = \text{id}_B$, so that a lift is a section of $p: X \rightarrow B$. If $\dim W \leq n$ and $\pi_i(F) = 0$ for $i < n$, every obstruction group $H^{i+1}(W; \pi_i F)$ vanishes — for $i < n$ because the coefficients do, and for $i \geq n$ because $H^{i+1}(W) = 0$ by dimension. So a section exists. This is the standard reason that a fibration with highly connected fibre has sections over low-dimensional complexes.

Exercise 17.9 (medium). Let (W, A) be a CW pair. Describe the cells of the pair

$$(W \times I, W \times \partial I \cup A \times I),$$

and show that a k -cell of $W \setminus A$ contributes a relative cell of dimension $k + 1$.

Hint. A homotopy is a map on a cylinder. Which parts of the cylinder is the map already defined on?

Solution. Give $W \times I$ the product CW structure. The subcomplex $W \times \partial I \cup A \times I$ contains both ends and the part of the cylinder over A . The cells not in it are exactly the products $e \times (0, 1)$ for cells e of $W \setminus A$, and $\dim(e \times (0, 1)) = \dim e + 1$.

Remark 17.10 (Existence versus uniqueness). Every existence statement in this subject has a uniqueness companion obtained by running the same argument on a cylinder. Given two solutions of a lifting problem that agree on A , a homotopy rel A between them is a lift over the pair of [Exercise 17.9](#); and by that exercise its relative cells sit one dimension higher than those of (W, A) . Feeding this into [Theorem 17.7](#) therefore produces obstruction groups shifted by one, with the coefficients unchanged.

Carrying that out — and being careful about which direction the shift goes, so that the answer lands in $H^n(W, A; \pi_n F)$ rather than H^{n+2} — is the natural exercise here. It is worth doing once explicitly, because the shifted statement is used far more often than the unshifted one: in practice one rarely asks whether a lift exists and almost always whether two given ones agree.

17.3 Principal fibrations

Definition 17.11. A fibration $E \rightarrow B$ is *principal* if it is the pullback of a path-loop fibration: there is a map $k: B \rightarrow C$ and a homotopy equivalence of fibrations

$$E \simeq B \times_C PC,$$

where $PC \rightarrow C$ is the path fibration. The fibre is then ΩC .

Example 17.12. By [Theorem 13.9](#) each stage $X^n \rightarrow X^{n-1}$ of a Postnikov tower is pulled back from the path fibration over $K(\pi_n X, n+1)$ along the k -invariant. So every Postnikov stage is principal, essentially by construction, and the fibre $K(\pi_n X, n)$ is $\Omega K(\pi_n X, n+1)$ as [Definition 17.11](#) requires.

Warning 17.13. Principal does not mean trivial. A principal fibration is a pullback of a *universal* fibration, not a product; it is trivial exactly when the classifying map k is nullhomotopic ([Exercise 13.12](#)). Nor does principal mean that the tower splits: each stage being principal says nothing about how the stages compose.

Exercise 17.14 (easy). Show that the loop functor preserves pullbacks: for maps $B \rightarrow C \leftarrow D$,

$$\Omega(B \times_C D) \cong \Omega B \times_{\Omega C} \Omega D.$$

Hint. $\Omega X = \text{Map}_*(S^1, X)$, and $\text{Map}_*(S^1, -)$ is a right adjoint.

Solution. $\Omega = \text{Map}_*(S^1, -)$ is right adjoint to $S^1 \wedge -$, and right adjoints preserve limits; a pullback is a limit. Concretely, a loop in $B \times_C D$ is a pair of loops in B and D agreeing after mapping to C , which is exactly a point of $\Omega B \times_{\Omega C} \Omega D$.

Exercise 17.15 (easy). Show that $\Omega K(\pi, n+1) \simeq K(\pi, n)$ for $n \geq 1$.

Solution. By [Example 14.5](#), $\pi_i(\Omega Y) \cong \pi_{i+1}(Y)$ for all $i \geq 0$. With $Y = K(\pi, n+1)$ this gives $\pi_n(\Omega Y) = \pi$ and $\pi_i(\Omega Y) = 0$ otherwise, so $\Omega K(\pi, n+1)$ is a $K(\pi, n)$; being a loop space it has the homotopy type of a CW complex, so the identification is a homotopy equivalence.

Remark 17.16 (What looping does to a tower). Put the two exercises together. Applying Ω to the pullback square that exhibits a Postnikov stage as principal produces another pullback square ([Exercise 17.14](#)), and the universal fibration at the corner is carried to the universal fibration one degree down ([Exercise 17.15](#)). The looped tower is therefore again a tower of the same shape, with the k -invariants replaced by their loopings.

Two features of the looped tower are worth anticipating. Its stages are principal for a structural reason rather than by construction — which matters, because [Example 17.12](#) was an artefact of how the tower was built and says nothing about towers arising otherwise. And Ωk is a loop map, hence an H-map, so the k -invariants of the looped tower respect the multiplications of [Chapter 10](#). Working out precisely what survives is the natural exercise, and both ingredients are on this page.

17.4 What obstruction theory is for

Remark 17.17. Three uses, all of which the book meets again.

Classifying maps. [Remark 13.15](#) observed that a map into a two-stage tower is a cohomology class plus a lift, with the k -invariant as the obstruction and a cohomology group as the indeterminacy. [Theorem 17.7](#) is what makes that precise, and [Remark 17.10](#) is what counts the lifts once they exist. When the k -invariant is a cup square, as in [Example 13.11](#), the obstruction is a quadratic function of the class — which is why linear invariants such as π_* and H_* cannot detect the answer.

Constructing operations. [Exercise 17.6](#) turns maps into Eilenberg–MacLane spaces into cohomology classes. [Chapter 18](#) inverts that: natural transformations of cohomology are computed by computing the cohomology of an Eilenberg–MacLane space.

Combinatorial lifting. The pattern “build over a skeleton, meet an obstruction in a homotopy group, kill it or record it” becomes, in [Chapter 23](#), the horn-filling conditions defining Kan complexes and quasi-categories. There the obstruction groups disappear because the lifts are required to exist outright, and what remains is exactly the combinatorics.

17.5 What to carry forward

- The obstruction cochain ([Construction 17.1](#)) and the difference between it and its class ([Warning 17.3](#)).
- Obstructions to extending a map live in $H^{n+1}(W, A; \pi_n Y)$, and to lifting one in $H^{n+1}(W, A; \pi_n F)$ ([Theorems 17.2](#) and [17.7](#)). The extension lemma is the case in which they all vanish ([Exercise 17.5](#)).
- Uniqueness is existence on a cylinder, with the degree shifted by one ([Exercise 17.9](#) and [Remark 17.10](#)).
- Postnikov stages are principal ([Example 17.12](#)), which is not the same as trivial ([Warning 17.13](#)); Ω preserves pullbacks and shifts Eilenberg–MacLane spaces ([Exercises 17.14](#) and [17.15](#)).

Chapter 18

Cohomology operations and the Steenrod algebra

Three times now the book has stopped at the same wall. [Chapter 4](#) found that homology cannot separate lens spaces; [Chapter 7](#) found that neither can the cohomology ring, and promised an operation that is not induced by any multiplication; [Chapter 13](#) added that even all the homotopy groups together fail on the same pair.

This chapter builds the missing invariant. The construction is in two steps: first, a formal identification of all natural transformations of cohomology with the cohomology of an Eilenberg–MacLane space — which converts a classification problem into a computation; second, the specific family of operations that fall out for mod-2 coefficients. [Section 18.3](#) then uses them to separate two spaces whose cohomology rings agree over every coefficient ring at once.

18.1 Operations, and why they are computable

Definition 18.1. A *cohomology operation* of type $(m, \pi; n, G)$ is a family of functions

$$\theta_X: H^m(X; \pi) \longrightarrow H^n(X; G),$$

one for each space X , natural in X : $\theta_Y(\alpha)$ pulls back to $\theta_X(f^*\alpha)$ for every $f: X \rightarrow Y$.

Nothing is assumed about additivity. That omission is deliberate and is the point of [Exercise 18.5](#).

Theorem 18.2 (Representability turns classification into computation). *Cohomology operations of type $(m, \pi; n, G)$ are in natural bijection with*

$$H^n(K(\pi, m); G),$$

the correspondence sending θ to $\theta(\iota_m)$, where $\iota_m \in H^m(K(\pi, m); \pi)$ is the fundamental class.

Exercise 18.3 (medium). Prove [Theorem 18.2](#).

Hint. Use $[X, K(\pi, m)] \cong H^m(X; \pi)$, natural in X . Given θ , apply it to the universal class; given a class $u \in H^n(K(\pi, m); G)$, define an operation by pulling u back.

Solution. Write $\Phi(\theta) = \theta(\iota_m)$. Conversely, given $u \in H^n(K(\pi, m); G)$ define $\Psi(u)$ on a class $\alpha \in H^m(X; \pi)$ by choosing $f_\alpha: X \rightarrow K(\pi, m)$ with $f_\alpha^*\iota_m = \alpha$ — which exists and is unique up

to homotopy by representability — and setting $\Psi(u)(\alpha) = f_\alpha^*u$. Naturality of $\Psi(u)$ follows from the naturality of the representing bijection.

These are mutually inverse. Given u , we have $\Phi(\Psi(u)) = \Psi(u)(\iota_m) = \text{id}^*u = u$. Given θ and $\alpha = f_\alpha^*\iota_m$, naturality of θ gives

$$\theta(\alpha) = \theta(f_\alpha^*\iota_m) = f_\alpha^*\theta(\iota_m) = \Psi(\Phi(\theta))(\alpha),$$

so $\Psi\Phi = \text{id}$ as well.

Example 18.4 (Operations that preserve the degree). Take $\pi = G = \mathbb{Z}$ and $n = m$. Since $K(\mathbb{Z}, m)$ is $(m-1)$ -connected, the Hurewicz theorem gives $H_m(K(\mathbb{Z}, m); \mathbb{Z}) \cong \pi_m = \mathbb{Z}$, and then $H^m(K(\mathbb{Z}, m); \mathbb{Z}) \cong \mathbb{Z}$ generated by ι_m . So by [Theorem 18.2](#) the only operations $H^m(-; \mathbb{Z}) \rightarrow H^m(-; \mathbb{Z})$ are the maps $\alpha \mapsto k\alpha$. Degree-preserving operations are therefore uninteresting, and everything of substance raises the degree.

Exercise 18.5 (easy). Exhibit a cohomology operation that is not a homomorphism.

Solution. Let $\theta_X(\alpha) = \alpha \smile \alpha$ for $\alpha \in H^m(X; R)$. This is natural because f^* is a ring map. It is not additive: in general $(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$, which differs from $\alpha^2 + \beta^2$ as soon as $2\alpha\beta \neq 0$ — for instance in $H^*(\mathbb{CP}^\infty \times \mathbb{CP}^\infty; \mathbb{Z})$ with α, β the two degree-2 generators.

18.2 The Steenrod squares

Theorem 18.6 (Steenrod squares; recalled). *There are cohomology operations*

$$\text{Sq}^i: H^n(X; \mathbb{F}_2) \longrightarrow H^{n+i}(X; \mathbb{F}_2), \quad i \geq 0,$$

which are homomorphisms and satisfy:

- (a) $\text{Sq}^0 = \text{id}$;
- (b) $\text{Sq}^n(x) = x^2$ if $|x| = n$, and $\text{Sq}^i(x) = 0$ if $i > |x|$;
- (c) the Cartan formula $\text{Sq}^k(xy) = \sum_{i+j=k} \text{Sq}^i(x) \text{Sq}^j(y)$;
- (d) stability: Sq^i commutes with the suspension isomorphism

$$\tilde{H}^n(X; \mathbb{F}_2) \cong \tilde{H}^{n+1}(\Sigma X; \mathbb{F}_2).$$

Axiom (b) is the source of everything: it says that in the top degree the operation is the cup square, and axiom (d) then propagates that square into degrees where no square exists.

Example 18.7 (The smallest computation). In $H^*(\mathbb{CP}^2; \mathbb{F}_2) = \mathbb{F}_2[x]/(x^3)$ with $|x| = 2$, axiom (b) gives

$$\text{Sq}^2 x = x^2 \neq 0.$$

There is nothing else to compute: $\text{Sq}^0 x = x$, $\text{Sq}^1 x$ lies in $H^3(\mathbb{CP}^2) = 0$, and $\text{Sq}^i x = 0$ for $i > 2$.

Warning 18.8 (Stability is what makes them useful). Axiom (d) is the axiom that does the work in [Section 18.3](#), and it is worth seeing why. On a suspension every cup product of positive-degree classes vanishes ([Exercise 16.15](#)), so the cohomology ring of a suspension carries no information beyond its groups. But the Steenrod operations survive suspension unchanged. So on suspensions — where the ring is guaranteed to be useless — the operations are exactly as informative as they were before.

18.3 Where the ring is not enough

Theorem 18.9. *Over every commutative coefficient ring, $\Sigma\mathbb{CP}^2$ and $S^3 \vee S^5$ have isomorphic cohomology rings. They are nevertheless not homotopy equivalent.*

Proof. The rings agree. $\tilde{H}^*(\mathbb{CP}^2; R)$ is R in degrees 2 and 4, so $\tilde{H}^*(\Sigma\mathbb{CP}^2; R)$ is R in degrees 3 and 5; and $\tilde{H}^*(S^3 \vee S^5; R)$ is the same. In both cases every product of positive-degree classes vanishes: on $\Sigma\mathbb{CP}^2$ because it is a suspension (Exercise 16.15), and on $S^3 \vee S^5$ because the classes come from different wedge summands and each squares into a zero group (Exercise 7.3). Two graded rings which agree additively and have identically zero products are isomorphic.

The spaces differ. Work mod 2. Write σ for the suspension isomorphism and let x generate $H^2(\mathbb{CP}^2; \mathbb{F}_2)$. By stability and Example 18.7,

$$\text{Sq}^2(\sigma x) = \sigma(\text{Sq}^2 x) = \sigma(x^2) \neq 0 \quad \text{in } H^5(\Sigma\mathbb{CP}^2; \mathbb{F}_2),$$

since σ is an isomorphism and x^2 generates $H^4(\mathbb{CP}^2; \mathbb{F}_2)$.

On $S^3 \vee S^5$, let a generate H^3 . Restricting to the two summands is injective on $\tilde{H}^5$, and Sq^2 is natural, so it suffices to compute Sq^2 of the restrictions. On S^3 the class restricts to the generator of $H^3(S^3; \mathbb{F}_2)$ and Sq^2 of it lies in $H^5(S^3; \mathbb{F}_2) = 0$; on S^5 the restriction of a is 0. Hence $\text{Sq}^2 a = 0$.

A homotopy equivalence would induce an isomorphism of mod-2 cohomology commuting with Sq^2 , carrying a generator of H^3 to a generator of H^3 . On one side Sq^2 of such a generator is nonzero and on the other it is zero, so no such isomorphism exists. $\square$

Remark 18.10 (The debt, paid). Theorem 18.9 is what Remark 7.12 promised: a pair of spaces with isomorphic cohomology rings, over every coefficient ring at once, that are not homotopy equivalent. The invariant separating them is natural in the space and commutes with induced maps — so it is an invariant of homotopy type — and it is not induced by any product, since both products are zero.

The structural reason is Warning 18.8. A suspension has no ring structure worth the name, and the Steenrod operations are precisely the part of the multiplicative structure that survives suspension: Sq^n on a class of degree n is the cup square, and stability carries the information about that square into a space where the square itself is unavailable. The operation remembers a product that the ring has forgotten.

Exercise 18.11 (medium). Let X be any space with $\tilde{H}^*(X; \mathbb{F}_2)$ concentrated in degrees n and $2n$, with the top class equal to the square of the bottom one. Show that ΣX is not homotopy equivalent to $S^{n+1} \vee S^{2n+1}$.

Solution. Let x generate $H^n(X; \mathbb{F}_2)$, so x^2 generates $H^{2n}(X; \mathbb{F}_2)$ by hypothesis. By Theorem 18.6(b), $\text{Sq}^n x = x^2 \neq 0$, and by stability $\text{Sq}^n(\sigma x) = \sigma(x^2) \neq 0$ in $H^{2n+1}(\Sigma X; \mathbb{F}_2)$.

On $S^{n+1} \vee S^{2n+1}$ the generator a of H^{n+1} restricts to the generator on S^{n+1} , where $\text{Sq}^n a$ lands in $H^{2n+1}(S^{n+1}; \mathbb{F}_2) = 0$, and to 0 on S^{2n+1} ; restriction being injective on $\tilde{H}^{2n+1}$ of a wedge, $\text{Sq}^n a = 0$. As in Theorem 18.9 the two spaces are therefore not homotopy equivalent.

18.4 The Steenrod algebra

Definition 18.12. The *Steenrod algebra* $\mathcal{A}$ is the graded $\mathbb{F}_2$ -algebra generated by the Sq^i under composition, modulo the relations they satisfy. By Theorem 18.2 its degree- n part is $\varprojlim_m H^{m+n}(K(\mathbb{F}_2, m); \mathbb{F}_2)$: the stable operations.

Theorem 18.13 (Adem relations; recalled). *For $a < 2b$,*

$$\mathrm{Sq}^a \mathrm{Sq}^b = \sum_c \binom{b-c-1}{a-2c} \mathrm{Sq}^{a+b-c} \mathrm{Sq}^c,$$

binomial coefficients being read mod 2. Consequently $\mathcal{A}$ has an $\mathbb{F}_2$ -basis of admissible monomials $\mathrm{Sq}^{i_1} \cdots \mathrm{Sq}^{i_k}$, those with $i_j \geq 2i_{j+1}$ for all j .

Example 18.14. Take $a = b = 1$, which satisfies $a < 2b$. The only value of c giving a term is $c = 0$, with coefficient $\binom{b-c-1}{a-2c} = \binom{0}{1} = 0$. The sum is therefore empty and the relation reads

$$\mathrm{Sq}^1 \mathrm{Sq}^1 = 0.$$

So Sq^1 is a differential — and [Chapter 9](#) has already met a natural operation of degree one squaring to zero.

Remark 18.15 (Identifying Sq^1). [Example 18.14](#) invites a comparison, and the ingredients for making it are now all available.

The group of stable operations $H^q(-; \mathbb{F}_2) \rightarrow H^{q+1}(-; \mathbb{F}_2)$ is small: by [Theorem 18.2](#) it is $H^{q+1}(K(\mathbb{F}_2, q); \mathbb{F}_2)$, which is $\mathbb{F}_2$ for $q \geq 1$. So there is exactly one nonzero stable operation of degree one, and any two ways of producing one must agree as soon as both are shown to be nonzero. [Chapter 9](#) produced one — the Bockstein β , which is natural, stable, and squares to zero by [Exercise 9.12](#) — and this chapter produces another.

Checking that both are nonzero requires a single example, and [Example 9.16](#) has already computed β on $H^*(\mathbb{RP}^\infty; \mathbb{F}_2)$. Carrying out the comparison, and being explicit about why one example suffices once uniqueness is known, is the natural exercise here; the trap is to verify agreement on one space and conclude agreement in general without the uniqueness statement.

Remark 18.16 (What is left). Three continuations. [Chapter 26](#) uses the degree-one operation of [Remark 18.15](#) together with the Wu formula to pin down an integral characteristic class from mod-2 data — the triangle [Remark 12.12](#) promised. [Chapter 27](#) computes the action of $\mathcal{A}$ on a polynomial generator and shows where the exponents 2^j in the admissible basis come from. And [Chapter 28](#) finds the operations obstructing an algebraic structure on a spectrum.

For odd primes there is a parallel family, the reduced powers P^i raising degree by $2i(p-1)$, together with a Bockstein; the formal theory is the same and we shall use it without redevelopment.

18.5 What to carry forward

- Operations $\leftrightarrow H^n(K(\pi, m); G)$ ([Theorem 18.2](#)); classification becomes computation.
- Operations need not be additive ([Exercise 18.5](#)), and degree-preserving ones are trivial ([Example 18.4](#)).
- The axioms ([Theorem 18.6](#)); the top square is the cup square, and stability carries it where no square exists ([Warning 18.8](#)).
- $\Sigma \mathbb{C}P^2 \not\cong S^3 \vee S^5$ although their rings agree over every coefficient ring ([Theorem 18.9](#)) — the invariant promised in [Chapters 4, 7 and 13](#), and the mechanism extracted in [Exercise 18.11](#).

- $\mathcal{A}$, Adem relations, admissible monomials (Definition 18.12 and Theorem 18.13); $Sq^1 Sq^1 = 0$ (Example 18.14) and the uniqueness of degree-one stable operations (Remark 18.15).

Chapter 19

Rational homotopy: the first computations

Over $\mathbb{Q}$ the machinery of the last four chapters becomes decisive. Torsion disappears, the universal coefficient theorem loses its Ext term, and the Serre spectral sequence — whose differentials are usually determined only up to extension — becomes a computation in graded-commutative algebra over a field. Whole homotopy types become finite pieces of algebraic data, and [Chapter 24](#) will make that statement precise.

This chapter does the computations that come first: the rational cohomology of Eilenberg–MacLane spaces, the transgression technique that produces them, and the Whitehead product, which is where the even-dimensional spheres reveal that they carry more rational homotopy than one expects.

19.1 Eilenberg–MacLane spaces over $\mathbb{Q}$

Theorem 19.1 (Recalled). *For $n \geq 1$,*

$$H^*(K(\mathbb{Z}, n); \mathbb{Q}) = \begin{cases} \mathbb{Q}[x_n], & n \text{ even}, \\ \Lambda(x_n), & n \text{ odd}, \end{cases} \quad |x_n| = n.$$

Moreover $H^(K(\pi, n); \mathbb{Q}) = \mathbb{Q}$ for every finite group π .*

The parity is not a curiosity: it is graded commutativity. A generator of even degree may be raised to arbitrary powers, while one of odd degree squares to zero, and the two cases produce a polynomial algebra and an exterior algebra respectively. Almost every parity condition in rational homotopy theory traces back to this.

Example 19.2 (Degree two). By [Exercise 14.6](#) we have $K(\mathbb{Z}, 2) \simeq \mathbb{CP}^\infty$, and [Example 16.2](#) computed $H^*(K(\mathbb{Z}, 2); \mathbb{Z}) = \mathbb{Z}[c]$. Tensoring with $\mathbb{Q}$ leaves $\mathbb{Q}[c]$, which is the even case of [Theorem 19.1](#).

Exercise 19.3 (medium). Compute $H^*(K(\mathbb{Z}, 3); \mathbb{Q})$ from [Example 19.2](#), using the path-loop fibration.

Hint. *The fibration is $K(\mathbb{Z}, 2) \rightarrow PK(\mathbb{Z}, 3) \rightarrow K(\mathbb{Z}, 3)$, with contractible total space. Only one differential can act on the generator c of $H^2(K(\mathbb{Z}, 2); \mathbb{Q})$; call its value x and use the Leibniz rule on c^k .*

Solution. Write $B = K(\mathbb{Z}, 3)$. The fibre has $H^*(K(\mathbb{Z}, 2); \mathbb{Q}) = \mathbb{Q}[c]$ with $|c| = 2$, so the E_2 -page is $H^*(B; \mathbb{Q}) \otimes \mathbb{Q}[c]$. The class c lies in $E_2^{0,2}$ and the only differential that can be nonzero on it is d_3 ; put $x = d_3(c) \in H^3(B; \mathbb{Q})$.

By the Leibniz rule $d_3(c^k) = k c^{k-1}x$, and over $\mathbb{Q}$ the factor k is invertible, so $d_3: \mathbb{Q}\{c^k\} \rightarrow \mathbb{Q}\{c^{k-1}x\}$ is an isomorphism for every $k \geq 1$. Everything in positive total degree therefore cancels, as it must, provided $H^*(B; \mathbb{Q})$ is spanned by 1 and x with $x^2 = 0$; any further generator would survive to E_∞ and contradict contractibility of the total space. Hence

$$H^*(K(\mathbb{Z}, 3); \mathbb{Q}) = \Lambda(x), \quad |x| = 3,$$

and $x^2 = 0$ automatically, x being of odd degree.

Exercise 19.4 (hard). Compute $H^*(\Omega S^{2m}; \mathbb{Q})$ as a ring.

Hint. Use the path-loop fibration over S^{2m} . Let x be the class transgressing to the fundamental class ι of the base; what degree is x , and what is left over after d_{2m} has acted on x ?

Hint. Compare degrees: $x\iota$ has degree $4m - 1$, so a class killing it would sit in degree $4m - 2$. Postulate such a class y and check that $\Lambda(x) \otimes \mathbb{Q}[y]$ is consistent with a contractible total space.

Solution. By [Example 15.3](#), $H_*(\Omega S^{2m}; \mathbb{Q})$ is $\mathbb{Q}$ in each degree $(2m - 1)k$. Let x generate degree $2m - 1$; being of odd degree, $x^2 = 0$, so $\mathbb{Q}[x]$ is not available and further generators are needed.

Run the spectral sequence of $\Omega S^{2m} \rightarrow P \rightarrow S^{2m}$. Only d_{2m} can act, and $d_{2m}(x) = \iota$. Then $d_{2m}(x \cdot z) = \iota z$ for any z in the fibre, and $d_{2m}(\iota) = 0$. So x and ι cancel, but the class $x\iota$ in $E^{2m, 2m-1}$ is not yet killed — nothing maps to it from $E^{0, 4m-2}$ unless the fibre has a class in degree $4m - 2$.

Postulate y of degree $4m - 2$ with $d_{2m}(y) = x\iota$; the degrees match, since d_{2m} raises total degree by one. Then $d_{2m}(y^k) = k y^{k-1}x\iota$, which over $\mathbb{Q}$ is again an isomorphism at each stage, and everything cancels. The resulting fibre cohomology,

$$H^*(\Omega S^{2m}; \mathbb{Q}) = \Lambda(x) \otimes \mathbb{Q}[y], \quad |x| = 2m - 1, \quad |y| = 4m - 2,$$

has one basis element in each degree $(2m - 1)k$ — namely y^j and xy^j — which matches the additive answer of [Example 15.3](#), so no further generators are possible.

Warning 19.5. Compare [Exercise 16.3](#) with [Exercise 19.4](#). For an odd sphere the loop space has polynomial rational cohomology on one even generator; for an even sphere it needs an exterior generator *and* a polynomial one. The extra generator is not an artefact of the method: it is the first sign that even spheres carry rational homotopy that odd spheres do not, and [Section 19.2](#) identifies what it is.

19.2 Whitehead products

Definition 19.6. Let $\alpha \in \pi_m(X)$ and $\beta \in \pi_n(X)$. The product $S^m \times S^n$ has a CW structure with one cell in each of the dimensions $0, m, n, m + n$, and the attaching map of the top cell is a map $S^{m+n-1} \rightarrow S^m \vee S^n$. Composing it with the map $S^m \vee S^n \rightarrow X$ given by α and β yields the *Whitehead product*

$$[\alpha, \beta] \in \pi_{m+n-1}(X).$$

Exercise 19.7 (medium). Verify the assertion in [Definition 19.6](#): $S^m \times S^n$ is obtained from $S^m \vee S^n$ by attaching a single $(m + n)$ -cell.

Hint. Give each sphere the CW structure with one 0-cell and one top cell, and take the product structure.

Solution. With $S^m = e^0 \cup e^m$ and $S^n = e^0 \cup e^n$, the product CW structure on $S^m \times S^n$ has cells $e^0 \times e^0$, $e^m \times e^0$, $e^0 \times e^n$ and $e^m \times e^n$, of dimensions 0, m , n and $m+n$. The union of all but the top one is $S^m \vee S^n$, so the product is that wedge with one $(m+n)$ -cell attached, along a map $S^{m+n-1} \rightarrow S^m \vee S^n$.

Exercise 19.8 (easy). In $H^*(S^n \times S^n; \mathbb{Z})$ let a and b be the pullbacks of a generator along the two projections. Show

$$(a+b)^2 = (1 + (-1)^n) ab,$$

so that the square is $2ab$ for n even and 0 for n odd.

Solution. By Künneth, $a^2 = b^2 = 0$, both being pullbacks of classes squaring to zero in $H^*(S^n)$, where $H^{2n}(S^n) = 0$. Graded commutativity gives $ba = (-1)^{n \cdot n} ab = (-1)^n ab$, since $n^2 \equiv n \pmod{2}$. Hence

$$(a+b)^2 = a^2 + ab + ba + b^2 = ab + (-1)^n ab = (1 + (-1)^n) ab.$$

Remark 19.9 (The Whitehead square). Take $\alpha = \beta = \iota$, the identity class of $\pi_n(S^n)$: the resulting $[\iota, \iota] \in \pi_{2n-1}(S^n)$ is the *Whitehead square*. By [Exercise 19.7](#) its mapping cone is S^n with a $2n$ -cell attached along it, and there is a natural map from $S^n \times S^n$ to that cone — fold the two copies of S^n together and extend over the top cell.

That map carries the generator of H^n of the cone to $a+b$, so [Exercise 19.8](#) computes the cup square in the cone, which by [Definition 16.13](#) is the Hopf invariant of $[\iota, \iota]$. Working this out is the natural exercise here, and the answer explains [Warning 19.5](#): for n even the Whitehead square has infinite order, and it is the source of the extra rational homotopy class.

Note that the parity in [Exercise 19.8](#) matches the parity in [Exercise 16.16](#): for odd n the Hopf invariant of anything vanishes, so no information can be extracted this way, and the even case is the whole content.

19.3 The rational homotopy of spheres

Theorem 19.10 (Serre; recalled, not proved here). For $n \geq 2$,

$$\pi_k(S^n) \otimes \mathbb{Q} = \begin{cases} \mathbb{Q}, & k = n, \\ \mathbb{Q}, & k = 2n - 1 \text{ and } n \text{ even}, \\ 0, & \text{otherwise.} \end{cases}$$

Warning 19.11. The book does not prove [Theorem 19.10](#), and the reader should not take the surrounding computations as a proof of it. The standard argument compares S^n with $K(\mathbb{Z}, n)$ and, for n even, kills the class x_n^2 by passing to the homotopy fibre of a map $K(\mathbb{Z}, n) \rightarrow K(\mathbb{Z}, 2n)$; computing the rational cohomology of that fibre is exactly the step this chapter leaves out. Everything needed for it is nevertheless in place: [Theorem 19.1](#) gives the two cohomologies, [Exercise 19.3](#) shows how a transgression kills a polynomial generator, and [Example 16.2](#) shows how multiplicativity propagates a single differential through a whole ring.

Remark 19.12 (Where the extra class comes from). For n even, [Theorem 19.10](#) produces a rational homotopy class in degree $2n - 1$ that odd spheres do not have. Three computations in this chapter have now pointed at it from different directions: [Warning 19.5](#) found an extra polynomial generator in $H^*(\Omega S^{2m}; \mathbb{Q})$ in degree $4m - 2 = 2(2m) - 2$, which is one less than $2n - 1$ for $n = 2m$ — exactly the shift $\pi_k(\Omega X) = \pi_{k+1}(X)$; [Exercise 19.8](#) found that a certain square is nonzero precisely for n even; and [Remark 19.9](#) identified the class producing it.

[Chapter 24](#) recomputes all of this from a two-generator algebraic model of S^n , where the parity appears as the difference between $(\Lambda(x), 0)$ with $|x|$ odd and $(\Lambda(x, y), dy = x^2)$ with $|x|$ even — and where the degree $2n - 1$ of the extra class is visible as the degree of y .

19.4 What to carry forward

- $H^*(K(\mathbb{Z}, n); \mathbb{Q})$ is polynomial for n even and exterior for n odd ([Theorem 19.1](#)); the parity is graded commutativity and nothing else.
- Transgression plus the Leibniz rule computes a whole ring from one differential ([Exercises 19.3](#) and [19.4](#)); over $\mathbb{Q}$ the integer factors that obstructed the integral computation become invertible.
- An even sphere's loop space needs an extra generator ([Warning 19.5](#)), and the Whitehead square is what produces it ([Definition 19.6](#) and [Remark 19.9](#)).
- $(a + b)^2 = (1 + (-1)^n)ab$ ([Exercise 19.8](#)) — two lines of graded commutativity that decide the whole parity story.
- Serre's theorem ([Theorem 19.10](#)) is quoted, not proved ([Warning 19.11](#)).

Part IV

Abstract homotopy theory

Chapter 20

Adjunctions, Kan extensions, and nerve–realisation

Part IV changes the level of description. Until now a construction was given by a formula and its properties proved by inspection; from here on constructions are characterised by universal properties, and their properties are consequences of those characterisations rather than of the formulas.

The gain is not elegance but transferability. Geometric realisation, the nerve of a category, the classifying space of a group, the simplicial chain complex — all are instances of one construction, and a proof given once at the general level covers all of them. This chapter sets up that construction. [Chapters 21](#) and [22](#) then redo the homotopy theory of Parts I–III in the same language.

20.1 Adjunctions

Definition 20.1. Functors $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{D} \rightarrow \mathcal{C}$ are *adjoint*, written $F \dashv G$, if there are bijections

$$\mathcal{D}(Fc, d) \cong \mathcal{C}(c, Gd)$$

natural in c and d . Equivalently, there are a unit $\eta: \text{id} \Rightarrow GF$ and a counit $\varepsilon: FG \Rightarrow \text{id}$ satisfying the triangle identities.

Exercise 20.2 (easy). Show that a left adjoint preserves all colimits that exist.

Hint. Compare the two sets of maps out of a colimit into a fixed object, using the adjunction to move F across.

Solution. Let $D: J \rightarrow \mathcal{C}$ have a colimit. For any $d \in \mathcal{D}$,

$$\mathcal{D}(F(\text{colim } D), d) \cong \mathcal{C}(\text{colim } D, Gd) \cong \lim_j \mathcal{C}(Dj, Gd) \cong \lim_j \mathcal{D}(FDj, d) \cong \mathcal{D}(\text{colim}_j FDj, d),$$

naturally in d ; by Yoneda the two objects agree.

Warning 20.3. A left adjoint preserves colimits, and in general nothing else. Products are limits, so a left adjoint has no reason to preserve them, and [Chapter 29](#) exhibits a left adjoint of importance that does not.

20.2 Kan extensions

Definition 20.4. Let $K: \mathcal{D} \rightarrow \mathcal{E}$ and $F: \mathcal{D} \rightarrow \mathcal{C}$. A *left Kan extension* of F along K is a functor $\text{Lan}_K F: \mathcal{E} \rightarrow \mathcal{C}$ together with a natural transformation $\eta: F \Rightarrow (\text{Lan}_K F)K$ that is universal: any $\gamma: F \Rightarrow GK$ factors as $\gamma = (\delta K) \circ \eta$ for a unique $\delta: \text{Lan}_K F \Rightarrow G$.

Proposition 20.5 (Pointwise formula; recalled). *Suppose $\mathcal{D}$ is small and $\mathcal{C}$ is cocomplete. Then $\text{Lan}_K F$ exists, and*

$$(\text{Lan}_K F)(e) = \text{colim}_{(K \downarrow e)} F,$$

the colimit being taken over the comma category whose objects are the pairs consisting of $d \in \mathcal{D}$ and a map $Kd \rightarrow e$.

Example 20.6 (Colimits are Kan extensions). Take $\mathcal{E} = \mathbf{1}$, the terminal category. Then $(K \downarrow *)$ is all of $\mathcal{D}$, and [Proposition 20.5](#) reads $\text{Lan}_K F = \text{colim}_{\mathcal{D}} F$. So a colimit is the simplest kind of Kan extension, and conversely a Kan extension is a colimit computed one object at a time.

Definition 20.7. A left Kan extension is *absolute* if it is preserved by every functor out of $\mathcal{C}$: for every $H: \mathcal{C} \rightarrow \mathcal{C}'$, the pair $(H \text{Lan}_K F, H\eta)$ is a left Kan extension of HF along K .

Exercise 20.8 (easy). Show that if $\text{Lan}_K F$ is absolute and $\text{Lan}_{K'}(\text{Lan}_K F)$ is absolute, then the composite is an absolute left Kan extension of F along $K'K$.

Solution. Absoluteness of $\text{Lan}_K F$ makes $H \text{Lan}_K F$ a left Kan extension of HF for every H ; taking $H = \text{Lan}_{K'}$ -composites and using the universal property twice, a transformation $F \Rightarrow GK'K$ corresponds first to $\text{Lan}_K F \Rightarrow GK'$ and then to $\text{Lan}_{K'} \text{Lan}_K F \Rightarrow G$, each step uniquely. The composite therefore has the universal property of $\text{Lan}_{K'K} F$, and it is absolute because each stage is preserved by every functor.

20.3 Density

Presheaves are colimits of representables, and that single fact is what makes the next section work.

Theorem 20.9 (Density). *Let $\mathcal{D}$ be small and let $\mathcal{Y}: \mathcal{D} \rightarrow [\mathcal{D}^{\text{op}}, \mathbf{Set}]$ be the Yoneda embedding. Every presheaf X is canonically the colimit of representables:*

$$X \cong \text{colim}_{\int X} \mathcal{Y},$$

where $\int X$ denotes the category of elements: its objects are the pairs (d, x) with $x \in X(d)$.

Exercise 20.10 (medium). Prove [Theorem 20.9](#).

Hint. Compute maps out of the colimit into an arbitrary presheaf Z , and use the Yoneda lemma on each term.

Solution. For any presheaf Z ,

$$[\mathcal{D}^{\text{op}}, \mathbf{Set}]\left(\text{colim}_{\int X} \mathcal{Y}, Z\right) \cong \lim_{\int X} [\mathcal{D}^{\text{op}}, \mathbf{Set}](\mathcal{Y}d, Z) \cong \lim_{(d,x) \in \int X} Z(d),$$

the last step by the Yoneda lemma. An element of that limit is a choice, for each $x \in X(d)$, of an element of $Z(d)$, compatible with the maps of $\int X$ — which is exactly a natural transformation $X \Rightarrow Z$. So the colimit represents the same functor as X , and Yoneda identifies them.

Remark 20.11 (Coend notation). The colimit of [Theorem 20.9](#) is written as a coend,

$$X \cong \int^{d \in \mathcal{D}} X(d) \cdot \mathcal{Y}d,$$

where $S \cdot c$ denotes the coproduct of S copies of c . The notation is worth adopting because the formula it abbreviates is the one that appears in [Section 20.4](#): replace $\mathcal{Y}d$ by an object of some other category and the same expression defines a realisation.

20.4 Nerve and realisation

Construction 20.12. Let $\mathcal{D}$ be small, $\mathcal{C}$ cocomplete, and $F: \mathcal{D} \rightarrow \mathcal{C}$ any functor. Define

$$\mathrm{Sing}_F: \mathcal{C} \longrightarrow [\mathcal{D}^{\mathrm{op}}, \mathbf{Set}], \quad \mathrm{Sing}_F(c)(d) = \mathcal{C}(Fd, c),$$

and let $|\cdot|_F = \mathrm{Lan}_{\mathcal{Y}} F$, which by [Proposition 20.5](#) exists and is given by the coend

$$|X|_F = \int^d X(d) \cdot Fd.$$

Example 20.13 (Geometric realisation). Take $\mathcal{D} = \Delta$, $\mathcal{C} = \mathbf{Top}$, and $F([n]) = |\Delta^n|$ the standard topological simplex. Then

$$\mathrm{Sing}_F(Y)_n = \mathbf{Top}(|\Delta^n|, Y)$$

is the singular simplicial set of Y , and the coend

$$|X|_F = \int^n X_n \cdot |\Delta^n| = \left(\coprod_n X_n \times |\Delta^n| \right) / \sim$$

is the usual geometric realisation: one topological simplex for each simplex of X , glued along the face and degeneracy identifications, which is exactly what the coend's coequaliser imposes.

Example 20.14 (The nerve of a category). Take $\mathcal{D} = \Delta$, $\mathcal{C} = \mathbf{Cat}$, and $F([n]) = [n]$ the poset $0 \rightarrow 1 \rightarrow \cdots \rightarrow n$ viewed as a category. Then $\mathrm{Sing}_F(\mathcal{A})_n = \mathbf{Cat}([n], \mathcal{A})$ is the set of strings of n composable morphisms: this is the nerve $N\mathcal{A}$. The construction $|\cdot|_F$ is its left adjoint, sending a simplicial set to the category it generates.

Example 20.15 (The pattern in its smallest case). Take $\mathcal{D} = \mathbf{1}$ and let F pick out an object $c_0 \in \mathcal{C}$. Presheaves on $\mathbf{1}$ are sets, $\mathrm{Sing}_F(c) = \mathcal{C}(c_0, c)$, and the coend collapses to $|S|_F = S \cdot c_0$, the coproduct of S copies of c_0 . The adjunction to be proved reads

$$\mathcal{C}\left(\coprod_S c_0, c\right) \cong \mathbf{Set}(S, \mathcal{C}(c_0, c)),$$

which is precisely the universal property of a coproduct: a map out of a coproduct is a family of maps, indexed by S . Nothing else is happening in the general case either — the coend is a coproduct glued along $\mathcal{D}$, and the naturality in d is what turns the family into a natural transformation.

Remark 20.16 (What remains to be checked). [Construction 20.12](#) produces, for every F , a pair of functors; the statement that they are adjoint,

$$\mathcal{C}(|X|_F, c) \cong [\mathcal{D}^{\text{op}}, \mathbf{Set}](X, \text{Sing}_F(c)),$$

is not yet proved here. Every ingredient is: the Yoneda lemma; the universal property of a left Kan extension ([Definition 20.4](#)); the coend formula ([Proposition 20.5](#) and [Remark 20.11](#)); and density ([Theorem 20.9](#)), which reduces the general X to the representables, where the Yoneda lemma applies directly. [Example 20.15](#) shows the shape of the argument in the case where the indexing category contributes nothing.

Proving it is worth doing once, because of what it then yields for free. The special case of [Example 20.13](#) is the adjunction $|\cdot| \dashv \text{Sing}$ between spaces and simplicial sets — the one on which [Chapters 22](#) and [23](#) depend, and which is usually verified by an explicit and uninformative computation with simplices. Proved at this level, it is a corollary rather than a computation, and the same corollary covers [Example 20.14](#) and every other choice of F .

Exercise 20.17 (medium). Grant the adjunction of [Remark 20.16](#). Deduce that geometric realisation preserves all colimits, and that Sing preserves all limits.

Solution. $|\cdot|$ is a left adjoint, so it preserves colimits by [Exercise 20.2](#). Dually Sing , being a right adjoint, preserves limits: the same computation with the arrows reversed.

Exercise 20.18 (medium). Let G be a group, regarded as a one-object category. Identify $|NG|$, the realisation of the nerve, and show it is a $K(G, 1)$.

Hint. Describe $(NG)_n$ explicitly, and then identify the realisation with the standard bar construction. For the homotopy type, exhibit a contractible space with a free G -action.

Solution. By [Example 20.14](#), $(NG)_n = \mathbf{Cat}([n], G)$ is the set of strings of n composable arrows, i.e. G^n , with faces given by multiplying adjacent entries and dropping outer ones. Realising gives the bar construction BG .

There is a parallel simplicial set EG with $(EG)_n = G^{n+1}$ and G acting freely by diagonal multiplication, whose realisation is contractible — it has an extra degeneracy, so its identity map is homotopic to a constant — and $|EG|/G = |NG| = BG$. Hence $|EG| \rightarrow BG$ is a covering with contractible total space, so $\pi_1(BG) \cong G$ by [Proposition 3.12](#) and $\pi_n(BG) \cong \pi_n(|EG|) = 0$ for $n \geq 2$ by [Exercise 13.7](#).

20.5 What to carry forward

- Left adjoints preserve colimits ([Exercise 20.2](#)) and nothing else ([Warning 20.3](#)).
- Left Kan extensions ([Definition 20.4](#)), their pointwise formula ([Proposition 20.5](#)), and absoluteness ([Definition 20.7](#) and [Exercise 20.8](#)) — the last is what [Chapter 22](#) needs.
- Density ([Theorem 20.9](#)) and coend notation ([Remark 20.11](#)).
- [Construction 20.12](#): one construction producing geometric realisation, the nerve, and the classifying space ([Examples 20.13](#) and [20.14](#) and [Exercise 20.18](#)); what remains to be verified about it is set out in [Remark 20.16](#).

Chapter 21

Homotopy limits and colimits

Strict limits and colimits are not homotopy invariant: replace one of the objects in a diagram by a homotopy equivalent one and the answer can change completely. Since every construction in Parts I–III was only ever defined up to homotopy, this is a serious defect, and the repair — replacing the construction by one that is invariant by design — is the subject of this chapter.

The two model cases are the pushout and the pullback. Their homotopy versions are the double mapping cylinder and the space of paths-with-lifts, and every construction the book has used so far — mapping cone, suspension, homotopy fibre — is one of them.

21.1 Why strict colimits are the wrong ones

Example 21.1 (The same diagram, twice). Consider the two pushout diagrams

$$* \longleftarrow S^1 \longrightarrow *, \quad D^2 \longleftarrow S^1 \longrightarrow D^2,$$

which differ only in that $*$ has been replaced by the homotopy equivalent D^2 , and the maps by homotopic ones. The first pushout is a point. The second is S^2 .

So the strict pushout is not an invariant of the homotopy types of the diagram: it distinguishes $*$ from D^2 , which no homotopy-theoretic construction may do.

Remark 21.2. The failure is not an accident of the example. A pushout identifies points on the nose, and the identification is sensitive to exactly which points — information that a homotopy destroys. The repair is to glue not along X but along a cylinder on X , so that the gluing is made only up to a recorded path. That is the definition below, and everything else in the chapter follows the same principle.

21.2 Homotopy pushouts

Definition 21.3. The *homotopy pushout* of $Y \xleftarrow{f} X \xrightarrow{g} Z$ is the double mapping cylinder

$$Y \sqcup (X \times I) \sqcup Z \big/ ((x, 0) \sim f(x), (x, 1) \sim g(x)).$$

Theorem 21.4 (Invariance; recalled). *A map of diagrams which is a homotopy equivalence at each of the three objects induces a homotopy equivalence of homotopy pushouts.*

Example 21.5 (Suspension). The homotopy pushout of $* \leftarrow X \rightarrow *$ is $* \sqcup (X \times I) \sqcup *$ with each end collapsed, which is the unreduced suspension ΣX . Compare [Example 21.1](#): the strict pushout of that diagram was a point, and the homotopy pushout is $\Sigma S^1 = S^2$ in the case considered there — which is also the strict pushout of the second diagram. The homotopy pushout gives the answer that the well-behaved representative of the diagram gave.

Example 21.6 (Mapping cylinder). The homotopy pushout of $Y \xleftarrow{f} X \xrightarrow{\text{id}} X$ is the mapping cylinder M_f , which deformation retracts onto Y . So a diagram in which one leg is the identity has homotopy pushout the other object, as it must.

Exercise 21.7 (medium). Show that if $f: X \rightarrow Y$ is a cofibration then the natural map from the homotopy pushout of $Y \xleftarrow{f} X \xrightarrow{g} Z$ to the strict pushout is a homotopy equivalence.

Hint. The double mapping cylinder maps to the strict pushout by collapsing the cylinder coordinate. Use the homotopy extension property to build an inverse up to homotopy.

Solution. Collapsing $X \times I$ onto $X \times \{0\}$ and applying f defines a map from the homotopy pushout to $Y \sqcup_X Z$. Since f is a cofibration, so is the inclusion of Z into the double mapping cylinder along $X \times \{1\}$, and the collapse $X \times I \rightarrow X$ is a homotopy equivalence rel the ends. The HEP allows that homotopy equivalence of the pieces to be pushed through the gluing: the induced map of pushouts is then a homotopy equivalence, by the gluing lemma for cofibrations.

Remark 21.8 (Which case is left). Two of the three special cases have been done: taking both legs to a point gives the suspension ([Example 21.5](#)), and taking one leg to be an identity gives the other object ([Example 21.6](#)). The remaining case — one leg a general map f and the other a constant — is the one that produces the constructions of [Chapter 11](#), and identifying it is a short computation with [Definition 21.3](#): glue a cylinder on X to Y at one end and collapse the other. In the based setting the result is the reduced mapping cone Cf of [Exercise 11.6](#), and working out the identification is the natural exercise here.

21.3 Homotopy pullbacks

Everything above dualises. Where a homotopy pushout glues along a cylinder, recording a path in the source, a homotopy pullback matches up along a path space, recording a path in the target.

Definition 21.9 (The case we shall use). Let $f: X \rightarrow Y$ be a map of based spaces. Its *homotopy fibre* is

$$F_f = \{(x, \gamma) \in X \times Y^I : \gamma(0) = f(x), \gamma(1) = y_0\}.$$

A point of F_f is a point of X together with a *path* witnessing that its image is the basepoint — not a point whose image *is* the basepoint. That replacement of an equation by a path is the whole idea.

Proposition 21.10. $F_f \rightarrow X$ is a fibration with fibre ΩY , and if f is already a fibration then the inclusion of the strict fibre $f^{-1}(y_0) \hookrightarrow F_f$ is a homotopy equivalence.

Example 21.11. Taking $X = *$, the homotopy fibre of $* \rightarrow Y$ is the space of paths from y_0 to y_0 , that is ΩY . Taking f to be a fibration recovers the long exact sequence of [Corollary 14.3](#), now available for an arbitrary map: every map has a homotopy fibre, and hence a long exact sequence, even when it is not a fibration.

Exercise 21.12 (easy). Give a strict pullback that is not a homotopy pullback, and state a condition under which the two agree.

Solution. Pull back $* \rightarrow S^1 \leftarrow *$ where both maps hit the basepoint: the strict pullback is a point. The homotopy version replaces the equation $* = *$ by a path in S^1 between the two images, giving ΩS^1 , which has $\pi_0 = \mathbb{Z}$ and is not a point.

The two agree when one of the maps is a fibration: then the strict pullback is already a homotopy pullback, which is the pullback half of [Exercise 21.7](#) and the reason [Proposition 21.10](#) says what it does.

Remark 21.13 (The general formula). [Definition 21.9](#) is the special case $Z = *$ of a general construction. For a diagram $Y \xrightarrow{p} X \xleftarrow{q} Z$, dualising [Definition 21.3](#) means replacing the condition “ $p(y)$ equals $q(z)$ ” by “a path in X from $p(y)$ to $q(z)$ is given”, and assembling the triples into a subspace of $X^I \times Y \times Z$. Writing down the conditions those triples must satisfy, and checking that the result has the invariance property of [Theorem 21.4](#), is the natural exercise here; the homotopy fibre is the case in which one of the two objects is a point, and it is worth checking that the general formula does specialise to [Definition 21.9](#).

21.4 Exactness

The point of these constructions is that they turn into exact sequences.

Proposition 21.14. Let $F \rightarrow E \xrightarrow{p} B$ be a fibration of based spaces and Z any based space. Then

$$[Z, F] \longrightarrow [Z, E] \xrightarrow{p_*} [Z, B]$$

is exact as a sequence of pointed sets: the image of the first map is the preimage of the basepoint under the second.

Proof. A map $Z \rightarrow F \rightarrow E$ composes to a constant, so the image is contained in the kernel. Conversely, let $g: Z \rightarrow E$ with $p \circ g$ nullhomotopic, and let $H: Z \times I \rightarrow B$ be a nullhomotopy, $H_0 = pg$ and H_1 constant at b_0 . By the homotopy lifting property, H lifts to $\tilde{H}: Z \times I \rightarrow E$ with $\tilde{H}_0 = g$. Then $\tilde{H}_1$ maps Z into $p^{-1}(b_0) = F$, and $g \simeq \tilde{H}_1$, so $[g]$ is in the image. $\square$

Warning 21.15. Exactness here means image = kernel and nothing more. These are pointed sets, not groups, so there is no reason for either map to be injective, and no “exact sequence” reasoning beyond that one identity is available. When Z is a suspension the sets become groups and more can be said, which is what makes the Puppe sequence useful.

Remark 21.16 (The dual statement). Everything in [Proposition 21.14](#) dualises to a cofibration $X \xrightarrow{f} Y \rightarrow Cf$, giving exactness of

$$[Cf, Z] \longrightarrow [Y, Z] \longrightarrow [X, Z]$$

for every Z ; the proof replaces the lifting of a nullhomotopy by the extension of one over a cone, and [Example 17.4](#) already observed that a nullhomotopy of a composite *is* an extension over the cone. Writing out that dual argument is the natural exercise, and combined with [Remark 21.8](#) it turns a cofibration into an exact sequence of pointed sets.

Iterating either version produces the Puppe sequence, in which cones are replaced by suspensions and the pointed sets eventually become groups.

21.5 General homotopy colimits

Definition 21.17. For a diagram $D: J \rightarrow \mathbf{Top}$ the *homotopy colimit* is the realisation of the simplicial replacement

$$\mathrm{hocolim}_J D = \left| [n] \mapsto \coprod_{j_0 \rightarrow \cdots \rightarrow j_n} D(j_0) \right|,$$

the coproduct being over strings of n composable morphisms of J . Dually, the homotopy limit is the totalisation of the cosimplicial replacement.

Example 21.18. For J the category $\bullet \leftarrow \bullet \rightarrow \bullet$, unwinding [Definition 21.17](#) in low degrees gives exactly the double mapping cylinder of [Definition 21.3](#): the 0-simplices contribute the three objects and the 1-simplices contribute the two cylinders.

Remark 21.19 (Forward). [Definition 21.17](#) is the definition that [Chapter 29](#) works with. There the question is when a homotopy colimit can be computed over a smaller indexing category — a cofinality statement — and the answer turns a computation into a statement about the contractibility of certain comma categories. [Chapter 22](#) supplies the general framework in which “homotopy colimit” becomes “derived functor of colimit”, which is the conceptual explanation of everything in this chapter.

21.6 What to carry forward

- Strict colimits are not homotopy invariant ([Example 21.1](#) and [Remark 21.2](#)); glue along a cylinder instead ([Definition 21.3](#)).
- Suspension is a homotopy pushout ([Example 21.5](#)), and so is the mapping cylinder ([Example 21.6](#)).
- Along a cofibration, strict and homotopy pushouts agree ([Exercise 21.7](#)).
- The homotopy fibre replaces an equation by a path ([Definition 21.9](#)), so every map has a long exact sequence ([Example 21.11](#)). A strict pullback along a fibration is already a homotopy pullback ([Exercise 21.12](#)).
- Exactness of pointed sets, and only that ([Proposition 21.14](#) and [Warning 21.15](#)).
- Homotopy colimits in general are realisations of simplicial replacements ([Definition 21.17](#)).

Chapter 22

Model categories and derived functors

Chapter 21 repaired individual constructions one at a time. A model structure repairs all of them at once: it is a package of three classes of maps from which homotopy limits, homotopy colimits, derived functors and the homotopy category can all be extracted formally.

The axioms look unmotivated on first reading. They are best understood as the minimum needed for two things: that every object can be replaced by a well-behaved one, and that a functor which behaves correctly on the well-behaved objects has a derived functor. Section 22.3 is where that pays off.

22.1 Lifting and factorisation

Definition 22.1. A map $i: A \rightarrow B$ has the *left lifting property* against $p: X \rightarrow Y$ if every commutative square

$$\begin{array}{ccc} A & \longrightarrow & X \\ i \downarrow & \nearrow & \downarrow p \\ B & \longrightarrow & Y \end{array}$$

admits a diagonal filler making both triangles commute. A *weak factorisation system* on $\mathcal{C}$ is a pair of classes $(\mathcal{L}, \mathcal{R})$ such that $\mathcal{L}$ consists exactly of the maps with the left lifting property against all of $\mathcal{R}$, $\mathcal{R}$ consists exactly of the maps with the right lifting property against all of $\mathcal{L}$, and every map factors as a map in $\mathcal{L}$ followed by one in $\mathcal{R}$.

Exercise 22.2 (medium). Prove the *retract argument*: if $f = p \circ i$ with $i \in \mathcal{L}$ and $p \in \mathcal{R}$, and if f has the left lifting property against all of $\mathcal{R}$, then f is a retract of i .

Hint. Lift in the square whose top edge is i and whose right edge is p .

Solution. Write $f: A \rightarrow B$ as $A \xrightarrow{i} C \xrightarrow{p} B$ and consider the square with top edge i , left edge f , right edge p and bottom edge id_B . It commutes, since $pi = f$. As f has the left lifting property against $p \in \mathcal{R}$, there is a filler $r: B \rightarrow C$ with

$$r \circ f = i, \quad p \circ r = \text{id}_B.$$

Now the two squares

$$\begin{array}{ccccc} A & \xrightarrow{\text{id}} & A & \xrightarrow{\text{id}} & A \\ f \downarrow & & \downarrow i & & \downarrow f \\ B & \xrightarrow{r} & C & \xrightarrow{p} & B \end{array}$$

commute — the left one is $rf = i$, the right one is $pi = f$ — and the two horizontal composites are id_A and $pr = \text{id}_B$. That is exactly the statement that f is a retract of i in the arrow category.

Example 22.3. On **Set**, the monomorphisms and the epimorphisms form a weak factorisation system: every function factors through its image as an injection followed by a surjection, and an injection lifts against a surjection because a surjection of sets splits. On **Top**, the relative CW inclusions and the maps with the right lifting property against them — the trivial Serre fibrations — form another.

22.2 The axioms

Definition 22.4. A *model structure* on a complete and cocomplete category $\mathcal{M}$ consists of three classes of maps — weak equivalences $\mathcal{W}$, cofibrations $\mathcal{C}$, fibrations $\mathcal{F}$ — such that

- (a) $\mathcal{W}$ satisfies two-out-of-three: if two of f, g, gf lie in $\mathcal{W}$ then so does the third;
- (b) all three classes are closed under retracts;
- (c) $(\mathcal{C} \cap \mathcal{W}, \mathcal{F})$ and $(\mathcal{C}, \mathcal{F} \cap \mathcal{W})$ are weak factorisation systems.

An object is *cofibrant* if $\emptyset \rightarrow X$ is a cofibration, and *fibrant* if $X \rightarrow *$ is a fibration.

Example 22.5 (Two structures on spaces). **Top** carries at least two model structures.

In the *Serre* structure the weak equivalences are the weak homotopy equivalences, the fibrations are the Serre fibrations, and the cofibrant objects are the retracts of CW complexes. Every object is fibrant.

In the *Hurewicz* structure the weak equivalences are the honest homotopy equivalences and the fibrations are the Hurewicz fibrations. Every object is both fibrant and cofibrant.

The two differ, and the reason they differ is [Example 22.6](#).

Example 22.6 (Weak equivalence is weaker). The Warsaw circle — the closed topologist's sine curve with its two ends joined by an arc — has vanishing homotopy groups in every degree, so the inclusion of a point into it is a weak homotopy equivalence. It is not a homotopy equivalence: the space is not contractible.

So the two classes of [Example 22.5](#) are genuinely different, and Whitehead's theorem ([Theorem 13.1](#)) is precisely the statement that they agree on CW complexes.

Exercise 22.7 (medium). Show that in a model category the cofibrations are closed under cobase change: if $i: A \rightarrow B$ is a cofibration and $A \rightarrow A'$ is any map, then the induced map $A' \rightarrow A' \sqcup_A B$ is a cofibration. Dually, fibrations are closed under base change.

Hint. A cofibration is exactly a map with the left lifting property against every trivial fibration. Given a lifting problem for the pushout, restrict it along $A \rightarrow B$ to get one for i .

Solution. Let $p: X \rightarrow Y$ be a trivial fibration and consider a square with left edge $A' \rightarrow A' \sqcup_A B$ and right edge p . Composing the top edge with $B \rightarrow A' \sqcup_A B$ gives a square with left edge i and right edge p , which by hypothesis has a filler $B \rightarrow X$. That filler agrees on A with the composite $A \rightarrow A' \rightarrow X$, because both are obtained from the original square, so by the universal property of the pushout the two together define a map $A' \sqcup_A B \rightarrow X$. It is a filler for the original square, again by the universal property. Hence $A' \rightarrow A' \sqcup_A B$ lifts against every trivial fibration and is a cofibration.

The dual statement is the same argument with all arrows reversed.

22.3 Derived functors

A functor between model categories rarely preserves weak equivalences. The repair is to apply it only to well-behaved objects, and the theorem that makes this legitimate is the following.

Theorem 22.8 (Ken Brown's lemma). *Let $F: \mathcal{M} \rightarrow \mathcal{N}$ be a functor from a model category to a category with weak equivalences. If F sends trivial cofibrations between cofibrant objects to weak equivalences, then it sends all weak equivalences between cofibrant objects to weak equivalences.*

Exercise 22.9 (medium). Prove Theorem 22.8.

Hint. Given a weak equivalence $f: A \rightarrow B$ between cofibrant objects, factor the map $(f, \text{id}): A \sqcup B \rightarrow B$ as a cofibration followed by a trivial fibration.

Solution. Factor $A \sqcup B \rightarrow B$ as $A \sqcup B \xrightarrow{j} C \xrightarrow{q} B$ with j a cofibration and q a trivial fibration. The two inclusions give maps $j_A: A \rightarrow C$ and $j_B: B \rightarrow C$, both cofibrations because A and B are cofibrant and cofibrations are closed under cobase change. Since $qj_B = \text{id}_B$ is a weak equivalence and q is one, two-out-of-three makes j_B a trivial cofibration; likewise $qj_A = f$ is a weak equivalence, so j_A is a trivial cofibration. By hypothesis $F(j_A)$ and $F(j_B)$ are weak equivalences, and $F(q)F(j_B) = \text{id}$ gives that $F(q)$ is one by two-out-of-three; then $F(f) = F(q)F(j_A)$ is a composite of weak equivalences.

Definition 22.10. The *total left derived functor* of $F: \mathcal{M} \rightarrow \mathcal{N}$ is the left Kan extension of $\mathcal{M} \rightarrow \mathcal{N} \rightarrow \text{Ho } \mathcal{N}$ along the localisation $\mathcal{M} \rightarrow \text{Ho } \mathcal{M}$, written $\mathbb{L}F$.

Theorem 22.11 (Existence; recalled). *Let F be a left Quillen functor — a left adjoint preserving cofibrations and trivial cofibrations. Then $\mathbb{L}F$ exists, is computed by $\mathbb{L}F(X) = F(QX)$ for a cofibrant replacement $QX \rightarrow X$, and is an absolute left Kan extension in the sense of Definition 20.7.*

Warning 22.12. Definition 22.10 defines $\mathbb{L}F$ by a universal property; Theorem 22.11 says one way of computing it. The two must not be confused. A derived functor built by some other construction — a different replacement, a bar resolution, an explicit formula — is the same functor, because both satisfy the same universal property, and no comparison of the constructions is needed. That is the entire practical value of Definition 22.10.

Remark 22.13 (Deriving an adjunction). Theorem 22.11 supplies more than existence: it supplies *absoluteness*, and absoluteness is what makes derived functors compose and transport (Exercise 20.8).

That is the ingredient needed for the following question. Suppose $F \dashv G$ is an adjunction between categories with weak equivalences, and suppose $\mathbb{L}F$ and $\mathbb{R}G$ exist as absolute Kan extensions. Are the derived functors adjoint? The answer should not depend on how either was constructed — by [Warning 22.12](#) it cannot — so the proof must consist of transporting the unit and counit through the universal properties and checking the triangle identities.

Working that out is the natural exercise here. Its interest is not the conclusion, which everyone expects, but the fact that no model structure is needed for it: the argument is about Kan extensions, and [Theorem 22.11](#) enters only to guarantee the hypothesis.

22.4 Quillen adjunctions and equivalences

Definition 22.14. A *Quillen adjunction* is an adjunction $F \dashv U$ between model categories with F left Quillen. It is a *Quillen equivalence* if the derived functors $\mathbb{L}F \dashv \mathbb{R}U$ form an equivalence of homotopy categories.

Example 22.15. The adjunction $|\cdot| \dashv \text{Sing}$ of [Example 20.13](#), between simplicial sets with the Kan–Quillen structure and spaces with the Serre structure, is a Quillen equivalence. This is the precise form of the statement that simplicial sets model spaces, and it is what licenses the combinatorial approach of [Chapter 23](#).

Remark 22.16 (The criterion used in practice). [Definition 22.14](#) is not directly checkable: verifying that an equivalence of homotopy categories exists requires knowing the homotopy categories. What one checks instead is a pointwise condition on cofibrant and fibrant objects — that a map $Fm \rightarrow n$ is a weak equivalence exactly when its adjunct $m \rightarrow Un$ is.

That the pointwise condition is equivalent to [Definition 22.14](#) is a theorem, and proving it is the natural exercise at this stage. [Theorem 22.8](#) is the tool for both directions, and the hypothesis that the objects be cofibrant and fibrant respectively is not removable: without it the adjoints need not be comparable at all. A reader who proves it will thereafter be able to verify Quillen equivalences, including the one quoted in [Example 22.15](#), without ever computing a homotopy category.

Remark 22.17 (Mixing two structures). [Example 22.5](#) left two model structures on the same category, one with the weak equivalences one wants (weak homotopy equivalences, so that [Theorem 13.1](#) applies) and one with the fibrations one wants (Hurewicz fibrations, which have the lifting property against all spaces rather than only against disks). The obvious question is whether the two can be combined: weak equivalences from the coarse structure, fibrations from the fine one.

They can, and the resulting *mixed* structure has an attractive property — its cofibrant objects are exactly the spaces of CW homotopy type, which is the class of spaces the whole of Parts I–III implicitly worked with. Constructing it means verifying [Definition 22.4](#), and the work is in the factorisation axiom: factor first in one structure, then correct in the other. The same construction applies verbatim to chain complexes, mixing quasi-isomorphisms with the fibrations of the Hurewicz structure.

The trap, for whoever carries this out, is to guess the cofibrations. They are determined by the other two classes and are neither the union nor the intersection of the two given classes of cofibrations.

22.5 What to carry forward

- Weak factorisation systems and the retract argument ([Definition 22.1](#) and [Exercise 22.2](#)).
- The axioms ([Definition 22.4](#)); cofibrations are closed under cobase change and fibrations under base change ([Exercise 22.7](#)).
- Weak equivalence is strictly weaker than homotopy equivalence ([Example 22.6](#)); Whitehead's theorem says where they agree.
- Ken Brown's lemma ([Theorem 22.8](#)) — the reason cofibrancy is the right hypothesis.
- A derived functor is defined by a universal property and merely *computed* by replacement ([Definition 22.10](#) and [Warning 22.12](#)); it is an absolute Kan extension ([Theorem 22.11](#)), which is what lets adjunctions be derived ([Remark 22.13](#)).

Chapter 23

Simplicial sets and quasi-categories

A simplicial set is a combinatorial object that behaves like a space, and the translation is close enough that homotopy theory can be done entirely without topology. This chapter sets up the dictionary and then takes one step past it: weakening a single condition in the definition of a Kan complex produces quasi-categories, in which composition exists but is determined only up to homotopy.

The step is worth taking slowly, because it is where the obstruction-theoretic pattern of [Chapter 17](#) becomes purely combinatorial. [Section 23.4](#) says exactly how.

23.1 Simplicial sets

Definition 23.1. Let Δ be the category of nonempty finite ordinals $[n] = \{0 < \dots < n\}$ and order-preserving maps. A *simplicial set* is a functor $X: \Delta^{\text{op}} \rightarrow \mathbf{Set}$; we write $X_n = X([n])$, with face maps d_i and degeneracy maps s_i induced by the injections and surjections of Δ .

The representable $\Delta^n = \Delta(-, [n])$ has, by Yoneda, the property that maps $\Delta^n \rightarrow X$ are exactly n -simplices of X . Two subobjects matter:

$$\partial\Delta^n = \bigcup_i d_i\Delta^{n-1}, \quad \Lambda_k^n = \bigcup_{i \neq k} d_i\Delta^{n-1},$$

the boundary and the k -th *horn*, the latter being the boundary with one face removed.

Example 23.2. Two constructions from [Chapter 20](#) produce simplicial sets. The singular complex $\text{Sing}(Y)_n = \mathbf{Top}(|\Delta^n|, Y)$ of a space, and the nerve $N(\mathcal{C})_n = \mathbf{Cat}([n], \mathcal{C})$ of a category, whose n -simplices are strings of n composable morphisms.

23.2 Kan complexes

Definition 23.3. A simplicial set X is a *Kan complex* if every horn extends: for all $0 \leq k \leq n$, every map $\Lambda_k^n \rightarrow X$ factors through Δ^n .

Exercise 23.4 (medium). Show that $\text{Sing}(Y)$ is a Kan complex for every space Y .

Hint. By adjunction, a map $\Lambda_k^n \rightarrow \text{Sing}(Y)$ is a map $|\Lambda_k^n| \rightarrow Y$. Is there a retraction $|\Delta^n| \rightarrow |\Lambda_k^n|$?

Solution. By [Example 20.13](#) and the adjunction $|\cdot| \dashv \text{Sing}$, a horn $\Lambda_k^n \rightarrow \text{Sing}(Y)$ corresponds to a map $|\Lambda_k^n| \rightarrow Y$, and a filler corresponds to an extension over $|\Delta^n|$.

The horn $|\Lambda_k^n|$ is a deformation retract of the simplex $|\Delta^n|$ — project from a point placed just outside the missing face, exactly as in the retraction of [Exercise 11.2](#). Composing a retraction $|\Delta^n| \rightarrow |\Lambda_k^n|$ with the given map extends it. So every horn fills.

Theorem 23.5 (Recalled). *Every simplicial group is a Kan complex. The proof is an explicit induction: given a horn, one corrects its faces one at a time using the group operation, each correction leaving the already-matched faces unchanged.*

Remark 23.6. [Theorem 23.5](#) is the reason homological algebra is easier than homotopy theory. A simplicial abelian group is automatically fibrant, so no replacement is ever needed; under Dold–Kan such objects correspond to chain complexes, and this is why chain complexes can be manipulated naively while spaces cannot.

The same argument applies to any simplicial object with enough algebraic structure to correct one face at a time, so it applies in particular to the free simplicial module generated by an arbitrary simplicial set — an observation worth carrying out, since that construction is how one passes from a space to its homology in this language.

Example 23.7 (Not everything is Kan). $\Delta^1 = N([1])$ is not a Kan complex. Its 1-simplices are the arrows of the poset $0 \leq 1$, and the horn $\Lambda_0^2 \rightarrow \Delta^1$ that assigns the arrow $0 \rightarrow 1$ to one leg and the identity to the other would require a 1-simplex $1 \rightarrow 0$ to fill, which does not exist.

More generally, $N(\mathcal{C})$ is a Kan complex if and only if $\mathcal{C}$ is a groupoid: filling outer horns is exactly the ability to invert arrows. Fibrancy of a nerve is invertibility.

23.3 Quasi-categories

[Example 23.7](#) isolates the difficulty: outer horns demand inverses, which most categories do not have. Dropping only those conditions leaves something new.

Definition 23.8. A simplicial set X is a *quasi-category* if every *inner* horn fills: every map $\Lambda_k^n \rightarrow X$ with $0 < k < n$ extends over Δ^n .

Example 23.9. Every Kan complex is a quasi-category. Every nerve $N(\mathcal{C})$ is a quasi-category: an inner horn $\Lambda_k^n \rightarrow N(\mathcal{C})$ is a string of composable arrows with one composite unspecified, and the composite exists and is unique. So quasi-categories generalise both spaces and categories, which is their point.

Remark 23.10 (Composition, up to homotopy). In a quasi-category the filling of a Λ_1^2 horn is a composite: given $f: x \rightarrow y$ and $g: y \rightarrow z$ as 1-simplices, a filler is a 2-simplex whose remaining face h is “ $g \circ f$ ”. The filler is not unique, so h is determined only up to the relation generated by 2-simplices — a homotopy relation on 1-simplices.

Quotienting by it gives the *homotopy category* hX : objects the vertices, morphisms the homotopy classes of 1-simplices, composition induced by 2-simplices. That composition is well defined and associative is not formal; it requires filling a horn one dimension up, and identifying which horn is instructive — the associativity of three composable arrows is a statement about a 3-simplex.

Exercise 23.11 (easy). Show that the nerve $N: \mathbf{Cat} \rightarrow \mathbf{sSet}$ is fully faithful, and that $N(\mathcal{C})$ is 2-coskeletal: an n -simplex, for $n \geq 3$, is uniquely determined by its faces.

Hint. In the nerve, an n -simplex records n arrows placed end to end. How much of that data is already visible in the 2-skeleton?

Solution. A simplicial map $N(\mathcal{C}) \rightarrow N(\mathcal{D})$ is determined by its effect on 0- and 1-simplices, i.e. on objects and morphisms, and the compatibility with 2-simplices says exactly that composition is preserved; so N is fully faithful.

For coskeletality, an n -simplex is a string $c_0 \rightarrow \cdots \rightarrow c_n$, which is determined by its edges $c_i \rightarrow c_{i+1}$, all of which appear in the 1-skeleton; conversely a compatible family of faces of a potential n -simplex specifies exactly such a string, and the 2-simplices impose precisely the condition that the composites agree. So for $n \geq 3$ no new information is carried.

Exercise 23.12 (easy). Show that $hN(\mathcal{C}) \cong \mathcal{C}$ for every small category $\mathcal{C}$.

Solution. Vertices of $N(\mathcal{C})$ are objects of $\mathcal{C}$ and 1-simplices are morphisms. Two morphisms are homotopic in the sense of [Remark 23.10](#) when some 2-simplex exhibits one as the composite of the other with an identity — that is, when they are equal. So the morphisms of $hN(\mathcal{C})$ are the morphisms of $\mathcal{C}$, and composition agrees because a 2-simplex of a nerve is a commuting triangle.

Remark 23.13 (The comparison in general). [Exercise 23.12](#) is one half of a comparison, and the other half is the statement that h and N are adjoint, $h \dashv N$. Setting up that adjunction means producing, for a quasi-category X and a category $\mathcal{C}$, a natural bijection between simplicial maps $X \rightarrow N\mathcal{C}$ and functors $hX \rightarrow \mathcal{C}$. [Exercise 23.11](#) is what makes it tractable: a map into a 2-coskeletal object is determined by its 2-skeleton, so only vertices, edges and triangles are in play, and those are exactly the data out of which hX was built.

The trap is to assume the composition in hX is associative without producing the 3-simplex that proves it. Anyone constructing the adjunction should say which horn is being filled, and why it is an inner one.

23.4 Filling is obstruction theory

Remark 23.14. [Remark 17.17](#) promised that the pattern of [Chapter 17](#) would reappear here, and this is the place.

In [Chapter 17](#) a map was built over a CW complex skeleton by skeleton; at each stage the obstruction to continuing over an $(n+1)$ -cell was the class of the composite $S^n \rightarrow X^n \rightarrow Y$, and the map extended precisely when that class vanished. The obstruction lived in $\pi_n(Y)$ and was recorded, when nonzero, as a cohomology class.

Here a map is built over a simplicial set simplex by simplex, and the question at each stage is whether a horn $\Lambda_k^n \rightarrow X$ extends over Δ^n . The structural difference is that no obstruction *group* appears: the definition of a Kan complex simply demands that every such extension exist, and the definition of a quasi-category demands it for the inner horns.

So the two theories differ in what they do with the obstruction. Obstruction theory measures it; the fibrancy conditions of this chapter abolish it by hypothesis, and the price is that only the objects satisfying the hypothesis may be used. That is exactly the role that “cofibrant” and “fibrant” play in [Chapter 22](#), and it is why replacement is unavoidable there: one has to *move* to an object where the obstructions vanish.

23.5 Left and right fibrations

Definition 23.15. A map $p: X \rightarrow Y$ of simplicial sets is a *left fibration* if it has the right lifting property against $\Lambda_k^n \hookrightarrow \Delta^n$ for $0 \leq k < n$, and a *right fibration* if it has it for $0 < k \leq n$. It is a *Kan fibration* if it has it for all $0 \leq k \leq n$.

Warning 23.16 (Handedness is not a convention). The two classes are genuinely different, and no argument transfers from one to the other by symmetry alone: reversing the direction of the simplices exchanges them, but that operation also reverses the direction of every arrow in sight, so a statement about right fibrations becomes a statement about left fibrations over the opposite quasi-category.

The practical consequence is that in any argument one must record which horns are available. An argument that fills Λ_0^n is not available for a right fibration, and the commonest error in this area is to use a lifting property that the hypothesis does not supply.

Definition 23.17. A 1-simplex f of a quasi-category X is an *isomorphism* if its class in $\mathrm{h}X$ is invertible.

Remark 23.18 (What one-sidedness buys). Isomorphisms admit a lifting characterisation — a 1-simplex is invertible exactly when certain horns involving it can be filled — and that characterisation is what converts the one-sidedness of [Definition 23.15](#) into homotopy-theoretic content. Two statements of that kind are worth knowing: that a right fibration reflects isomorphisms (if $p(f)$ is invertible then so is f), and that an isomorphism in the base with a prescribed lift of its source lifts to an isomorphism upstairs.

Both are proved by choosing the right horn and filling it, and both fail for the opposite handedness, which makes them a good test of [Warning 23.16](#). Anyone working them out should write down, for each filling, which Λ_k^n is being used and check it against [Definition 23.15](#).

23.6 What to carry forward

- Simplicial sets, horns, and the two fibrancy conditions ([Definitions 23.1](#), [23.3](#) and [23.8](#)).
- $\mathrm{Sing}(Y)$ is Kan by the same retraction that proved the HEP ([Exercise 23.4](#)); simplicial groups are Kan for free ([Theorem 23.5](#) and [Remark 23.6](#)).
- A nerve is Kan exactly for a groupoid, and always a quasi-category ([Examples 23.7](#) and [23.9](#)).
- Composition in a quasi-category is defined up to homotopy by inner horn filling ([Remark 23.10](#)); the nerve is 2-coskeletal ([Exercise 23.11](#)).
- Horn filling is obstruction theory with the obstructions abolished by hypothesis rather than measured ([Remark 23.14](#)).
- Left and right fibrations are not interchangeable; see [Warning 23.16](#) and [Remark 23.18](#).

Chapter 24

Sullivan models

Rationally, a simply connected space is a piece of algebra. This chapter makes that precise: to every space one associates a commutative differential graded algebra, and for a large class of spaces a particularly economical model of that algebra — the minimal Sullivan model — determines the rational homotopy type completely, with the rational homotopy groups readable off its generators.

Chapter 19 promised that the parity distinguishing even from odd spheres would reappear as a feature of such a model. Example 24.5 delivers it, and the reader should compare that example with Theorem 19.10 before going further.

24.1 Commutative models

Theorem 24.1 (Polynomial de Rham theory; recalled). *There is a functor A_{PL} from spaces to commutative differential graded algebras over $\mathbb{Q}$, with a natural isomorphism*

$$H^*(A_{\text{PL}}(X)) \cong H^*(X; \mathbb{Q})$$

of graded algebras. Two spaces are rationally equivalent if their A_{PL} 's are connected by a chain of quasi-isomorphisms.

The point is that $A_{\text{PL}}(X)$ is commutative on the nose, unlike the singular cochain algebra. Over $\mathbb{Q}$ that rigidity is available, and everything below exploits it.

Definition 24.2. A *Sullivan algebra* is a cdga of the form $(\Lambda V, d)$, free as a graded commutative algebra, whose space of generators admits an increasing filtration

$$V(0) \subseteq V(1) \subseteq \cdots, \quad V = \bigcup_k V(k),$$

with $d(V(k)) \subseteq \Lambda(V(k-1))$. It is *minimal* if in addition $d(V) \subseteq \Lambda^{\geq 2}V$: the differential of a generator has no linear term.

Warning 24.3 (Free is not the condition). Being free as a graded commutative algebra is only half of Definition 24.2. The filtration is the other half, and it is the half that matters: it says the generators can be adjoined in an order such that each new differential is expressible in the ones already present. Without it there is no induction to run, and every argument in this chapter is an induction over that filtration.

A useful consequence to keep in hand: if all generators sit in degree one, then $d(V(0)) \subseteq \Lambda(V(-1)) = \mathbb{Q}$, which has nothing in degree two, so d vanishes on $V(0)$

— some generator must be a cocycle. Any candidate algebra in which no generator is closed therefore fails to be Sullivan, however free it may be.

Theorem 24.4 (Existence and uniqueness; recalled). *Every simply connected cdga of finite type admits a minimal Sullivan model, unique up to isomorphism: a quasi-isomorphism $(\Lambda V, d) \rightarrow A$ with $(\Lambda V, d)$ minimal. For a simply connected space X of finite $\mathbb{Q}$ -type, the minimal model of $A_{\text{PL}}(X)$ is called the minimal model of X .*

Example 24.5 (Spheres, and the promise of Chapter 19). Let $n \geq 2$.

n odd. Take $(\Lambda(x), 0)$ with $|x| = n$. Since n is odd, $x^2 = 0$ automatically, so the algebra is $\mathbb{Q}\{1, x\}$ with zero differential and its cohomology is $H^*(S^n; \mathbb{Q})$. It is minimal, d being zero.

n even. Now $(\Lambda(x), 0)$ will not do: x has even degree, so $\Lambda(x) = \mathbb{Q}[x]$ is a polynomial algebra with $x^2 \neq 0$, whereas $H^{2n}(S^n; \mathbb{Q}) = 0$. The remedy is to adjoin a second generator to kill x^2 :

$$(\Lambda(x, y), dx = 0, dy = x^2), \quad |x| = n, |y| = 2n - 1.$$

The degree of y is forced — d raises degree by one and $|x^2| = 2n$.

Exercise 24.6 (easy). Verify that the algebra of Example 24.5 for n even has the cohomology of S^n , and that it is minimal.

Hint. y has odd degree, so $y^2 = 0$ and every element is $p(x) + q(x)y$. Compute d of such an element.

Solution. Since $|y|$ is odd, $y^2 = 0$ and the algebra has basis $\{x^k, x^k y\}$. The differential is determined by $dx = 0$, $dy = x^2$ and the Leibniz rule:

$$d(x^k) = 0, \quad d(x^k y) = x^{k+2}.$$

So the cocycles are spanned by the x^k , and the coboundaries by $x^2, x^3, \dots$. Hence

$$H^*(\Lambda(x, y), d) = \mathbb{Q}\{1, x\} = \mathbb{Q}[x]/(x^2),$$

which is $H^*(S^n; \mathbb{Q})$ for n even. Minimality holds because $dy = x^2 \in \Lambda^{\geq 2}$ and $dx = 0$, so no differential has a linear term.

Remark 24.7 (The promise, kept). Compare Example 24.5 with Theorem 19.10. For n odd the model has one generator, in degree n ; for n even it has two, in degrees n and $2n - 1$. Those are exactly the degrees in which $\pi_*(S^n) \otimes \mathbb{Q}$ is nonzero.

That is not a coincidence but the content of Theorem 24.16 below: the generators of the minimal model are dual to the rational homotopy groups. So the extra rational homotopy class of an even sphere, which Warning 19.5 detected as an extra generator in the cohomology of a loop space and Remark 19.9 identified as a Whitehead square, is here simply the generator y that had to be adjoined to kill x^2 . Three descriptions of one phenomenon, and the algebraic one is the shortest.

24.2 Relative models and fibrations

Definition 24.8. A *relative Sullivan algebra* over a cdga B is a cdga of the form $(B \otimes \Lambda V, d)$ with V filtered as in Definition 24.2 and $d(V(k)) \subseteq B \otimes \Lambda(V(k-1))$.

Relative Sullivan algebras are the cofibrations of the model structure on cdgas, and they model fibrations: a fibration $F \rightarrow E \rightarrow B$ of simply connected spaces gives rise to a relative Sullivan algebra over a model of B , whose quotient by the augmentation ideal of B models F .

Exercise 24.9 (medium). Let B be a cdga with zero differential and let $f \in B$ be a non-zero-divisor of even degree. Show that

$$(B \otimes \Lambda(w), dw = f), \quad |w| = |f| - 1,$$

is quasi-isomorphic to $B/(f)$.

Hint. $|w|$ is odd, so every element is $b_0 + b_1w$. Compute the differential and read off cocycles and coboundaries.

Solution. Since $|w|$ is odd, $w^2 = 0$ and a general element is $b_0 + b_1w$ with $b_i \in B$. The differential is $d(b_0 + b_1w) = \pm b_1f$. Hence the cocycles are the elements with $b_1f = 0$, which by the non-zero-divisor hypothesis means $b_1 = 0$: the cocycles are B . The coboundaries are $\{bf\} = (f)$. So the cohomology is $B/(f)$, and the quotient map is the required quasi-isomorphism.

Warning 24.10 (Cofibrancy is what makes the quotient correct). The statement that the fibre is modelled by $\mathbb{Q} \otimes_B (B \otimes \Lambda V)$ requires the model to be a *relative Sullivan algebra* — a cofibration — and not merely some algebra mapping to another. The tensor product is a non-derived functor; applied to a non-cofibrant map it computes the wrong answer, in exactly the way that the strict pushouts of [Example 21.1](#) computed the wrong homotopy type.

This is the same phenomenon as [Warning 22.12](#): the correct object is defined by a universal property, and only cofibrant representatives compute it.

Remark 24.11 (Building a relative model). The recipe for modelling a map $B \rightarrow A$ is the one [Example 24.5](#) followed. Adjoin generators to B to build a relative Sullivan algebra mapping quasi-isomorphically to A : first generators to hit the cohomology of A that B misses, then generators to kill the relations that B imposes and A does not, the differential of each new generator being an element already present. [Exercise 24.9](#) is the case where a single relation has to be killed.

Two things are then worth computing for practice: the cohomology of the resulting relative algebra, which must reproduce $H^*(A)$; and the cohomology of its quotient by the augmentation ideal of B , which by [Warning 24.10](#) is a model for the fibre of whichever map of spaces the algebra came from. Doing this for a map of polynomial algebras — say one sending a generator to a power of another — is the natural exercise, and the answer should be recognisable: it will be the cohomology of a space met in [Chapter 19](#).

24.3 Spatial realisation

The passage from algebra back to topology is by a simplicial construction.

Definition 24.12. Let $\Omega^*(\Delta^n)$ denote Sullivan's cdga of polynomial differential forms on the standard n -simplex. For a cdga A , the *spatial realisation* is the simplicial set

$$\langle A \rangle_n = \text{Hom}_{\text{cdga}}(A, \Omega^*(\Delta^n)).$$

Proposition 24.13. $\langle A \rangle$ is a Kan complex.

Proof sketch. The restriction $\Omega^*(\Delta^n) \rightarrow \Omega^*(\Lambda_k^n)$ is surjective — polynomial forms extend from a horn to the simplex — so a horn in $\langle A \rangle$, which is a compatible family of cdga maps into the forms on the faces, extends to a cdga map into the forms on Δ^n . That is precisely the horn-filling condition of [Definition 23.3](#). $\square$

Remark 24.14. [Proposition 24.13](#) is the third appearance of one argument. In [Exercise 23.4](#) fibrancy of Sing came from a retraction of the topological simplex onto a horn; in [Theorem 23.5](#) it came from an algebraic structure allowing faces to be corrected; here it comes from surjectivity of a restriction map. In each case fibrancy is the statement that some extension problem always has a solution.

Example 24.15 (The realisation of a single generator). Let $A = (\Lambda(x), 0)$ with $|x| = k \geq 2$. A cdga map $A \rightarrow \Omega^*(\Delta^n)$ is determined by a closed form of degree k , so

$$\langle A \rangle_n = \{\text{closed } k\text{-forms on } \Delta^n\},$$

a simplicial abelian group; its homotopy is the degree- k cohomology of a simplex in each dimension, and $\langle A \rangle \simeq K(\mathbb{Q}, k)$. The single generator of V in degree k has produced a space with a single rational homotopy group, in degree k .

Theorem 24.16 (Sullivan; recalled). *Let X be simply connected of finite $\mathbb{Q}$ -type with minimal model $(\Lambda V, d)$. Then for every $k \geq 2$*

$$\pi_k(X) \otimes \mathbb{Q} \cong \text{Hom}_{\mathbb{Q}}(V^k, \mathbb{Q}).$$

Remark 24.17 (What has to be checked, and where minimality enters). [Theorem 24.16](#) is quoted here; proving it in the form $\pi_k \langle A \rangle \cong \text{Hom}(V^k, \mathbb{Q})$ is the natural exercise, and the ingredients are all in place: [Proposition 24.13](#) makes $\langle A \rangle$ fibrant, so its homotopy groups may be computed simplicially; and [Example 24.15](#) is the case of a single generator, which is the base of the induction over the Sullivan filtration.

Minimality is not decoration. Without it a generator may have a linear term in its differential, and such a generator contributes nothing to homotopy while still enlarging V ; the statement is then simply false. Anyone proving the theorem should locate the step where $d(V) \subseteq \Lambda^{\geq 2}V$ is used, and check what the induction produces without it.

Example 24.18 (Spheres, one last time). Apply [Theorem 24.16](#) to [Example 24.5](#). For n odd, $V = \mathbb{Q}$ concentrated in degree n , so $\pi_*(S^n) \otimes \mathbb{Q}$ is $\mathbb{Q}$ in degree n and zero elsewhere. For n even, V has one dimension in degree n and one in degree $2n - 1$, so the rational homotopy is $\mathbb{Q}$ in those two degrees.

That is [Theorem 19.10](#), obtained by writing down a two-generator algebra. Compare the effort with the spectral sequence argument sketched in [Warning 19.11](#).

24.4 What to carry forward

- A_{PL} makes rational cohomology into a commutative algebra ([Theorem 24.1](#)); minimal models exist and are unique ([Theorem 24.4](#)).
- Sullivan means free *plus* a filtration ([Definition 24.2](#) and [Warning 24.3](#)); in degree one the filtration forces some generator to be closed.
- The minimal models of spheres, and the parity ([Example 24.5](#), [Exercise 24.6](#), and [Remark 24.7](#)) — the promise of [Chapter 19](#).

- Relative Sullivan algebras model fibrations, and the quotient computes the fibre only because they are cofibrant ([Definition 24.8](#) and [Warning 24.10](#)); the Koszul complex is the basic step ([Exercise 24.9](#)).
- Rational homotopy is dual to the generators ([Theorem 24.16](#)), with minimality essential ([Remark 24.17](#)).

Chapter 25

Operads and homotopy Lie algebras

An operad records the operations of an algebraic structure together with the ways they can be composed. The formalism costs something to set up and repays it twice over: it makes “algebra up to homotopy” definable, and it makes the rational homotopy of a space into an algebraic object with its own structure theory.

This chapter does the minimum of the first and the beginning of the second. The target is the homotopy Lie algebra of a space — $\pi_*(\Omega X) \otimes \mathbb{Q}$ with the Whitehead product — and the dichotomy it produces: a simply connected finite complex is either *elliptic*, with finite-dimensional rational homotopy, or *hyperbolic*, with rational homotopy growing exponentially. There is nothing in between.

25.1 Operads

Definition 25.1. An *operad* $\mathcal{P}$ in vector spaces consists of vector spaces $\mathcal{P}(n)$ with an action of the symmetric group $\mathbb{S}_n$, a unit $\text{id} \in \mathcal{P}(1)$, and composition maps

$$\circ_i: \mathcal{P}(m) \otimes \mathcal{P}(n) \longrightarrow \mathcal{P}(m+n-1), \quad 1 \leq i \leq m,$$

— plug the second operation into the i -th input of the first — satisfying associativity, unitality and equivariance axioms.

Think of $\mathcal{P}(n)$ as the space of n -ary operations of some kind of algebra, with $\mathbb{S}_n$ permuting the inputs.

Example 25.2 (The two basic operads). $\text{Ass}(n) = k[\mathbb{S}_n]$, the regular representation, with $\circ_i$ given by substitution of words: an algebra over Ass is an associative algebra, and the $n!$ basis elements record the $n!$ ways of ordering the inputs of an n -fold product.

$\text{Com}(n) = k$, the trivial representation: an algebra over Com is a commutative associative algebra, and there is only one n -ary operation because the order of the inputs does not matter.

Example 25.3 (Comparing them). There is a morphism of operads $\text{Ass} \rightarrow \text{Com}$: in each arity it is the map $k[\mathbb{S}_n] \rightarrow k$ sending every permutation to 1. It is compatible with $\circ_i$, because substituting words and then forgetting the order gives the same result as forgetting the order throughout.

That is the direction that works, and it corresponds to the evident fact that a commutative algebra is in particular an associative one.

Warning 25.4 (Arity-wise maps are not operad morphisms). The opposite direction is tempting and wrong. In each arity there is an evident equivariant map $\text{Com}(n) \rightarrow \text{Ass}(n)$, embedding the trivial representation in the regular one by sending the generator to the sum of all permutations. It is a map of $\mathbb{S}$ -modules, so it looks like a candidate.

It is not an operad morphism, and the reason is a counting mismatch: composing two such sums produces fewer terms than the sum attached to the composite arity, because the sum over $\mathbb{S}_m$ composed with the sum over $\mathbb{S}_n$ does not exhaust $\mathbb{S}_{m+n-1}$. Checking this on the smallest interesting case — two binary operations composed into a ternary one — is a short computation and a good corrective: it shows that being a morphism of $\mathbb{S}$ -modules in every arity is strictly weaker than respecting $\circ_i$.

25.2 The homotopy Lie algebra

Definition 25.5. For a simply connected space X , the graded vector space

$$L_X = \pi_*(\Omega X) \otimes \mathbb{Q}$$

carries a graded Lie bracket, induced by the Whitehead product of [Definition 19.6](#) under the isomorphism $\pi_k(\Omega X) \cong \pi_{k+1}(X)$. It is the *homotopy Lie algebra* of X .

Example 25.6. For $X = S^n$ with n odd, L_X is one-dimensional, spanned by a class in degree $n-1$, with zero bracket. For n even it is two-dimensional, spanned by classes in degrees $n-1$ and $2n-2$, and the bracket of the first with itself is the second — this is the Whitehead square of [Remark 19.9](#), appearing now as a Lie bracket. [Example 24.5](#) recorded the same two cases as one generator versus two.

Theorem 25.7 (Milnor–Moore; recalled). *For X simply connected of finite $\mathbb{Q}$ -type there is an isomorphism of graded Hopf algebras*

$$U(L_X) \cong H_*(\Omega X; \mathbb{Q}),$$

where U is the universal enveloping algebra.

So the homology of a loop space is determined by the homotopy Lie algebra, and conversely — L_X is the space of primitives, in the sense of [Definition 10.6](#). Counting one therefore counts the other, and that is the mechanism behind the dichotomy.

Proposition 25.8 (Free Lie algebras). *For a graded vector space V there is a free graded Lie algebra $\mathbb{L}(V)$, and*

$$U\mathbb{L}(V) \cong T(V),$$

the free associative algebra on V .

Exercise 25.9 (medium). Prove [Proposition 25.8](#), and explain why the proof says nothing about the size of $\mathbb{L}(V)$ itself.

Hint. U is left adjoint to the functor sending an associative algebra to its commutator Lie algebra, and $\mathbb{L}$ is left adjoint to the forgetful functor to graded vector spaces.

Solution. Both $U\mathbb{L}$ and T are left adjoints of the same functor: for an associative algebra A ,

$$\text{Alg}(U\mathbb{L}(V), A) \cong \text{Lie}(\mathbb{L}(V), A_{\text{Lie}}) \cong \text{Vect}(V, A) \cong \text{Alg}(T(V), A),$$

naturally in A ; a left adjoint is unique up to isomorphism, so $U\mathbb{L}(V) \cong T(V)$.

The computation determines the enveloping algebra completely and says nothing directly about $\mathbb{L}(V)$, which sits inside $T(V)$ as the sub-Lie-algebra generated by V — a much smaller and less obvious object.

Theorem 25.10 (Graded Poincaré–Birkhoff–Witt; recalled). *Work over a field of characteristic zero, and let L be a graded Lie algebra. Then $U(L)$ agrees, as a graded vector space, with the free graded commutative algebra generated by L : a polynomial algebra on the generators of even degree, tensored with an exterior algebra on those of odd degree. In terms of Poincaré series, writing $L_k = \dim L_k$,*

$$\sum_n \dim U(L)_n t^n = \prod_{k \text{ even}} (1 - t^k)^{-L_k} \cdot \prod_{k \text{ odd}} (1 + t^k)^{L_k}.$$

Warning 25.11 (The parity is not optional). The two products in [Theorem 25.10](#) differ, and using the ungraded form of the theorem — polynomial on everything — gives wrong dimensions in every odd degree. This is the same parity that separated the two cases of [Example 24.5](#) and produced the extra generator of [Warning 19.5](#); by now it should be automatic.

Remark 25.12 (Counting a free Lie algebra). Combining [Proposition 25.8](#) and [Theorem 25.10](#) gives a way to compute the dimensions of a free graded Lie algebra. The Poincaré series of $T(V)$ is elementary: $\sum_n (\dim V)^n t^n$ suitably graded, a geometric series. The Poincaré series of $U\mathbb{L}(V)$ is that same series, by [Proposition 25.8](#). And [Theorem 25.10](#) expresses it as a product over the unknown dimensions $\dim \mathbb{L}(V)_k$. Equating the two and inverting determines those dimensions one degree at a time.

Carrying this out for V of dimension two — the case $X = S^n \vee S^m$, where the homotopy Lie algebra is free on two generators — is the natural exercise. Two things are worth noticing while doing it. The geometric series has radius of convergence less than one as soon as $\dim V \geq 2$, which is what forces the answer to grow; and a lower bound suffices for the qualitative conclusion, so there is no need for a closed formula.

25.3 The dichotomy

Definition 25.13. A simply connected finite CW complex X is *elliptic* if $\dim_{\mathbb{Q}} \pi_*(X) \otimes \mathbb{Q} < \infty$, and *hyperbolic* otherwise.

Example 25.14. Spheres are elliptic, by [Example 24.18](#): their rational homotopy has dimension one or two. So are $\mathbb{CP}^n$, Lie groups, and homogeneous spaces — all of them have minimal models with finitely many generators, and [Theorem 24.16](#) then bounds the rational homotopy.

Theorem 25.15 (Dichotomy; recalled). *A simply connected finite complex is either elliptic, or hyperbolic with*

$$\sum_{k \leq r} \dim_{\mathbb{Q}} \pi_k(X) \otimes \mathbb{Q}$$

growing exponentially in r . No intermediate growth occurs.

Remark 25.16 (Where the simplest hyperbolic example is). A wedge of two spheres has free homotopy Lie algebra — freeness is what a wedge does, exactly as π_1 of a wedge of circles is free — so [Remark 25.12](#) applies to it directly. Working out the growth for $S^n \vee S^m$ shows that the simplest finite complex beyond a single sphere is already hyperbolic, and that its rational homotopy is enormous: a two-cell complex with an infinite-dimensional rational homotopy, growing exponentially.

That is worth pausing on. [Chapter 14](#) could compute $\pi_3(S^2)$ and stopped; here one gets the full rational homotopy of a whole family of spaces, at the cost of tensoring with $\mathbb{Q}$ and of the machinery of Parts III and IV. The trade is characteristic of the subject.

Remark 25.17 (Forward). Three threads continue into Part V. The composition product of [Definition 25.1](#) is made explicit in [Chapter 30](#), where it becomes a computation with induced representations. The notion of an algebra “up to homotopy” over an operad is constructed there too, by a duality that turns $\mathcal{P}$ into a cooperad. And [Chapter 31](#) returns to [Theorem 25.7](#) and to U , where the four standard properties of the universal enveloping algebra are derived from the single adjunction used in [Exercise 25.9](#).

25.4 What to carry forward

- Operads ([Definition 25.1](#)); Ass and Com ([Example 25.2](#)), with $\text{Ass} \rightarrow \text{Com}$ the direction that works ([Example 25.3](#) and [Warning 25.4](#)).
- The homotopy Lie algebra ([Definition 25.5](#)), and Milnor–Moore identifying its enveloping algebra with $H_*(\Omega X; \mathbb{Q})$ ([Theorem 25.7](#)).
- $U\mathbb{L}(V) = T(V)$ from adjunctions ([Exercise 25.9](#)); graded PBW converts that into dimensions ([Theorem 25.10](#), [Warning 25.11](#), and [Remark 25.12](#)).
- The elliptic/hyperbolic dichotomy ([Definition 25.13](#) and [Theorem 25.15](#)); wedges are the simplest hyperbolic examples ([Remark 25.16](#)).

Part V

Research preparation

Chapter 26

Integral characteristic classes

Chapter 12 worked mod 2, where the arithmetic is easy and the conclusions are about stable triviality. This chapter works over $\mathbb{Z}$, where signs and torsion both matter and the conclusions are correspondingly sharper.

The organising object is the Euler class. It reduces mod 2 to the top Stiefel–Whitney class, so it refines the mod-2 theory; it is a torsion class in odd rank, so it is detected by a Bockstein; and it is multiplicative, so it can be computed on sums of line bundles. Those three facts are the three edges of the triangle Remark 12.12 promised, and Section 26.2 draws them.

26.1 The Euler class

Definition 26.1. Let $E \rightarrow B$ be an oriented real vector bundle of rank k . Its *Euler class* $e(E) \in H^k(B; \mathbb{Z})$ is the restriction of the Thom class (Theorem 16.12) to the zero section; equivalently it is the obstruction appearing in the Gysin sequence of Theorem 16.9.

Proposition 26.2 (Basic properties; recalled). (a) *Naturality:* $e(f^*E) = f^*e(E)$.

(b) *Multiplicativity:* $e(E \oplus F) = e(E) \smile e(F)$.

(c) *If E admits a nowhere-zero section then $e(E) = 0$.*

(d) *The mod-2 reduction of $e(E)$ is the top Stiefel–Whitney class: $\rho e(E) = w_k(E)$.*

Part (c) is the reason for the name: for the tangent bundle of a closed oriented manifold, $e(TM)$ evaluated on the fundamental class is the Euler characteristic, and it vanishes exactly when a nowhere-zero vector field exists.

Example 26.3 (Line bundles). A complex line bundle L over B is classified by a map $B \rightarrow \mathbb{CP}^\infty$, and by Exercise 14.6 that is the same as a class in $H^2(B; \mathbb{Z})$. Under this correspondence

$$e(L) = c_1(L) \in H^2(B; \mathbb{Z}),$$

and every class in $H^2(B; \mathbb{Z})$ arises: line bundles over B are in bijection with $H^2(B; \mathbb{Z})$. So in rank one the realisation question has the simplest possible answer.

Over $\mathbb{CP}^\infty$ itself the tautological line bundle has $e = c$, the generator of H^2 , and $H^*(\mathbb{CP}^\infty; \mathbb{Z}) = \mathbb{Z}[c]$ by Example 16.2.

Exercise 26.4 (easy). Show that $e(L \otimes L') = e(L) + e(L')$ for complex line bundles.

Hint. Tensor product of line bundles corresponds to a map $\mathbb{CP}^\infty \times \mathbb{CP}^\infty \rightarrow \mathbb{CP}^\infty$. Which map, given that $\mathbb{CP}^\infty = K(\mathbb{Z}, 2)$?

Solution. Line bundles are classified by $[B, \mathbb{CP}^\infty]$, and tensor product makes this set into a group, corresponding to a map $\mu: \mathbb{CP}^\infty \times \mathbb{CP}^\infty \rightarrow \mathbb{CP}^\infty$. Since $\mathbb{CP}^\infty = K(\mathbb{Z}, 2)$ (Exercise 14.6), that classifying set is $H^2(B; \mathbb{Z})$ and μ induces its addition — this is the H-space structure of Example 10.2 and Remark 10.11. Hence the class classifying $L \otimes L'$ is the sum of those classifying L and L' , which by Example 26.3 says $e(L \otimes L') = e(L) + e(L')$.

Exercise 26.5 (medium). Let E be an oriented real vector bundle of odd rank. Show $2e(E) = 0$.

Hint. Find a bundle automorphism of E that reverses the orientation, and apply naturality.

Solution. Let $\sigma: E \rightarrow E$ be fibrewise multiplication by -1 . It is a bundle automorphism covering the identity of B , and on each fibre it has determinant $(-1)^{\text{rank}} = -1$, since the rank is odd. So σ carries the given orientation to the opposite one.

Reversing the orientation of a bundle negates its Euler class, directly from Definition 26.1: the Thom class changes sign. On the other hand σ is an isomorphism of bundles, so by naturality it preserves the Euler class. Combining, $e(E) = -e(E)$, that is $2e(E) = 0$.

Example 26.6 (Chern classes). Let E be a complex bundle of rank n . By Example 16.7 the ring $H^*(\mathbb{P}(E); \mathbb{Z})$ is free over $H^*(B; \mathbb{Z})$ on the classes $1, x, \dots, x^{n-1}$, so there is a unique relation

$$x^n = -c_1(E)x^{n-1} - \dots - c_n(E),$$

and its coefficients define the Chern classes. The top one is the Euler class of the underlying oriented real bundle: $c_n(E) = e(E_{\mathbb{R}})$.

26.2 The triangle

Remark 12.12 promised a relation between three things: the mod-2 classes w_i , the Bockstein of Chapter 9, and the integral Euler class. Here are the three edges.

Edge one: e reduces to w_{top} . This is Proposition 26.2(d). So the mod-2 theory sees the Euler class, but only its reduction; anything of odd order or any sign information is lost.

Edge two: in odd rank e is 2-torsion. This is Exercise 26.5. A 2-torsion integral class is exactly what the Bockstein produces: the coefficient sequence $0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{F}_2 \rightarrow 0$ of Definition 9.11 has connecting map $\tilde{\beta}: H^n(B; \mathbb{F}_2) \rightarrow H^{n+1}(B; \mathbb{Z})$ whose image consists of classes killed by 2, and exactness identifies that image as the kernel of multiplication by 2. So $e(E) = \tilde{\beta}(u)$ for some mod-2 class u , and the question is which one.

Edge three: Wu's formula. The mod-2 classes are related to each other by a Steenrod operation:

$$\text{Sq}^1 w_i = w_1 w_i + (i+1) w_{i+1}, \quad (26.1)$$

with the coefficient read mod 2. For an *oriented* bundle $w_1 = 0$, so (26.1) reduces to $\text{Sq}^1 w_i = (i+1)w_{i+1}$, which is w_{i+1} exactly when i is even.

Exercise 26.7 (medium). Verify (26.1) for the tangent bundle of $\mathbb{RP}^n$, using Example 12.6 and the Cartan formula.

Hint. By Example 12.6, $w(\mathbb{RP}^n) = (1 + a)^{n+1}$, so $w_i = \binom{n+1}{i} a^i$. Compute $\text{Sq}^1(a^i)$ from the axioms of Theorem 18.6.

Solution. In $H^*(\mathbb{RP}^n; \mathbb{F}_2) = \mathbb{F}_2[a]/(a^{n+1})$ we have $\text{Sq}^1 a = a^2$, by Theorem 18.6(b) with $|a| = 1$. The Cartan formula gives $\text{Sq}^1(a^i) = i a^{i+1}$.

By Example 12.6, $w_i = \binom{n+1}{i} a^i$, so

$$\text{Sq}^1 w_i = \binom{n+1}{i} i a^{i+1}, \quad w_1 w_i = \binom{n+1}{1} \binom{n+1}{i} a^{i+1}, \quad w_{i+1} = \binom{n+1}{i+1} a^{i+1}.$$

So (26.1) asserts the mod-2 identity $\binom{n+1}{i} i = \binom{n+1}{1} \binom{n+1}{i} + (i+1) \binom{n+1}{i+1}$, which follows from $(i+1) \binom{n+1}{i+1} = (n+1-i) \binom{n+1}{i}$ and reducing coefficients modulo 2.

Remark 26.8 (Closing the triangle). The three edges now sit together. For an oriented bundle of odd rank $2k+1$: edge two says $e(E)$ is 2-torsion, hence $e(E) = \tilde{\beta}(u)$ for some $u \in H^{2k}(B; \mathbb{F}_2)$; edge one says $\rho e(E) = w_{2k+1}(E)$; and edge three, in the oriented case with $i = 2k$ even, says $\text{Sq}^1 w_{2k} = w_{2k+1}$.

Comparing the last two identifies $\rho \tilde{\beta}(u)$ with $\text{Sq}^1 w_{2k}$, and the natural candidate for u is w_{2k} itself. Turning that into a proof needs two further steps, and both have been set up but not carried out here. First, the relation between $\rho \tilde{\beta}$ and Sq^1 : these are two natural stable operations of degree one, and Remark 18.15 explained why there is only one such and what has to be checked to identify them. Second, injectivity: knowing two integral classes have equal reductions determines them only where ρ is injective on the relevant torsion, which for this problem means working in the classifying space $BSO(2k+1)$ and pulling back — the standard device for promoting a mod-2 identity to an integral one.

Assembling this is the natural exercise, and it is a genuine synthesis: it uses Chapter 9 for the Bockstein, Chapter 12 for the Wu formula, Chapter 18 for the uniqueness of Sq^1 , and this chapter for the torsion. A reader who completes it has connected four chapters.

26.3 Which classes are characteristic classes?

Example 26.9 (Rank one is unconstrained). By Example 26.3, every class in $H^2(B; \mathbb{Z})$ is the Euler class of a line bundle. There is no constraint at all in rank one.

Remark 26.10 (Higher rank). In rank k the multiplicativity of Proposition 26.2(b) makes the Euler class of a sum of line bundles a product of k classes of degree two. Over $\mathbb{CP}^\infty$, where $H^* = \mathbb{Z}[c]$, the question of which classes of H^{2k} arise this way is therefore a question about which multiples of c^k can be written as a product of k classes in $\mathbb{Z}c$ — and Exercise 26.4 supplies more freedom than one might expect, since $e(L^{\otimes m}) = m e(L)$.

Working out the answer is a short exercise once those two facts are in hand, and the conclusion is worth contrasting with Remark 26.11 below: the *top* class is unconstrained, while the total class is not.

Remark 26.11 (The total class is constrained). Not every formal expression $1 + c_1 + c_2 + \cdots$ is the total Chern class of a bundle. The standard obstruction is arithmetic: for a complex bundle over S^{2k} , the top Chern class c_k evaluated on the fundamental class is divisible by $(k-1)!$, a theorem of Bott. So realisability of a single class in the top

degree, which [Example 26.9](#) and [Remark 26.10](#) make easy, is a quite different question from realisability of a whole system of classes.

This is the integral counterpart of the contrast promised in [Remark 12.12](#), and the reason characteristic classes are studied through classifying spaces: $H^*(BU(n); \mathbb{Z}) = \mathbb{Z}[c_1, \dots, c_n]$ says exactly which *universal* classes exist, and the constraints appear only when one asks which of them survive pullback along a map from a particular space.

26.4 What to carry forward

- The Euler class ([Definition 26.1](#)) and its four properties ([Proposition 26.2](#)); it reduces to w_{top} .
- Line bundles are classified by H^2 , and e is additive under tensor product ([Example 26.3](#) and [Exercise 26.4](#)) — the H-space structure of $K(\mathbb{Z}, 2)$ at work.
- Odd rank forces $2e = 0$ ([Exercise 26.5](#)), so the odd-rank Euler class lives in the image of a Bockstein.
- Wu's formula ([Exercise 26.7](#)) relates the mod-2 classes to each other by Sq^1 ; with the previous two items it closes the triangle of [Remark 12.12](#) up to the identification discussed in [Remark 26.8](#).
- Top classes are unconstrained; total classes are not ([Remark 26.11](#)).

Chapter 27

The Steenrod algebra in earnest

Chapter 18 introduced the squares axiomatically and used one of them. This chapter treats $\mathcal{A}$ as an algebra: it has a basis, a coproduct, and relations that constrain what its elements can do. The payoff is a theorem no earlier chapter could reach — a numerical restriction on which spheres can carry a Hopf invariant one map, proved by nothing more than the Adem relations and a degree count.

27.1 The total square

Definition 27.1. The *total square* of $x \in H^*(X; \mathbb{F}_2)$ is $\text{Sq}(x) = \sum_{i \geq 0} \text{Sq}^i(x)$, a finite sum by Theorem 18.6(b).

Exercise 27.2 (easy). Show that Sq is a ring homomorphism: $\text{Sq}(xy) = \text{Sq}(x) \text{Sq}(y)$.

Hint. Collect terms of the Cartan formula by total degree.

Solution. The degree- k component of $\text{Sq}(x) \text{Sq}(y)$ is $\sum_{i+j=k} \text{Sq}^i(x) \text{Sq}^j(y)$, which by Theorem 18.6(c) is $\text{Sq}^k(xy)$, the degree- k component of $\text{Sq}(xy)$.

Example 27.3 (The squares on projective space). Let c generate $H^2(\mathbb{CP}^\infty; \mathbb{F}_2)$, so that $H^*(\mathbb{CP}^\infty; \mathbb{F}_2) = \mathbb{F}_2[c]$. By Theorem 18.6(b), $\text{Sq}^2 c = c^2$ and $\text{Sq}^i c = 0$ for $i > 2$; and $\text{Sq}^1 c$ lies in $H^3(\mathbb{CP}^\infty; \mathbb{F}_2) = 0$. Hence

$$\text{Sq}(c) = c + c^2 = c(1 + c).$$

Now Exercise 27.2 does the rest:

$$\text{Sq}(c^n) = (\text{Sq} c)^n = c^n(1 + c)^n = \sum_i \binom{n}{i} c^{n+i},$$

so that $\text{Sq}^{2i}(c^n) = \binom{n}{i} c^{n+i}$ and the odd squares vanish on c^n .

The entire action of $\mathcal{A}$ on a polynomial generator has been reduced to a table of binomial coefficients modulo 2 — which, by Exercise 12.10, is a statement about binary expansions.

Remark 27.4. Example 27.3 was carried out on a class of degree two. The same computation on a class of degree one — where $\text{Sq}(x) = x + x^2$ and no degree argument is needed to kill intermediate terms — is shorter still, and its consequences are sharper, because iterating it produces exactly the constraint that governs the admissible basis of Section 27.2. Carrying out that iteration, and identifying which sequences of squares survive, is the natural exercise once Section 27.2 has been read.

27.2 The admissible basis

Definition 27.5. A monomial $\text{Sq}^I = \text{Sq}^{i_1} \text{Sq}^{i_2} \cdots \text{Sq}^{i_k}$ is *admissible* if $i_j \geq 2i_{j+1}$ for all j . Its *excess* is $e(I) = i_1 - (i_2 + \cdots + i_k)$.

Theorem 27.6 (Serre–Cartan; recalled). *The admissible monomials form an $\mathbb{F}_2$ -basis of $\mathcal{A}$. Moreover $\text{Sq}^I(x) = 0$ whenever $e(I) > |x|$, and $\text{Sq}^I(x) = x^2$ when $e(I) = |x|$.*

Exercise 27.7 (medium). Use the Adem relations of [Theorem 18.13](#) to write $\text{Sq}^2 \text{Sq}^2$ in the admissible basis.

Hint. Take $a = b = 2$ and check the condition $a < 2b$. Which values of c give a nonzero binomial coefficient?

Solution. With $a = b = 2$ the condition $a < 2b$ holds, and [Theorem 18.13](#) gives

$$\text{Sq}^2 \text{Sq}^2 = \sum_c \binom{b-c-1}{a-2c} \text{Sq}^{a+b-c} \text{Sq}^c = \sum_c \binom{1-c}{2-2c} \text{Sq}^{4-c} \text{Sq}^c.$$

For $c = 0$ the coefficient is $\binom{1}{2} = 0$; for $c = 1$ it is $\binom{0}{0} = 1$. Hence

$$\text{Sq}^2 \text{Sq}^2 = \text{Sq}^3 \text{Sq}^1,$$

and $\text{Sq}^3 \text{Sq}^1$ is admissible since $3 \geq 2 \cdot 1$.

Remark 27.8 ($\mathcal{A}$ is a Hopf algebra). The Cartan formula says exactly that the map

$$\psi: \mathcal{A} \longrightarrow \mathcal{A} \otimes \mathcal{A}, \quad \psi(\text{Sq}^k) = \sum_{i+j=k} \text{Sq}^i \otimes \text{Sq}^j,$$

is compatible with the action on a product, and it makes $\mathcal{A}$ a Hopf algebra in the sense of [Chapter 10](#). The primitives of $\mathcal{A}$ are spanned by the Sq^{2^j} , which is the algebraic shadow of the decomposability statement below.

27.3 Decomposability, and Hopf invariant one

Proposition 27.9 (Recalled). *If n is not a power of 2 then Sq^n is decomposable: it lies in the square of the augmentation ideal, so*

$$\text{Sq}^n = \sum_{0 < i < n} a_i \text{Sq}^i$$

for suitable $a_i \in \mathcal{A}$ of positive degree. The Sq^{2^j} generate $\mathcal{A}$ as an algebra.

This is a consequence of the Adem relations, and [Exercise 27.7](#) is its smallest instance: Sq^3 , with 3 not a power of 2, is expressible using lower squares.

Theorem 27.10. *If there is a map $f: S^{2n-1} \rightarrow S^n$ with odd Hopf invariant, then n is a power of 2.*

Proof. Let $C_f = S^n \cup_f e^{2n}$ and work with $\mathbb{F}_2$ coefficients, so that $H^*(C_f; \mathbb{F}_2)$ is $\mathbb{F}_2$ in degrees 0, n and $2n$, spanned by 1, α and β . By [Definition 16.13](#) and the hypothesis, $\alpha^2 = H(f)\beta = \beta \pmod{2}$, so $\alpha^2 \neq 0$.

By [Theorem 18.6\(b\)](#), $\text{Sq}^n \alpha = \alpha^2 \neq 0$.

Suppose n were not a power of 2. By [Proposition 27.9](#) we could write $Sq^n = \sum_{0 < i < n} a_i Sq^i$, so

$$Sq^n \alpha = \sum_{0 < i < n} a_i (Sq^i \alpha).$$

But for $0 < i < n$ the class $Sq^i \alpha$ lies in $H^{n+i}(C_f; \mathbb{F}_2)$, and $0 < n + i < 2n$, a degree in which the cohomology of C_f vanishes. So every term is zero and $Sq^n \alpha = 0$, contradicting the previous paragraph.

Hence n is a power of 2. □

Remark 27.11 (How far this gets, and what it costs). [Theorem 27.10](#) narrows the possible dimensions from all n to $n \in \{1, 2, 4, 8, 16, \dots\}$, and it does so with the Adem relations and one degree count. The classical Hopf maps realise $n = 1, 2, 4, 8$ ([Example 16.14](#) and its quaternionic and octonionic analogues), so the question is whether $n = 16$ or beyond can occur.

They cannot, but that is Adams' theorem, proved with secondary operations or with K -theory, and the book does not prove it — as [Remark 16.17](#) already said. What is worth registering is how much of the way an elementary argument goes: everything in this chapter is a manipulation of the algebra $\mathcal{A}$, and it reduces an infinite family of possibilities to a logarithmically sparse one.

The same style of argument constrains division algebras, parallelisability of spheres, and the existence of certain fibre bundles — all of them, by [Remark 16.17](#), questions about the same integer.

27.4 What to carry forward

- The total square is a ring map ([Exercise 27.2](#)); every computation follows from that plus one value ([Example 27.3](#)).
- The admissible basis and excess ([Definition 27.5](#) and [Theorem 27.6](#)); Adem relations are used by rewriting, as in [Exercise 27.7](#).
- $\mathcal{A}$ is a Hopf algebra generated by the Sq^{2^j} ([Remark 27.8](#) and [Proposition 27.9](#)).
- Odd Hopf invariant forces n to be a power of 2 ([Theorem 27.10](#)) — an elementary argument that reaches within one step of Adams' theorem ([Remark 27.11](#)).

Chapter 28

Spectra: why symmetric, and where the signs come from

Stable homotopy theory studies what survives suspension. [Theorem 14.12](#) showed that homotopy groups of spheres stabilise; a spectrum is the object whose homotopy groups are the stable ones by construction, and the stable homotopy category is where such objects live.

The technical difficulty, and the reason this chapter exists, is the smash product. Sequential spectra carry one only up to homotopy and with no coherent symmetry, which is fatal for any discussion of ring objects. Symmetric spectra repair this by adding group actions, and the actions are not bookkeeping: forget them and the theory collapses in two different ways.

28.1 Spectra

Definition 28.1. A *spectrum* X is a sequence of based spaces X_n together with structure maps $\sigma_n: \Sigma X_n \rightarrow X_{n+1}$. Its *naive homotopy groups* are

$$\hat{\pi}_k(X) = \operatorname{colim}_n \pi_{k+n}(X_n),$$

the colimit being taken over suspension followed by σ_n .

Example 28.2. The *sphere spectrum* $\mathbb{S}$ has $\mathbb{S}_n = S^n$ with the identity structure maps. Its naive homotopy groups are $\hat{\pi}_k(\mathbb{S}) = \operatorname{colim}_n \pi_{k+n}(S^n)$. By [Theorem 14.12](#) this colimit stabilises, and its value is the stable stem π_k^s of [Example 14.13](#).

A based space Y gives a suspension spectrum $\Sigma^\infty Y$, with $(\Sigma^\infty Y)_n = \Sigma^n Y$; its naive homotopy is the stable homotopy of Y . The Eilenberg–MacLane spectrum $H\pi$ has $(H\pi)_n = K(\pi, n)$, its structure maps being adjoint to the equivalences $K(\pi, n) \simeq \Omega K(\pi, n+1)$ of [Exercise 17.15](#).

Definition 28.3. A *symmetric spectrum* is a spectrum in which each X_n carries an action of Σ_n , such that the iterated structure map

$$\sigma^m: X_n \wedge S^m \longrightarrow X_{n+m}$$

is $\Sigma_n \times \Sigma_m$ -equivariant. Here Σ_m permutes the coordinates of $S^m = (\mathbb{R}^m)^+$, and acts on X_{n+m} via the inclusion $\Sigma_m \subseteq \Sigma_{n+m}$ onto the last m letters.

Example 28.4. The sphere spectrum is symmetric: Σ_n acts on $S^n = (\mathbb{R}^n)^+$ by permuting coordinates, and the structure maps are equivariant because smashing S^n with S^m concatenates coordinates.

Warning 28.5 (The actions carry information). It is tempting to regard the symmetric group actions of [Definition 28.3](#) as bookkeeping introduced to make the smash product associative. They are not. Two independent defects appear when the naive homotopy groups of [Definition 28.1](#) are used in place of the correct stable homotopy groups, and both are visible on symmetric spectra:

- a symmetric spectrum whose group actions are trivial in infinitely many levels has *no* naive homotopy at all, so forgetting the actions destroys everything;
- naive homotopy groups fail to commute with infinite products, unlike the genuine ones.

Both are worth establishing for oneself, and the tools for each are supplied by [Exercises 28.6](#) and [28.7](#) below. Neither is a pathology invented for the purpose: the first is what forces the equivariance in [Definition 28.3](#) to be taken seriously, and the second is why the stable homotopy category cannot be built from $\hat{\pi}_*$ alone.

Exercise 28.6 (easy). Let $A_0 \rightarrow A_1 \rightarrow \cdots$ be a sequence of abelian groups in which, for cofinally many n , the map $A_n \rightarrow A_{n+1}$ is zero. Show $\operatorname{colim}_n A_n = 0$.

Solution. An element of the colimit is represented by some $a \in A_n$, two representatives being identified when they agree after mapping far enough along. Choose $m > n$ with $A_m \rightarrow A_{m+1}$ zero; the image of a in A_{m+1} is then 0. So every element of the colimit is zero.

Exercise 28.7 (easy). Let $\mathbb{Z}^n \hookrightarrow \mathbb{Z}^{n+1}$ be the inclusion of the first n coordinates. Show $\operatorname{colim}_n \mathbb{Z}^n = \bigoplus_{n \geq 1} \mathbb{Z}$, and contrast this with $\prod_{n \geq 1} \mathbb{Z}$.

Solution. Every element of the colimit comes from some $\mathbb{Z}^n$, hence has only finitely many nonzero coordinates; conversely every finitely supported sequence appears. So the colimit is $\bigoplus \mathbb{Z}$, which is countable, while $\prod \mathbb{Z}$ is uncountable — the same distinction as [Example 5.3](#).

28.2 Where the signs come from

Graded commutativity has been in force since [Chapter 7](#), and the Koszul sign rule has been used in [Chapter 25](#) without comment. Both come from one computation.

Lemma 28.8. Let $\tau \in \Sigma_k$ act on $S^k = (\mathbb{R}^k)^+$ by permuting coordinates. Then $\deg \tau = \operatorname{sgn}(\tau)$. In particular the shuffle exchanging a block of n coordinates with a block of m coordinates has degree $(-1)^{nm}$.

Proof. Permuting coordinates is the restriction to $\mathbb{R}^k$ of a linear automorphism whose matrix is the permutation matrix of τ , with determinant $\operatorname{sgn}(\tau)$. A linear automorphism of $\mathbb{R}^k$ extends to the one-point compactification S^k , and its degree is the sign of its determinant: through linear automorphisms it is homotopic to the identity when the determinant is positive and to a single reflection when it is negative.

For the block shuffle, moving each of the n coordinates of the first block past each of the m coordinates of the second requires nm transpositions, so its sign is $(-1)^{nm}$. $\square$

Remark 28.9 (The sign rule is a theorem). [Lemma 28.8](#) is the source of every Koszul sign in the book. In the stable category an element of degree n is represented by a map out of a sphere S^n , and interchanging two such elements requires interchanging the corresponding sphere coordinates; the resulting map differs from the original by the shuffle, hence by the factor $(-1)^{nm}$.

Making that precise means constructing a symmetric monoidal structure on a category of formal desuspensions — objects (X, n) with morphisms the stable maps, smash defined by $(X, n) \wedge (Y, m) = (X \wedge Y, n + m)$ — and checking the coherence axioms, with [Lemma 28.8](#) supplying the symmetry isomorphism. That construction is the natural exercise here, and its value is that afterwards the sign rule is never again a convention to be remembered.

Note also where the difficulty in [Definition 28.3](#) comes from. The symmetry has to be present at the level of spaces, coherently and for all permutations at once, which is precisely what the Σ_n -actions provide and what a merely sequential spectrum lacks.

28.3 Moore spectra

Definition 28.10. For an integer m , the *Moore spectrum* $\mathbb{S}/m$ is the cofibre of $m: \mathbb{S} \rightarrow \mathbb{S}$.

Exercise 28.11 (easy). Show that $H^*(\mathbb{S}/2; \mathbb{F}_2)$ is $\mathbb{F}_2$ in degrees 0 and 1, with Sq^1 carrying the degree-zero class to the degree-one class.

Hint. Use the long exact sequence of the cofibre sequence $\mathbb{S} \xrightarrow{2} \mathbb{S} \rightarrow \mathbb{S}/2$ in mod-2 cohomology, and recall from [Chapter 9](#) which operation the coefficient sequence $0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{F}_2 \rightarrow 0$ produces.

Solution. Mod 2, the map $2: \mathbb{S} \rightarrow \mathbb{S}$ induces zero on cohomology, so the long exact sequence of the cofibre sequence breaks into short exact pieces and gives $H^*(\mathbb{S}/2; \mathbb{F}_2) = \mathbb{F}_2$ in degrees 0 and 1 and zero elsewhere.

The connecting map of that cofibre sequence is, by construction, the operation associated with reduction modulo 2 — the Bockstein of [Definition 9.11](#). So it carries the degree-zero generator to the degree-one generator, and a degree-one operation acts nontrivially between them.

Exercise 28.12 (easy). Show that for a wedge of spectra, $H^*(X \vee Y; \mathbb{F}_2) \cong H^*(X; \mathbb{F}_2) \oplus H^*(Y; \mathbb{F}_2)$ as modules over the Steenrod algebra.

Solution. The inclusions and retractions of a wedge induce, exactly as in [Exercise 7.3](#), a splitting of $H^*(X \vee Y)$ as a direct sum. Both maps are induced by maps of spectra, and Steenrod operations are natural, so the splitting is one of $\mathcal{A}$ -modules: an operation applied to a class from one summand stays in that summand.

Remark 28.13 (An algebraic structure that does not exist). Put the two exercises together and a question presents itself. The smash square $\mathbb{S}/2 \wedge \mathbb{S}/2$ has mod-2 cohomology in degrees 0, 1, 1, 2, by Künneth. If it were equivalent to a wedge $\mathbb{S}/2 \vee \Sigma \mathbb{S}/2$ then by [Exercise 28.12](#) its cohomology would split as an $\mathcal{A}$ -module, with the degree-zero class and the degree-two class in different summands — so every operation carrying one to the other would have to vanish. In particular Sq^2 would vanish on the bottom class.

Whether it does is a computation, and it decides the question. A negative answer also settles a second one: a homotopy ring structure on $\mathbb{S}/2$ would force the splitting,

so its non-existence would follow. Carrying this out is the natural exercise, and it uses only [Exercise 28.11](#), [Exercise 28.12](#) and the Cartan formula.

The moral is worth stating in advance. The mod-2 Moore spectrum is the standing counterexample of stable homotopy theory: the cofibre of a self-map of the sphere, as simple an object as one can construct, and it is not a ring. Any theory that assumes such cofibres are well behaved is wrong.

28.4 What to carry forward

- Spectra and naive homotopy groups ([Definition 28.1](#) and [Example 28.2](#)); symmetric spectra add equivariance ([Definition 28.3](#)).
- The actions are not bookkeeping ([Warning 28.5](#)); the two colimit lemmas ([Exercises 28.6](#) and [28.7](#)) are what establish that.
- The Koszul sign is the degree of a permutation ([Lemma 28.8](#) and [Remark 28.9](#)), not a convention.
- $H^*(\mathbb{S}/2; \mathbb{F}_2)$ and the wedge splitting ([Exercises 28.11](#) and [28.12](#)) detect the failure of $\mathbb{S}/2$ to be a ring ([Remark 28.13](#)).

Chapter 29

Quasi-category technology

[Chapter 23](#) defined quasi-categories and stopped. This chapter assembles the three tools one needs to work with them: an induction that upgrades lifting against generators to lifting against all monomorphisms; an explicit description of the simplicial category associated with a simplex; and a cofinality criterion that replaces explicit contracting homotopies by a statement about comma categories.

None of the three is deep. All three are the kind of thing that, unstated, makes the literature unreadable.

29.1 Lifting against everything

Definition 29.1. A map of simplicial sets is a *trivial fibration* if it has the right lifting property against the boundary inclusions $\partial\Delta^n \hookrightarrow \Delta^n$ for all $n \geq 0$.

Exercise 29.2 (medium). Show that a trivial fibration lifts against *all* monomorphisms, not merely against the boundary inclusions of [Definition 29.1](#).

Hint. A monomorphism $A \hookrightarrow B$ is built from A by attaching nondegenerate simplices of B that are not in A , one dimension at a time. What is the pushout describing the attachment of a single such simplex?

Solution. Filter B by $A = B^{(-1)} \subseteq B^{(0)} \subseteq \dots$, where $B^{(n)}$ is A together with all simplices of B of dimension at most n . Passing from $B^{(n-1)}$ to $B^{(n)}$ attaches the nondegenerate n -simplices of B not in A , and each attachment is a pushout

$$\begin{array}{ccc} \coprod \partial\Delta^n & \longrightarrow & B^{(n-1)} \\ \downarrow & & \downarrow \\ \coprod \Delta^n & \longrightarrow & B^{(n)}. \end{array}$$

The right lifting property is preserved under cobase change — this is [Exercise 22.7](#) read on the other side — and under composition and transfinite composition. Since p lifts against each $\partial\Delta^n \hookrightarrow \Delta^n$, it lifts against every stage of the filtration, hence against $A \hookrightarrow B$.

Remark 29.3 (What the strong form is for). The typical application is to build an inverse. Given a trivial fibration $p: X \rightarrow Y$ and a quasi-category Z , one wants to invert the map $[Y, Z] \rightarrow [X, Z]$ induced by p ; constructing the inverse means solving a lifting problem whose left-hand map is $\emptyset \rightarrow Y$ or, for the homotopy statement, an inclusion

$A \times J \cup B \times \partial J \hookrightarrow B \times J$. Neither is a boundary inclusion, and neither is a generator; both are monomorphisms.

So [Exercise 29.2](#) is exactly what converts [Definition 29.1](#) into a statement about mapping spaces, and anyone attempting such an argument with only the generators available will stall at the first step. Establishing that trivial fibrations induce bijections on homotopy classes of maps into any quasi-category — that is, that they are equivalences in the appropriate sense — is the natural exercise here, and its only ingredient beyond this section is a description of homotopies of simplicial maps.

29.2 Rigidification

The comparison between quasi-categories and categories enriched in simplicial sets goes through an adjunction

$$\mathfrak{C} : \mathbf{sSet} \rightleftarrows \mathbf{sCat} : N_{\mathrm{hc}},$$

where N_{hc} is the homotopy coherent nerve. What makes it computable is an explicit formula for $\mathfrak{C}$ on simplices.

Proposition 29.4 (The hom-cube; recalled). *For the standard simplex,*

$$\mathfrak{C}\Delta^n(i, j) = N(P_{i,j}), \quad P_{i,j} = \{ S \subseteq \{i, i+1, \dots, j\} : i, j \in S \},$$

the nerve of that poset ordered by inclusion, and empty when $i > j$. It is a cube of dimension $j - i - 1$.

Example 29.5 (Small cases). For $n = 1$ and $(i, j) = (0, 1)$ the poset $P_{0,1}$ has the single element $\{0, 1\}$, so $\mathfrak{C}\Delta^1(0, 1) = \Delta^0$: a point, a cube of dimension $1 - 0 - 1 = 0$. Hence $\mathfrak{C}\Delta^1$ is the category $[1]$ with discrete hom-spaces.

For $n = 2$ and $(i, j) = (0, 2)$ the poset has two elements, $\{0, 2\} \subset \{0, 1, 2\}$, so $\mathfrak{C}\Delta^2(0, 2) = \Delta^1$: an interval, a cube of dimension 1. The two endpoints are the two ways of getting from 0 to 2 — directly, or via 1 — and the interval between them is the homotopy relating them.

For $n = 3$ and $(i, j) = (0, 3)$ the poset has four elements and the hom-space is a square.

Warning 29.6. [Example 29.5](#) is worth reading twice. Already for Δ^2 the hom-space from 0 to 2 is an interval and not a point: the composite is not unique, it is unique up to a specified homotopy. That is the entire difference between $\mathfrak{C}\Delta^n$ and the poset $[n]$, and it is why $\mathfrak{C}$ is called rigidification — it produces a genuinely simplicial category, not a discrete one.

Proposition 29.7 (Rigidifying a nerve; recalled). *Let $\mathcal{A}$ be an ordinary category. Then $\mathfrak{C}(N\mathcal{A})$ is the free simplicial resolution of $\mathcal{A}$. In the hom-space between two objects, a vertex is a chain of composable arrows, none of them an identity, running from the first object to the second; the simplices above dimension zero record iterated bracketings of such a chain.*

Remark 29.8 (A left adjoint that fails to preserve products). $\mathfrak{C}$ is a left adjoint, so by [Exercise 20.2](#) it preserves colimits. [Warning 20.3](#) warned that this says nothing about products, and $\mathfrak{C}$ is the promised example.

To see it, compare two computations of the same hom-space. On one side, $\mathfrak{C}\Delta^1 \times \mathfrak{C}\Delta^1$ is a product of two copies of $[1]$ with discrete homs, by [Example 29.5](#), so its hom-space

between the two extreme objects is a single point. On the other, $\Delta^1 \times \Delta^1$ is the nerve of the commuting square, so [Proposition 29.7](#) applies and the corresponding hom-space has one 0-simplex for each string of composable non-identity arrows from one corner to the opposite one.

Counting those strings settles the matter, and the count is the exercise. What should be noted afterwards is that the comparison map is nevertheless a weak equivalence: the failure is strict, not homotopical, which is exactly the situation [Chapter 22](#) was built to handle.

29.3 Cofinality, and extra degeneracies

Definition 29.9. A functor $u: J' \rightarrow J$ is *homotopy initial* if for every $j \in J$ the comma category $(u \downarrow j)$ has contractible nerve. Homotopy colimits may then be computed over J' instead of J .

Example 29.10 (The bar resolution, by hand). Let G be a group and let $E_\bullet G$ be the simplicial set with $E_n G = G^{n+1}$ and the evident faces and degeneracies. It is augmented over a point, and it admits an *extra degeneracy*

$$s_{-1}: E_n G \longrightarrow E_{n+1} G, \quad (g_0, \dots, g_n) \longmapsto (e, g_0, \dots, g_n),$$

satisfying the simplicial identities that make it a contracting homotopy for the augmented complex.

The classical consequence is that $|E_\bullet G|$ is contractible: the extra degeneracy provides, degree by degree, a homotopy from the identity to the constant map. That is what made $|E_\bullet G| \rightarrow BG$ a universal cover in [Exercise 20.18](#).

Exercise 29.11 (medium). Let $A_\bullet$ be an augmented simplicial abelian group with an extra degeneracy. Show that its associated chain complex is acyclic in positive degrees, with H_0 the augmentation.

Hint. Write $h = (-1)^{n+1} s_{-1}$ on degree n and compute $\partial h + h \partial$ using the simplicial identities.

Solution. With $\partial = \sum_i (-1)^i d_i$ and $h = \pm s_{-1}$, the simplicial identities $d_0 s_{-1} = \text{id}$ and $d_{i+1} s_{-1} = s_{-1} d_i$ give, after collecting signs,

$$\partial h + h \partial = \text{id}$$

in positive degrees, and in degree zero the same computation leaves the augmentation. A chain homotopy from the identity to zero forces the homology to vanish where it applies.

Remark 29.12 (Replacing the computation by a statement). [Example 29.10](#) and [Exercise 29.11](#) are the hands-on version, and they have to be redone every time the indexing changes. The conceptual version observes that an augmentation together with a contracting homotopy is exactly the data of an extension of the simplicial object over a larger category — the simplex category with a formally adjoined top or bottom element — and that the inclusion of Δ into that larger category is homotopy initial in the sense of [Definition 29.9](#).

Once that is established, the conclusion of [Example 29.10](#) is a corollary rather than a computation, and it applies verbatim to bar resolutions, to Čech nerves, and to descent situations, where the hands-on argument would have to be repeated each time. Establishing it is the natural exercise here, and the two things to be careful about are which of the two adjoined elements corresponds to which handedness of contracting homotopy, and which of homotopy initial and homotopy final is the condition relevant to colimits.

29.4 What to carry forward

- A trivial fibration lifts against every monomorphism, not only the generators ([Exercise 29.2](#)); the strong form is what applications use ([Remark 29.3](#)).
- The hom-cube of a simplex ([Proposition 29.4](#)), with its small cases worked out ([Example 29.5](#)).
- Already for Δ^2 the hom-space is an interval, not a point ([Warning 29.6](#)).
- Rigidifying a nerve gives the free simplicial resolution ([Proposition 29.7](#)).
- $\mathfrak{C}$ preserves colimits but not products ([Remark 29.8](#)).
- Extra degeneracies give contractibility ([Example 29.10](#) and [Exercise 29.11](#)); cofinality is how one stops redoing that computation ([Definition 29.9](#) and [Remark 29.12](#)).

Chapter 30

Operads, Koszul duality, and twisting morphisms

[Chapter 25](#) defined operads and used them lightly. This chapter makes the composition product explicit, introduces the duality that produces “algebras up to homotopy”, and identifies the equation governing all of it.

The organising observation is that one equation — Maurer–Cartan — appears in three disguises: as the condition defining a twisting morphism, as the condition that a perturbed differential squares to zero, and as the structure equations of a homotopy algebra. [Section 30.3](#) exhibits the first two as the same thing in the case everyone already knows.

30.1 $\mathbb{S}$ -modules and Schur functors

Definition 30.1. An $\mathbb{S}$ -module is a sequence $E = (E(n))_{n \geq 0}$ of representations of the symmetric groups $\mathbb{S}_n$. Its *Schur functor* is

$$\tilde{E}(V) = \bigoplus_{n \geq 0} E(n) \otimes_{\mathbb{S}_n} V^{\otimes n},$$

where $\mathbb{S}_n$ permutes the tensor factors.

Example 30.2 (The dictionary). Three cases fix the pattern.

- $E(n)$ the trivial representation: $\tilde{E}(V) = \bigoplus_n \text{Sym}^n(V)$, since coinvariants under the permutation action are the symmetric powers.
- $E(n)$ the sign representation: $\tilde{E}(V) = \bigoplus_n \Lambda^n(V)$.
- $E(n) = k[\mathbb{S}_n]$ the regular representation: $k[\mathbb{S}_n] \otimes_{\mathbb{S}_n} V^{\otimes n} = V^{\otimes n}$, so $\tilde{E}(V) = T(V)$, the tensor algebra.

Compare [Example 25.2](#): the three $\mathbb{S}$ -modules underlying Com , its signed variant, and Ass produce exactly the free commutative, exterior and associative algebras.

Definition 30.3. The *composition product* $E \circ F$ of two $\mathbb{S}$ -modules is the $\mathbb{S}$ -module whose Schur functor is the composite: $\widetilde{E \circ F} = \tilde{E} \circ \tilde{F}$. An operad is a monoid for $\circ$, with the multiplication encoding the $\circ_i$ of [Definition 25.1](#).

Remark 30.4 (Making $\circ$ explicit). [Definition 30.3](#) defines $E \circ F$ by a property rather than a formula, which is enough for the theory and useless for computation. Writing down the representation $(E \circ F)(N)$ explicitly means expanding $\tilde{E}(\tilde{F}(V))$ and collecting the terms of tensor degree N .

Two features of that expansion are worth predicting before doing it. Each term involves n copies of some $F(m)$ together with one element of $E(n)$, so $N = mn$ in the homogeneous case; and the symmetric group $\mathbb{S}_N$ acts by permuting the m inputs inside each block *and* permuting the n blocks, which is the action of a wreath product rather than of a direct product. Getting that second point wrong — using $\mathbb{S}_m^{\times n}$ where the wreath product belongs — is the standard error, and it produces a representation of the wrong dimension.

Carrying out the expansion is the natural exercise, and the resulting operation on representations is classical enough to have its own name.

30.2 Koszul duality and the weight grading

Definition 30.5. For a quadratic operad $\mathcal{P}$ — one presented by generators of arity two and relations of arity three — there is a *Koszul dual cooperad* $\mathcal{P}^!$, and the cobar construction on it,

$$\mathcal{P}_\infty = \Omega \mathcal{P}^!,$$

is a cofibrant resolution of $\mathcal{P}$. An algebra over $\mathcal{P}_\infty$ is a $\mathcal{P}$ -algebra up to homotopy.

The construction comes with a grading that makes everything computable.

Proposition 30.6 (The weight grading; recalled). *$\mathcal{P}^!$ is graded by weight, the number of generators used, and its weight- w component sits in homological degree $w - 1$. Consequently the operation of $\mathcal{P}_\infty$ coming from weight w raises degree by $w - 2$.*

Example 30.7 (A_∞ -algebras). Take $\mathcal{P} = \text{Ass}$. Then $\mathcal{P}_\infty$ is the A_∞ -operad, and an algebra over it is a graded module A with operations

$$m_k: A^{\otimes k} \longrightarrow A, \quad k \geq 1,$$

of degree $k - 2$, satisfying the relations $\sum \pm m_i(1^{\otimes a} \otimes m_j \otimes 1^{\otimes b}) = 0$. The first three are familiar: m_1 has degree -1 and squares to zero, so it is a differential; m_2 has degree 0 and is a product; m_3 has degree 1 and is a homotopy witnessing associativity of m_2 .

The degrees are forced by [Proposition 30.6](#), and they are the whole content of the definition: once one knows $|m_k| = k - 2$, the shape of the theory is determined.

Exercise 30.8 (easy). In [Example 30.7](#), write out the relation involving m_1 , m_2 and m_3 and identify what each term says.

Solution. The relation in arity two says $m_1 m_2 = m_2(m_1 \otimes 1 + 1 \otimes m_1)$ up to sign: m_1 is a derivation for m_2 , i.e. the Leibniz rule.

The relation in arity three says

$$m_2(m_2 \otimes 1 - 1 \otimes m_2) = \pm (m_1 m_3 + m_3(m_1 \otimes 1 \otimes 1 + \cdots)),$$

whose left side is the associator of m_2 and whose right side is the differential of m_3 . So m_2 is associative up to the explicit homotopy m_3 , and strictly associative exactly when that differential vanishes.

Remark 30.9 (Degree counting as a prediction). [Proposition 30.6](#) lets one predict the shape of a homotopy structure before computing anything. Suppose the underlying module is concentrated in a single degree, with zero differential; then its endomorphism operad is concentrated in degree zero, and an operation of nonzero degree has nowhere to land.

Reading that off for a general $\mathcal{P}$, and identifying exactly which operations survive, is the natural exercise here; [Example 30.7](#) supplies the case $\mathcal{P} = \text{Ass}$, where $|m_k| = k - 2$ makes the answer visible at a glance. The one thing to be careful about is the differential: an $\mathcal{P}_\infty$ -structure includes m_1 , and concluding anything requires accounting for it as well as for the higher operations.

30.3 Twisting morphisms and Maurer–Cartan

Definition 30.10. Let $\mathcal{C}$ be a coaugmented cooperad and A a dg module. The graded space $\text{Hom}_{\mathbb{S}}(\mathcal{C}, \text{End}_A)$ carries a *convolution* product $\star$, formed by comultiplying in $\mathcal{C}$, applying the two maps, and composing in End_A , together with a differential ∂ induced by those of $\mathcal{C}$ and A . A degree -1 element α with $\alpha \circ \eta = 0$ is a *twisting morphism* if it satisfies the *Maurer–Cartan equation*

$$\partial(\alpha) + \alpha \star \alpha = 0.$$

The equation is not exotic. It is the condition, familiar from deformation theory, under which a perturbed differential still squares to zero.

Exercise 30.11 (medium). Let $(\mathfrak{g}, d, [\cdot, \cdot])$ be a differential graded Lie algebra and $x \in \mathfrak{g}^1$. Show that

$$d_x := d + [x, -]$$

satisfies $d_x^2 = 0$ if and only if $dx + \frac{1}{2}[x, x] = 0$.

Hint. Expand $d_x^2(y)$ using the Leibniz rule $d[x, y] = [dx, y] - [x, dy]$, valid because $|x| = 1$, and then use the graded Jacobi identity to rewrite $[x, [x, y]]$.

Solution. For $y \in \mathfrak{g}$,

$$d_x^2(y) = d(dy + [x, y]) + [x, dy + [x, y]] = d[x, y] + [x, dy] + [x, [x, y]].$$

Since $|x| = 1$ the Leibniz rule reads $d[x, y] = [dx, y] - [x, dy]$, so the middle terms cancel and

$$d_x^2(y) = [dx, y] + [x, [x, y]].$$

The graded Jacobi identity for an odd element gives $[x, [x, y]] = \frac{1}{2}[[x, x], y]$. Hence

$$d_x^2(y) = \left[dx + \frac{1}{2}[x, x], y \right],$$

which vanishes for all y exactly when $dx + \frac{1}{2}[x, x] = 0$.

Remark 30.12 (The same statement, operadically). [Exercise 30.11](#) says that a Maurer–Cartan element is precisely a perturbation of a differential that preserves the property of squaring to zero. The operadic version of [Definition 30.10](#) says the same thing, and setting it up is a matter of bookkeeping: extend a twisting morphism α to a map $\tilde{\alpha}$ defined on all of $\mathcal{C}$ by declaring its value on the coaugmentation to be the differential d_A , and then compare $\tilde{\alpha} \star \tilde{\alpha}$ with $\partial(\alpha) + \alpha \star \alpha$.

Two things make that comparison delicate, and both are worth anticipating. The terms of $\tilde{\alpha} \star \tilde{\alpha}$ split according to where the coaugmentation lands, and only some of them reproduce $\partial\alpha$; and the signs are governed by the Koszul rule of [Lemma 28.8](#), so they have to be tracked rather than guessed. Carrying it out is the natural exercise, and the payoff is that twisting morphisms, the bar–cobar adjunction and ∞ -morphisms all become instances of one equation.

Remark 30.13 (Forward). Two threads run into [Chapter 31](#). The homotopy transfer theorem — that a $\mathcal{P}_\infty$ -structure can be pushed along a homotopy equivalence of underlying complexes — is proved by producing a twisting morphism, so it is an application of [Section 30.3](#). And the A_∞ -structures of [Example 30.7](#), transferred onto cohomology, turn out to be a classical construction in disguise.

30.4 What to carry forward

- Schur functors and the dictionary ([Definition 30.1](#) and [Example 30.2](#)).
- The composition product is composition of Schur functors ([Definition 30.3](#)); making it explicit involves a wreath product ([Remark 30.4](#)).
- Weight grading fixes the degrees of a homotopy structure ([Proposition 30.6](#)); for A_∞ , $|m_k| = k - 2$ ([Example 30.7](#) and [Exercise 30.8](#)).
- Maurer–Cartan is the condition for a perturbed differential to square to zero ([Exercise 30.11](#)), and the operadic form is the same statement ([Definition 30.10](#) and [Remark 30.12](#)).

Chapter 31

Lie models, the rational Hurewicz map, and A_∞ -structures

The last chapter of the book collects three things that have been converging since Part III: the algebraic model of a rational homotopy type, the map relating homotopy to homology in that model, and the structure that survives when an algebra is replaced by its cohomology.

The third is the deepest. Passing to cohomology destroys a differential graded algebra — but not entirely, and what remains is an A_∞ -structure whose higher operations are, in the classical language, Massey products.

31.1 Lie models

Theorem 31.1 (Quillen; recalled). *The rational homotopy category of simply connected spaces is equivalent to the homotopy category of connected differential graded Lie algebras over $\mathbb{Q}$. Under the equivalence a space X corresponds to a dg Lie algebra L_X with $H_*(L_X) \cong \pi_*(\Omega X) \otimes \mathbb{Q}$, the homotopy Lie algebra of [Definition 25.5](#).*

So there are two algebraic models of a rational homotopy type: the commutative one of [Chapter 24](#) and the Lie one here. They are related by duality, and each is convenient for different questions — the commutative model computes cohomology, the Lie model computes homotopy.

Exercise 31.2 (medium). Let L be a graded Lie algebra. Show that

$$\Delta(x) = x \otimes 1 + 1 \otimes x \quad (x \in L)$$

extends to a map of algebras $U(L) \rightarrow U(L) \otimes U(L)$, making $U(L)$ a Hopf algebra in which every element of L is primitive.

Hint. By the universal property of U , it is enough to check that $x \mapsto x \otimes 1 + 1 \otimes x$ is a map of graded Lie algebras into the commutator Lie algebra of $U(L) \otimes U(L)$.

Solution. Write $a = x \otimes 1$ and $b = 1 \otimes y$. In the graded tensor product of algebras $ab = x \otimes y$, while $ba = (1 \otimes y)(x \otimes 1) = (-1)^{|x||y|} x \otimes y$, so the graded commutator is

$$[a, b] = ab - (-1)^{|a||b|} ba = x \otimes y - (-1)^{|x||y|} (-1)^{|x||y|} x \otimes y = 0.$$

The cross terms therefore vanish, and

$$[\Delta x, \Delta y] = [x, y] \otimes 1 + 1 \otimes [x, y] = \Delta[x, y].$$

So Δ is a map of Lie algebras $L \rightarrow (U(L) \otimes U(L))_{\text{Lie}}$, and the universal property of U extends it uniquely to an algebra map on $U(L)$. Coassociativity and counitality are checked on L , where they are immediate, and hold on $U(L)$ because both sides are algebra maps agreeing on generators. Every $x \in L$ is primitive by construction.

Remark 31.3 (Other properties of U). [Exercise 31.2](#) is one of a family. The same style of argument — compare universal properties, never compute — identifies U of an abelian Lie algebra, describes U of a direct sum, and produces a presentation of $U(L)$ by generators and relations. [Exercise 25.9](#) was another member of the family, needed in [Chapter 25](#) for the growth estimate.

One of these deserves a warning in advance. The natural guess for $U(L \oplus M)$ is $U(L) \otimes U(M)$, and the guess is correct — but not because “left adjoints preserve coproducts”. The coproduct of associative algebras is the free product, not the tensor product; the tensor product is the coproduct in the category of cocommutative Hopf algebras, which is where [Exercise 31.2](#) places $U(L)$. Anyone deriving the identity should say which category the coproduct is being taken in.

31.2 The rational Hurewicz map

Definition 31.4. The *rational Hurewicz map* is the rationalisation of the classical one,

$$h: \pi_k(X) \otimes \mathbb{Q} \longrightarrow H_k(X; \mathbb{Q}).$$

By [Theorem 24.16](#) the source is $\text{Hom}(V^k, \mathbb{Q})$ for the minimal model $(\Lambda V, d)$, and the target is dual to $H^k(\Lambda V, d)$. So h is dual to a map $H^k(\Lambda V, d) \rightarrow V^k$, and there is only one natural candidate: the projection of ΛV onto its indecomposables $V = \Lambda^{\geq 1} V / \Lambda^{\geq 2} V$.

Example 31.5 (The two-sphere). Take $X = S^2$, with minimal model $(\Lambda(x, y), dy = x^2)$, $|x| = 2$, $|y| = 3$, from [Example 24.5](#).

On the topological side, $\pi_2(S^2) \otimes \mathbb{Q} = \mathbb{Q}$ maps isomorphically to $H_2(S^2; \mathbb{Q}) = \mathbb{Q}$ — the Hurewicz theorem in the bottom degree — while $\pi_3(S^2) \otimes \mathbb{Q} = \mathbb{Q}$, generated by the Hopf map, maps to $H_3(S^2; \mathbb{Q}) = 0$, so h vanishes in degree three.

On the algebraic side, [Exercise 24.6](#) computed $H^*(\Lambda V, d) = \mathbb{Q}\{1, x\}$. The projection onto indecomposables carries the class of x in H^2 to $x \in V^2$, an isomorphism; and $H^3 = 0$ maps to $V^3 = \mathbb{Q}y$ by the zero map.

The two computations agree degree by degree, and the vanishing in degree three records on one side that $H_3(S^2) = 0$ and on the other that y is not a cocycle, $dy = x^2 \neq 0$. The Hopf map is invisible to homology for exactly the reason that y had to be adjoined.

Remark 31.6 (Proving the duality, and what it gives). Establishing the duality in general is the natural exercise here. Both maps are natural, [Example 31.5](#) checks the smallest case, and the pairing between $H^*(\Lambda V, d)$ and $H_*(X; \mathbb{Q})$ comes from [Theorem 24.1](#); what has to be done is to check the identification on generators, and then to see that minimality is what makes the indecomposable quotient the right target.

The consequence worth having is the Cartan–Serre theorem. If X is elliptic in the sense of [Definition 25.13](#) then V is finite-dimensional, so only finitely many degrees carry rational homotopy at all; the dichotomy of [Theorem 25.15](#) becomes a statement about how many generators a minimal model can have. Rational homotopy theory turns a question about infinitely many homotopy groups into a question about the size of one graded vector space.

31.3 Transferred structures and Massey products

Theorem 31.7 (Homotopy transfer; recalled). *Let (A, d_A) be a dg algebra and let $(H, 0)$ be a complex equipped with a deformation retraction of A onto H : maps $i: H \rightarrow A$, $p: A \rightarrow H$ and a homotopy h with $pi = \text{id}$ and $ip - \text{id} = d_A h + h d_A$. Then H carries an A_∞ -structure whose m_2 is the transported product and whose higher operations are built from h , in such a way that i extends to an ∞ -quasi-isomorphism.*

Taking $H = H^(A)$ equips the cohomology of any dg algebra with such a structure.*

So cohomology remembers more than its ring structure: it remembers a family of higher operations $m_3, m_4, \dots$, of the degrees computed in [Example 30.7](#). The classical name for the first of them is older than the operadic language.

Definition 31.8. Let a, b, c be cohomology classes with $ab = 0$ and $bc = 0$. Choose cochain representatives together with u, v satisfying $du = ab$ and $dv = bc$. The *triple Massey product* is

$$\langle a, b, c \rangle = [uc \pm av],$$

well defined as a coset of $aH + Hc$.

Exercise 31.9 (medium). Verify that [Definition 31.8](#) produces a cocycle, and that changing u and v changes its class by an element of $aH + Hc$.

Hint. Differentiate $uc \pm av$ with the Leibniz rule and the defining equations. For the second part, note that two choices of u differ by a cocycle.

Solution. Since $du = ab$, $dv = bc$, and a and c are cocycles,

$$d(uc \pm av) = (du)c \pm (-1)^{|a|}a(dv) = abc \pm (-1)^{|a|}abc,$$

and with the sign chosen as in [Definition 31.8](#) the terms cancel, so $uc \pm av$ is a cocycle.

If u' also satisfies $du' = ab$ then $u' - u$ is a cocycle, so replacing u by u' alters the class by $[u' - u]c \in Hc$; likewise a different v alters it by an element of aH . Hence the product is well defined modulo $aH + Hc$.

Example 31.10 (Vanishing cup products, nonzero Massey product). Let X be the complement of the Borromean rings in S^3 . All cup products of one-dimensional classes vanish — any two of the three circles are unlinked — so the cohomology ring carries no information beyond its groups. The triple Massey product of the three generators is nevertheless nonzero, and it detects the linking that the pairwise products cannot.

This is the phenomenon of [Theorem 18.9](#) again: a ring with nothing in it, and a finer operation that sees the difference. Here the operation is not a Steenrod square but a transferred m_3 .

Remark 31.11 (What is left, and what it costs). Three statements complete the picture, and each is a reasonable exercise.

First, [Definition 31.8](#) is the transferred m_3 of [Theorem 31.7](#). Matching the two means comparing the defining equations of a Massey product with the A_∞ relations of [Exercise 30.8](#): the elements u, v chosen above are values of the homotopy h , and the indeterminacy of [Exercise 31.9](#) is the freedom in choosing h . Higher Massey products correspond in the same way to $m_4, m_5, \dots$, with signs governed by the Koszul rule.

Second, formality — the existence of a quasi-isomorphism between A and its cohomology with zero differential — forces the higher operations, hence all higher Massey products, to vanish.

Third, and this is the caution: the converse fails. Vanishing Massey products do not imply formality, because the transferred structure can be nontrivial in ways that no finite family of products detects. The book states this without proof; what it does supply is the reason the question is subtle, namely that m_k for large k carries information no m_3 can see.

31.4 Closing

Remark 31.12 (The arc). It is worth looking back along the route.

Chapter 2 could not distinguish two lens spaces; neither could Chapter 4 with homology, Chapter 7 with the cohomology ring, or Chapter 13 with homotopy groups. Chapter 18 produced an operation finer than all of them. Example 31.10 shows the same pattern once more, in a setting where the finer operation is a transferred product rather than a Steenrod square.

The recurring shape is this: an invariant is built, computed until it fails, and the failure names the next invariant. Cup products failed because they are degree-wise, and operations repaired that. Strict constructions failed because they are not homotopy invariant, and derived functors repaired that. Homology failed to see the universal cover, and local coefficients repaired that. In every case the repair was already visible in the shape of the failure, which is why the failures were worth studying carefully.

The webapp accompanying this book asks harder questions of the same kind. A reader who has worked the exercises here has the technique for them; what remains is the synthesis, and that is the part no book can do for anyone.

31.5 What to carry forward

- Two models of a rational homotopy type, commutative and Lie (Theorem 31.1).
- $U(L)$ is a Hopf algebra with L primitive (Exercise 31.2); derive properties of U from universal properties, minding which category a coproduct is taken in (Remark 31.3).
- The rational Hurewicz map is dual to the projection onto indecomposables (Example 31.5 and Remark 31.6), whence Cartan–Serre.
- Cohomology carries a transferred A_∞ -structure (Theorem 31.7); its m_3 is the triple Massey product (Definition 31.8), a coset and not an element (Exercise 31.9).
- Formality kills the higher operations; their vanishing does not imply formality (Remark 31.11).

References

How this list is organised

The first section lists the standard references, grouped by the part of this book they support, with a note on what each is good for. The second collects the results this book states without proving, so that each has a precise home. The third records the sources against which the exercise programme was designed, as [required](#) of any book whose problems are built from the problems of others.

Locators — page, section or exercise numbers — are given only where they have been checked against the source. Where a result is cited at the level of a whole book, that is deliberate: an unverified locator is worse than none, since it sends a reader to the wrong place with false confidence.

Standard references

Parts I and II: foundations and machinery

A. HATCHER, *Algebraic Topology*, Cambridge University Press, 2002. The default reference for everything in Parts I and II: van Kampen, covering spaces, cellular homology, universal coefficients, Künneth, cup products, Poincaré duality, Whitehead's theorems, Postnikov towers, obstruction theory and the Steenrod squares. Chapters 0–3 cover Part I of this book almost exactly; Chapter 4 and the additional sections 4.L and 3.H cover most of Part II. Freely posted by the author at

<https://pi.math.cornell.edu/~hatcher/AT/ATpage.html>

J. P. MAY, *A Concise Course in Algebraic Topology*, University of Chicago Press, 1999. Terse and structural where Hatcher is discursive and geometric. The treatment of cofibrations and the homotopy extension property (Chapter 6) is the one to read alongside [Chapter 11](#); the chapters on Postnikov towers and on characteristic classes are correspondingly compressed. Freely posted at

<https://math.uchicago.edu/~may/CONCISE/ConciseRevised.pdf>

J. W. MILNOR AND J. D. STASHEFF, *Characteristic Classes*, Annals of Mathematics Studies 76, Princeton University Press, 1974. The reference for [Chapters 12](#) and [26](#): Stiefel–Whitney classes and their axioms, the Whitney formula, Chern classes, the Thom isomorphism, the Euler class, and the Wu formula.

A. HATCHER, *Vector Bundles and K-Theory*, version 2.2, 2017; a book in progress. Complementary to Milnor–Stasheff, and closer in style to this book. Freely posted at

<https://pi.math.cornell.edu/~hatcher/VBKT/VBpage.html>

Part III: spectral sequences and operations

A. HATCHER, *Spectral Sequences in Algebraic Topology*; an unpublished manuscript. The reference for [Chapters 15](#) and [16](#): its first chapter is the best available account of how one actually runs a Serre spectral sequence, as opposed to how one constructs it. Freely posted at

<https://pi.math.cornell.edu/~hatcher/SSAT/SSATpage.html>

N. E. STEENROD AND D. B. A. EPSTEIN, *Cohomology Operations*, Annals of Mathematics Studies 50, Princeton University Press, 1962. The construction of the squares, the Adem relations and the admissible basis of [Chapters 18](#) and [27](#).

Part IV: abstract homotopy theory

E. RIEHL, *Categorical Homotopy Theory*, New Mathematical Monographs 24, Cambridge University Press, 2014. The reference for most of Part IV: homotopy limits and colimits, derived functors as Kan extensions, model structures, simplicial sets, quasi-categories, the homotopy coherent nerve and its left adjoint. [Chapters 20](#) to [23](#) and [29](#) all follow its treatment in outline. Freely posted at

<https://emilyriehl.github.io/files/cathtpy.pdf>

S. MAC LANE, *Categories for the Working Mathematician*, 2nd edition, Graduate Texts in Mathematics 5, Springer, 1998. Adjunctions, Kan extensions, ends and coends: the background [Chapter 20](#) assumes.

M. HOVEY, *Model Categories*, Mathematical Surveys and Monographs 63, American Mathematical Society, 1999; and D. G. QUILLEN, *Homotopical Algebra*, Lecture Notes in Mathematics 43, Springer, 1967. The axioms of [Chapter 22](#) and their consequences, in the modern and the original treatments respectively.

P. G. GOERSS AND J. F. JARDINE, *Simplicial Homotopy Theory*, Progress in Mathematics 174, Birkhäuser, 1999. Simplicial sets in detail, including the proof that simplicial groups are Kan complexes ([Theorem 23.5](#)) and the Dold–Kan correspondence.

J. LURIE, *Higher Topos Theory*, Annals of Mathematics Studies 170, Princeton University Press, 2009. The systematic development of quasi-categories, well beyond [Chapters 23](#) and [29](#).

Y. FÉLIX, S. HALPERIN AND J.-C. THOMAS, *Rational Homotopy Theory*, Graduate Texts in Mathematics 205, Springer, 2001. The standard reference for [Chapters 24](#), [25](#) and [31](#): Sullivan models, minimal models, spatial realisation, the homotopy Lie algebra, and the elliptic–hyperbolic dichotomy.

A. BERGLUND, *Rational Homotopy Theory*, lecture notes, Stockholm University, 2012. A shorter route to the same material. Posted at

<https://staff.math.su.se/alexb/rathom2.pdf>

Part V: research preparation

J.-L. LODAY AND B. VALLETTE, *Algebraic Operads*, Grundlehren der mathematischen Wissenschaften 346, Springer, 2012. The reference for [Chapter 30](#): S-modules, the composition product, Koszul duality, twisting morphisms, and the homotopy transfer theorem of [Theorem 31.7](#). Freely posted at

<https://www.math.univ-paris13.fr/~vallette/Operads.pdf>

S. SCHWEDE, *Symmetric Spectra*, version 3.0, 2012; a preliminary book draft. The reference for [Chapter 28](#), including the pathologies of naive homotopy groups that [Warning 28.5](#) describes. The document states on its first page that it is incomplete, so its

numbering is not stable between versions. Freely posted at

<https://www.math.uni-bonn.de/people/schwede/SymSpec-v3.pdf>

M. HOVEY, B. SHIPLEY AND J. SMITH, *Symmetric spectra*, Journal of the American Mathematical Society 13 (2000), 149–208. The original construction.

Results quoted without proof

This book states a number of theorems it does not prove. Most are standard and covered by the references above — van Kampen, excision, Poincaré duality, the construction of the Serre spectral sequence, the existence of minimal models. The following are the ones where the omission is substantial, and each is flagged in the text at the point of use.

- **The homotopy classification of lens spaces** (Example 7.10 and Warning 7.11). Classical, and due to J. H. C. Whitehead; the invariant is the linking form, and the surrounding theory is simple-homotopy theory. See M. M. Cohen, *A Course in Simple-Homotopy Theory*, Graduate Texts in Mathematics 10, Springer, 1973, and J. W. Milnor, *Whitehead torsion*, Bulletin of the American Mathematical Society 72 (1966), 358–426.
- **Serre’s computation of $\pi_*(S^n) \otimes \mathbb{Q}$** (Theorem 19.10 and Warning 19.11). J.-P. Serre, *Groupes d’homotopie et classes de groupes abéliens*, Annals of Mathematics 58 (1953), 258–294. Hatcher’s spectral sequences manuscript gives a modern account; Chapter 24 obtains the same answer by another route.
- **Adams’ theorem on Hopf invariant one** (Remarks 16.17 and 27.11). J. F. Adams, *On the non-existence of elements of Hopf invariant one*, Annals of Mathematics 72 (1960), 20–104. The later K -theoretic proof is due to Adams and Atiyah, *K-theory and the Hopf invariant*, Quarterly Journal of Mathematics (Oxford), 1966. Chapter 27 proves the elementary part — that the dimension must be a power of two.
- **Milnor–Moore** (Theorem 25.7). J. W. Milnor and J. C. Moore, *On the structure of Hopf algebras*, Annals of Mathematics 81 (1965), 211–264.
- **Bott’s integrality theorem** (Remark 26.11), constraining which total Chern classes occur. See Milnor–Stasheff.
- **Higher Massey products and formality** (Remark 31.11). The higher products are due to D. Kraines, *Massey higher products*, Transactions of the American Mathematical Society 124 (1966), 431–449; that formality forces them to vanish, and that the converse fails, is discussed in P. Deligne, P. Griffiths, J. Morgan and D. Sullivan, *Real homotopy theory of Kähler manifolds*, Inventiones Mathematicae 29 (1975), 245–274.

Sources of inspiration, and originality

This book was written to prepare a reader for a separate collection of exercises, whose problems are drawn from the eight sources listed above — Hatcher’s three books, May, Berglund, Riehl, Schwede, and Loday–Vallette. Those problems shaped this book: they

determined which techniques are developed, in what order, and to what depth. In that sense the exercise sets of those texts are the source of inspiration for everything in these pages, and the debt is acknowledged here rather than left implicit.

What the book does *not* do is reproduce them. No exercise in this book is an exercise from any of those sources, reworded or otherwise. Where a source exercise identified a technique worth teaching, the book develops the technique and then poses a different problem using it — usually on a different space, in a different dimension, or from the opposite direction. Each chapter’s source file opens with a comment recording precisely which problems it was written to prepare for and which it deliberately leaves alone; those comments are part of the manuscript and are not printed.

Three practices support that claim. First, the general theorems of the subject are proved here as theorems, not smuggled in as exercises: a reader is never asked to rediscover van Kampen or the Adem relations. Second, where the book would otherwise state a target result outright, it states the ingredients and names what remains — the passages beginning “the natural exercise here” are that device, and they are deliberate rather than evasive. Third, the manuscript is checked mechanically against the source problems: a script compares every printed sentence of this book with the statements, hints and solutions of all eighty-four, and reports any run of ten consecutive words in common, with mathematical notation excluded from the comparison so that shared symbols cannot trigger or mask a match. At the time of writing it reports one accepted collision, the statement of the Wang sequence, which is a standard theorem this book is obliged to state.

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